Debt Indexation, Determinacy, and Inflation*

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Abstract

Contrary to popular belief, inflation-indexed government debt can boost inflation in response to deficit shocks, conditional on a lack of sufficient future fiscal backing. I formalize this insight through a calibrated HANK model with multiple asset types, showing that the cumulative first-year price increase following an unfunded government spending shock of 1% of quarterly GDP is 0.23 percentage points larger when 27% of the government debt stock is indexed to inflation, as is the case in the United Kingdom, relative to a baseline case without indexed debt. Inflation-indexed debt makes the price level partially backward-looking through the government debt valuation equation, thereby causing additional inflationary pressure. Empirical evidence from a large, narratively identified fiscal deficit shock supports this finding, which has additional implications for the distinction between 'fiscally-led' mechanisms and 'HANK-type' mechanisms surpassing Ricardian Equivalence, as well as for the distributional consequences of fiscally-driven inflation.

Keywords: Debt Indexation, Government Debt, Fiscal-Monetary Interactions.
JEL Codes: E52, E63, H63

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1 Introduction

Debates over the causes of the post-2020 inflationary episode highlight rival theories, such as labor-market tightness, supply-chain disruptions, or fiscal deficits and the associated monetary response. This paper adds a novel, yet overlooked aspect to this ongoing debate: the *composition* of government debt, with a particular distinction between purely nominal debt and *inflation-indexed debt*, which differs from the commonly modeled government obligations as its face value changes with the gross rate of inflation. As the government budget balance plays a crucial role for aggregate demand, and thereby co-determines the price level, an interesting feedback loop arises when the face value of a part of government debt itself changes with gross inflation. In this paper, I show the qualitative and quantitative importance of this mechanism, which is powerful conditional on fiscal deficits not being completely backed by future expected surpluses.

Indexed debt is not a mere theoretical curiosity. Figure 1 shows the share of inflation-indexed debt as part of the overall sovereign debt stock over time in a number of countries. While there is considerable heterogeneity across countries, indexed bonds are present across the board, and have been so for the past three decades. This paper mostly focuses on the United Kingdom (U.K.) and the United States (U.S.), since these two indexed debt markets are among the largest ones in both absolute and relative terms.

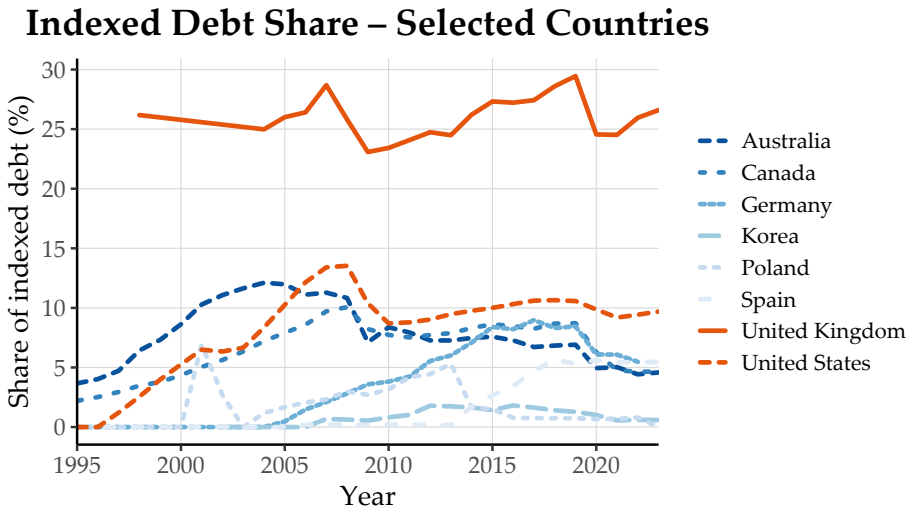


Figure 1: The market value-weighted share of inflation-indexed debt in sovereign debt portfolios over time. Data source: BIS (2024).

Analyzing the role of the government debt structure for inflation requires a delicate treatment of the interactions between fiscal and monetary policy. A *fiscally-led policy mix* (defined here through a fiscal authority committing to deficit-financed expenditures without plans for sufficient stabilizing taxation, which amounts to at least partially *unfunded deficits* for the remainder of this paper) enhances the role of fiscal policy as a driver of inflationary dynamics in macroeconomic general equilibrium models (Leeper, 1991; Sims, 2011; Ascari et al., 2023; Rachel and Ravn, 2025). I therefore motivate this paper further with a real-world example of fiscal policy committing to unfunded

deficits and plausibly forcing the hand of the monetary policy authority (and thus informing a possibly fiscally-led policy mix): the 'U.K. mini-budget' episode in September 2022.[1]

Figure 2 plots average market-implied expectations of the 'Bank rate', the main monetary policy rate set by the Bank of England, in the window around the 'mini-budget' announcement and its subsequent cancellation. The first solid line depicts the date of a Bank of England Monetary Policy Committee (MPC) meeting, which occurred ahead of the announcement of the 'mini-budget' fiscal policy measure, with the MPC minutes being released on the 22nd of September 2022, one day ahead of the fiscal policy announcement. This timing is useful for the argument insofar as the meeting likely communicated the Bank of England's stance on future rate changes clearly, taking all available information up to that point into account (Braun et al., 2024). Nonetheless, implied policy rates rose sharply in the subsequent days, just after the announcement of the mini-budget (denoted by the first dotted line), with the shift amounting to a 120bps peak increase in expected policy rates for the upcoming twelve months ahead. After the scraping of the mini-budget (second dotted line), expected policy rates swiftly returned to their 'pre-shock' levels.[2]

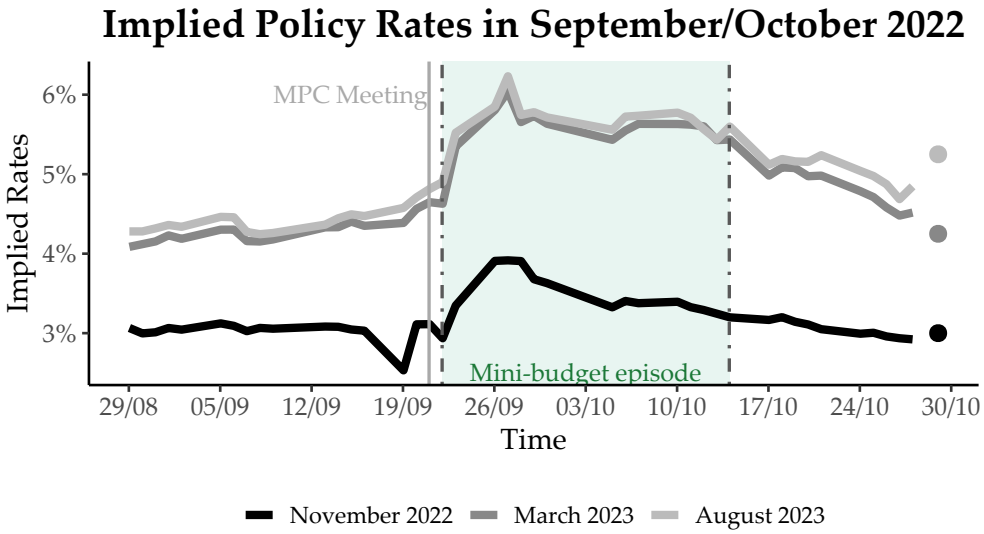


Figure 2: Expectations of Bank rates in the United Kingdom for three scheduled MPC meetings after the 'mini-budget' announcement in September 2022, defined as the break-even rates under which Overnight Indexed Swaps are fairly priced at the prevailing forward interest rate curve (Bloomberg data series GB0BFR). The dots at the end depict the factual policy rates after each of the meetings.

This event resonates well with the possible idea of unfunded deficits forcing the hand of the monetary policy authority, which I label as a (at least partially) fiscally-led policy mix: financial market participants clearly expected changes to the monetary policy stance beyond the very short term, plausibly in response to an announced fiscal policy measure, although risk or liquidity premiums

[1]For a more detailed argument related to this fiscal shock, see Leeper (2023), NIESR (2022), and section 2.1 of this paper.

[2]The expected monetary policy response was partially driven by a concurrent funding mismatch in liability-driven investment strategies of defined-benefit pension funds that were closely tied to movements in yields of sovereign bonds. See Pinter (2023) for a detailed exposition of this point.

likely also contributed to the observed movements.

I motivate the relevance of indexed debt for inflation further, using data to narratively pin down the true shock component of the 'U.K. mini-budget', tracking it through market expectations on sovereign deficits and government bond price revaluations. I estimate a sizable *deficit-inflation multiplier*, which grows with the share of inflation-indexed debt. My second empirical finding, based on an exercise with local projections using exogenous news about future U.S. federal deficits, shows that inflation-indexed debt boosts observed consumer price inflation in response to unfunded deficits when the share of inflation-indexed debt increases alongside the deficit news.

Next, I introduce inflation-indexed debt in a one-equation model to pin down the main mechanism by which indexed debt matters for the price level. It does so through the government debt valuation equation, in which the price level becomes partially backwards-looking. Thus, previous price levels matter for the determination of the current price level, even without further sources of stickiness in the economy. For a corresponding general equilibrium economy with inflation-indexed debt, I prove some conditions supporting the uniqueness of the corresponding dynamic stationary general equilibrium.

I then analyze the effects of fiscal deficit shocks in a setting featuring (i) non-Ricardian fiscal policy through imperfect risk-sharing among households and (ii) the presence of inflation-indexed debt. This framework is a heterogeneous-agent New Keynesian model á la Kaplan et al. (2018), which I solve making use of the methods pioneered in Auclert et al. (2021) to solve heterogeneous-agent models up to first-order in aggregate variables, while preserving heterogeneity with respect to the individual agents in the economy. I additionally pay attention to the different insurance properties of inflation-indexed debt in models with incomplete markets, accounting for the possible windfall gains that can be earned by governments, following Brunnermeier et al. (2024). Inflation-indexed debt matters quantitatively: the standard deviation of inflation relative to that of government spending increases from 0.199 without indexed debt to 0.494 with the U.K. debt portfolio. In terms of the level impact, for an economy with a 27% share of inflation-indexed debt in the sovereign debt portfolio (as observed in the U.K.), the cumulative first-year price increase following an unfunded government spending shock of 1% of quarterly GDP is 0.23 percentage points larger than in the baseline case without indexed debt, which is quantitatively highly relevant. I furthermore establish that the classic notions of 'active/passive monetary/fiscal policy', as derived by Leeper (1991), do not directly translate into the world with inflation-indexed debt, even though similarities in the determination of saddle path-stable equilibria prevail.

As a final corollary, I prove that inflation-indexed debt allows to discriminate to some degree between various mechanisms overcoming Ricardian Equivalence. Once *long-term* inflation-indexed debt is present, the mechanisms through which fiscal policy induces real effects under incomplete markets generally work differently than mechanisms arising through fiscally-led policy mixes, and explicitly accounting for deficit-driven inflation dynamics cannot be sidelined even when households respond to fiscal policy in a non-Ricardian way irrespective of the perceived fiscal backing.

Literature Review

This paper contributes to the burgeoning literature on fiscal-monetary interactions, pioneered in Sargent and Wallace (1981) and formalized through Leeper (1991). Initial contributions focusing on the possibility of a fiscally-led policy mix include Sims (1994) and Woodford (1995).[3] More succinct summaries of the literature are provided by Leeper and Leith (2016) and Cochrane (2023). Leeper et al. (2017), Bassetto and Cui (2018), Miao and Su (2024), Ascari et al. (2023), and Bianchi et al. (2023) provide model-based advances in analyzing the effects of fiscally-led policy configurations, while empirical support for the possibility of fiscally driven inflation has been developed in Jacobson et al. (2024), Cochrane (2022a), Cochrane (2022b), Chen et al. (2022), Cloyne et al. (2023), Ascari et al. (2024), and Barro and Bianchi (2026).[4] A narrative example of a specific fiscal shock informing inflation rates is provided by Hazell and Hobler (2024), who focus on the 2021 Georgia Senate election runoff.

The focus of models emphasizing the link between fiscal-monetary policy interactions and the price level recently shifted towards models with explicitly *non-Ricardian* fiscal policy, thereby making real interest rates endogenous to fiscal frameworks. This endogeneity is important insofar as fiscal-monetary interactions ultimately constitute criteria that constrain the transversality condition on government debt to hold for a candidate price level, but the transversality condition itself depends on the real interest rate.[5,6]

Many such models fall under the category of heterogeneous-agent New Keynesian (HANK) frameworks. Brunnermeier et al. (2022), Kaplan et al. (2026), Campos et al. (2024), and Kwicklis (2025) have all considered the effects of fiscal policy on inflation in rich heterogeneous-agent frameworks. A second type of such models considers New Keynesian models with mortality frictions, exemplified by Angeletos et al. (2024), Dupraz and Rogantini Picco (2025), and Rachel and Ravn (2025). Angeletos et al. (2024) negate the need to consider fiscally-led policy specifications in quantitative analyses of inflationary dynamics, finding identical responses of inflation to expansionary fiscal shocks in such NK-OLG models under monetary-led policy specifications. An overview over the state-of-the-art is provided by Kaplan (2025b), and a complementing succinct mathematical description of the link between non-Ricardian properties of households and fiscal-monetary interactions can be found in Kaplan (2025a).

[3] To avoid confusion, I do not explicitly define the term 'Fiscal Theory of the Price Level' (FTPL) in this paper, which has initially been coined by Woodford (1995). Instead, I follow Brunnermeier et al. (2022), who define 'core aspects', or themes, commonplace in the FTPL literature, which broadly apply in this paper as well. These aspects are: (a) governments issue some (nominal) liabilities in their own currency, (b) the government faces a debt valuation equation over which it has some agency, (c) the price level plays a key role in ensuring debt sustainability in equilibrium, (d) additional assumptions, e.g., on fiscal-monetary policy mixes, complement equilibrium uniqueness, and (e) the framework could deliver price level uniqueness even without nominal frictions.

[4] The effects of monetary policy shocks, in turn, are also constrained and informed by the underlying fiscal reaction function, as shown by Bigio et al. (2024) and Caramp and Silva (2023, 2026).

[5] Brunnermeier et al. (2022) and Kaplan et al. (2026) provide conditions under which such models admit (unique) forward-looking equilibria expressed through the price level.

[6] Niepelt (2004) highlights the possible pitfalls in specifying a model with *real* obligations and the fiscal authority ultimately pinning down the price level. Appendix F discusses these concerns in detail and provides a concrete remedy.

My contribution is to introduce a second type of assets (inflation-indexed debt) with a feedback loop between asset holdings and the price level, quantifying the importance that such indexed debt has for inflation dynamics in a calibrated state-of-the-art macroeconomic model. In doing so, I specifically pay attention to the different insurance properties borne by the two types of debt, thereby mitigating concerns related to misspecifications of aggregate transversality conditions (Brunnermeier et al., 2024). The specific insurance properties of inflation-related financial market instruments are also highlighted by Bahaj et al. (2023).

I also contribute explicitly to the literature on inflation-linked government bonds. Such bonds were introduced in economic and financial research long ago, especially in relation to the introduction of TIPS in the U.S. in 1997. One of the earliest contributions in this field is Fischer (1975), who derives household demand for such assets in a multi-asset framework. The special insurance properties of such inflation-linked debt are extensively discussed in Campbell and Shiller (1996), Barr and Campbell (1997), Garcia and van Rixtel (2007), Gürkaynak et al. (2010) and Andreasen et al. (2021), while Barro (2003) characterizes the optimal indexed-debt issuance strategy for a fiscal authority operating under a tax-smoothing objective, emphasizing the relevance of indexed consol bonds. Notably, Sims (2013) briefly mentions the possible detrimental consequences of indexed debt in fiscally-led policy frameworks. This paper builds on his remarks, providing a rigorous framework nesting his intuitions. Schmid et al. (2026) provide a systematic analysis of inflation-indexed debt as a policy tool, emphasizing its role as an ex-ante commitment device against inflation. In this paper, I leverage the unique properties of inflation-indexed debt, which express themselves mostly through the induction of a backward-looking component in the government budget equilibrium condition and through the insurance premia they bear. This paper's focus thus effectively rests on the 'ex-post' effects that inflation-indexed debt can have in the face of unfunded increases in government purchases.

In the later sections of the paper, I rely on modern computational methods to efficiently solve and estimate heterogeneous-agent models, as in Kaplan et al. (2018) and Bayer and Luetticke (2020). In particular, I leverage the efficient computation algorithms of Auclert et al. (2021) and Auclert et al. (2024) to solve a model with heterogeneous households, two types of assets, and fiscal-monetary interactions.[7]

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The rest of the paper is structured as follows. Section 2 exposes the relevance of indexed debt for materialized and expected inflation in the face of fiscal shocks, after which I introduce inflation-indexed debt in a one-equation framework in section 3. I specify the main quantitative model in section 4. Section 5 discusses the calibration and the estimation of the model, and I present the quantitative findings in section 6. Section 7 isolates the main mechanism from the quantitative model in a simpler supplementary framework with long-term indexed debt. Section 8 concludes.

[7]To motivate the relevance of household heterogeneity applied to holdings of sovereign debt, figure A.1 in the appendix provides evidence on the skew of household holdings of such debt, sorted by their respective income decile. This skew is even more pronounced for inflation-indexed debt.

2 Empirical evidence: indexed debt and inflationary dynamics

To motivate the relevance of indexed debt as a possible driver of the net present value of government debt and, therefore, of price level dynamics through the government debt valuation equation, I provide two pieces of evidence. First, I build on Hazell and Hobler (2024) and provide a narrative analysis of a large fiscal shock, but in an environment with high levels of indexed debt. My headline estimate gives a larger inflation-compensation multiplier in response to deficit news, although the horizons and inflation measures differ. Second, I employ a long-running series of exogenous news about future U.S. federal deficits in a local projection to pin down the effects that inflation-indexed debt has on inflation itself when unexpected deficit-increasing policy occurs.

2.1 Narrative evidence on the effect of deficits under indexed debt: the 2022 U.K. 'mini-budget'

I now provide narrative evidence on the effects of indexed debt through a cleanly identified fiscal policy shock: the September 2022 U.K. fiscal policy announcement, commonly dubbed the 'mini-budget'. I focus on this specific shock for two reasons: first, the event was largely unexpected in terms of its magnitude, allowing a clear identification of the effects of fiscal shortfalls on inflation. Second, this exercise is a complement to Hazell and Hobler (2024), who exploit probabilistic variation on Democrat Senate control around the 2021 Georgia Senate run-off election to infer the expected effects of expansionary fiscal policy on the price level. This paper provides a similar exercise in an environment with high levels of inflation-indexed debt, complementing their estimates.

The institutional setup of U.K. fiscal policy serves as an excellent device for identifying the 2022 'mini-budget' episode as a clear fiscal shock. Fiscal policy in the U.K. is shaped by regular fiscal announcements, which set up the broad guidelines for expected sovereign income and spending in a given fiscal year. From 1980 to 2016, the larger 'budget announcement' usually occurred in early spring (coinciding with the beginning of a new fiscal year), supplemented by shorter budget statements in the fall of the same year. Between 2017 and 2019, the regular budget announcement was moved to fall, with the spring season being used usually for supplementary statements. Beginning in 2020, the main budget announcement started to take place in early spring again.

In spring 2022, Chancellor Rishi Sunak provided a budget statement, scheduled to be followed up by a full budget announcement in November 2022. In-between, and therefore outside of the usual bi-annual statement/announcement cycle, Chancellor Kwasi Kwarteng (who had since been appointed) presented a Ministerial Statement dubbed "The Growth Plan", with fiscal policy measures amounting to 150 Billion GBP, or approximately 5% of the GDP of the United Kingdom (NIESR, 2022). This statement did *not* constitute a budget announcement in the usual sense, being placed outside of the bi-annual statement cycle. The release of all budget statements made by the British government is usually supplemented with a concurrently released report by the Office for Budget Responsibility (OBR), an independent auditor supervising budgetary questions in the United Kingdom. In the case of the 'mini-budget', no such independent forecast of the budgetary

consequences of the statement was publicly released, as the ruling government denied the release of the forecast created by the OBR.[8]

The episode of early fall 2022 is characterized by this fiscal policy announcement and its expected effects. In particular, the effects of the fiscal policy announcement are (in the very short-term) plausibly shielded from unrelated monetary policy news (both in terms of the interest rate level and in terms of the signaling of the state of the economy), since the preceding Bank of England Monetary Policy Committee decision was released one day *before* the announcement of the fiscal policy measure, on September 22, 2022.

2.1.1 The size of the shock

The most important question is the *size* of the fiscal policy shock, which is *not* equal to the overall size of the fiscal package, as the policy announcement had been expected ahead of the budget statement. Ignoring this would bias the estimated effects of the policy announcement downwards by assuming a larger fiscal 'shock' than what has factually been observed. Additionally, the probability of the fiscal policy measures being implemented upon announcement need not equal 100%, which might also contribute to a bias of the estimates.

To address these points, I follow the lead of Hazell and Hobler (2024), albeit with some limitations related to data availability. First, I establish the expected degree of debt-financing of the announced fiscal measures through their impact on the budget balance. This provides benchmarks for the overall financing needs against which the unexpected component can be assessed. Second, I calculate the share of debt-financing that is unexpected. For the first element, the overall increase in government debt issuance, two estimates are plausible:

- The first is based on a direct reading of the corresponding budget statement.[9] A reading of the implied policy measures yields an increase of the Debt Management Office's Net Financing Requirement from GBP 161.7 billion to GBP 234.1 billion in 2022-23, such that the corresponding increase in borrowing requirements for that fiscal year would be GBP 72.4 billion.
- The second is an analysis by the *Institute of Fiscal Studies*, which predicted a GBP ~60 billion funding shortfall.[10]

The *shock* impacting the expected path of debt through the fiscal announcement is not equivalent to the sum of additional deficits, since the policy package had been expected, but its full extent was simply not known. To isolate the shock component, I exploit private sector forecasts on *Public*

[8]The forecast made by the OBR at that time has since been released, although it is only of limited relevance with respect to the eventual policy measures announced as the report was made 18 days ahead of the budget announcement, thus not capturing the full extent of the fiscal policy proposals. I therefore sideline this report for my analysis. The report can be found under: https://obr.uk/docs/dlm_uploads/FOI-Information-on-preparatory-work-for-the-mini-budget.pdf.

[9]The report is available under: https://www.gov.uk/government/publications/the-growth-plan-2022-documents.

[10]The report is available under: https://ifs.org.uk/articles/mini-budget-response.

Sector Net Borrowing, which are aggregated on a monthly basis and released by the U.K. Treasury.[11] These are forecasts of the factual borrowing requirement of the U.K. government in each fiscal year, provided both by financial market participants as well as other independent forecasters. I collect data on the forecasts provided in the period between September 1, 2022, and September 22, 2022 (i.e., until the day before the shock) and compare these forecasts with the ones collected between October 1, 2022, and October 10, 2022. Unfortunately, data is not collected at narrower time intervals around the 'mini-budget' announcement. This limitation can bias the estimated deficit-inflation multiplier in either direction.[12]

For October 2022 (i.e., after the announcement of the fiscal package), forecasts were provided by Barclays Capital, Goldman Sachs, JP Morgan, Beacon Economic Forecasting, CEBR, Heteronomics, ICAEW, Kern Consulting, and Oxford Economics. The total mean forecast revision of Public Sector Net Borrowing for the 2022-23 and 2023-24 Fiscal Years lies at GBP 47.4 billion, vastly exceeding all other non-crisis forecast revisions.[13] This confirms the initial intuition that the 'mini-budget' shock was indeed economically significant and largely unexpected. The following analysis therefore works with this two-year forecast revision of GBP 47.4 billion as the estimated funding shortfall. With annual nominal U.K. GDP at current market prices in 2022 having been GBP 2,581 billion (ONS series YBHA), the fiscal shock can be sized to around $\frac{47.4}{2581} \times 100 \approx 1.8365\%$ of annual GDP.

2.1.2 Linking the deficit shock to expected inflation

I now introduce data capturing changes to expected inflation through a high-frequency identification strategy. The analysis thus follows Hazell and Hobler (2024), postulating that around the 'mini-budget announcement' the dynamics of asset prices y_t can be summarized by the process:

$$y_t = \begin{cases} \varepsilon_t & \text{if } t < T, \\ \varepsilon_t + \alpha_t & \text{if } t \geq T, \end{cases} \tag{1}$$

where T denotes the time period at which fiscal stimulus occurred. ε_t is an arbitrary process describing asset price movements, and α_t is the effect induced by the fiscal package for all $t \geq T$. I set the shock period T to September 23, 2022, 09.30am, coinciding with the beginning of the budget statement in parliament. Denoting by j a counter of periods after the event, $\hat{\alpha}_{T+j} = y_{T+j} - \mathbb{E}_T \left[y_{T+j} | \alpha_{T+j} = 0 \right]$ is the estimate of the causal effect of the shock in the narrow time window around the announcement.

[11] The forecast summaries are available under: https://www.gov.uk/government/collections/data-forecasts.

[12] The multiplier is larger for a given change in inflation compensation when the measured borrowing shock is smaller. If October forecasters already expected the package to unwind, this could reduce the measured shock and raise the multiplier, while other intervening forecast news could work in the opposite direction. See appendix A.2 for further details on the narrative around the 'mini-budget'.

[13] The only periods with larger absolute adjustments in the expected two-year budget deficit forecast were April 2020 (GBP 147.4 billion), May 2020 (GBP 114.9 billion), and May 2009 (GBP 50 billion). Outside of the GFC and Covid periods, the largest absolute month-on-month average forecast revision was GBP 20.8 billion in October 2019, less than half of the size of the forecast change in October 2022.

The main quantity of interest is one-year ahead inflation compensation implied by GBP inflation swaps traded at the London Stock Exchange, which I use as a high-frequency proxy for expected inflation.[14] Since inflation swaps operate with a two-month indexation lag in the U.K., I adjust the prices of the swaps to reflect this lag, as done by Hazell and Hobler (2024). Figure 3 summarizes the movements of expected one-year ahead inflation, derived through inflation swaps, around the 'mini-budget' shock on September 23, 2022.

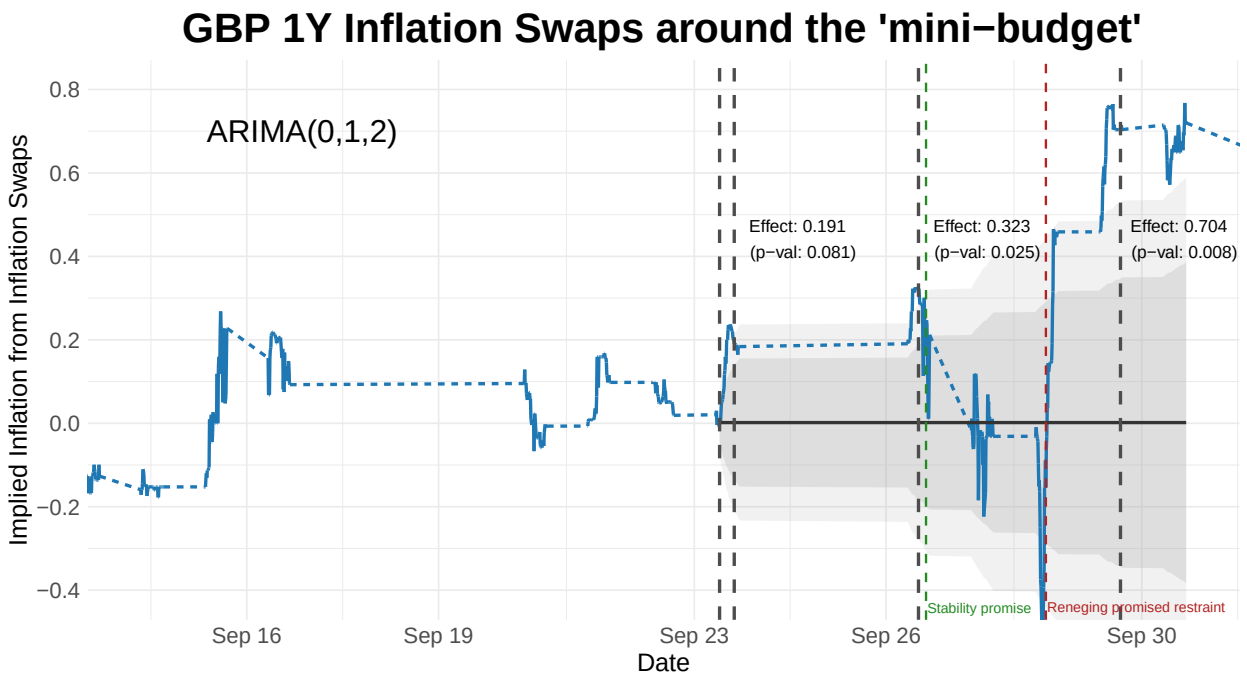


Figure 3: One-year GBP inflation compensation in the period around the 'mini-budget' shock, with data normalized to 0 on September 23, 2022, at 09:30am. The gray fan chart depicts 68% and 95% pointwise confidence intervals for a forecast of the swap rate from the moment of the shock onward, with the model being chosen in accordance with the Bayesian Information Criterion.

The gray dashed vertical lines are points at which meaningful estimates of changes to one-year ahead inflation expectations are recovered. The first vertical line depicts the beginning of the shock, as implied by the beginning of the budget speech announcing the 'mini-budget' measures in detail. The second line measures one-year ahead inflation expectations on the same day at 3:00pm, 5.5 hours after the budget speech commenced. Even though markets can credibly be expected to take a couple of days to incorporate fiscal news into forming inflation expectations (Bahaj et al., 2023), there is a significant response of implied inflation on the day of the 'mini-budget' announcement. Moving forward to September 26, i.e., the next trading day, the effect magnifies further, yielding an implied year-on-year inflationary response of 0.323% to the narratively identified shock.

Between the third and the fourth vertical line, implied inflation drops sharply. This drop is consistent with the expectation that the fiscal spending announcement might end up being unraveled,

[14] Inflation swaps do not provide a perfect measure of expected CPI inflation. Inflation compensation can also contain inflation-risk and liquidity premia, and indexed debt contracts in the U.K. are linked to RPI and therefore also reflect the RPI–CPI spread as well as institutional-demand components.

indicated by the green dashed line labeled 'Stability Promise'. I provide a narrative description of the events in this period in appendix A.2, including a brief description of the role played by the troubles on LDI markets.

On September 28 & 29, expected inflation increased significantly again. While by then other events might contaminate the evolution of inflation swap prices, the observed sharp appreciation perfectly coincides with statements of the Treasury that *despite* the market turmoil, the proposed fiscal package will be followed through, superseding previous statements of a release of a stabilizing medium-term fiscal plan. This event is depicted by the red dashed line. The elevated levels of expected inflation then continued to persist well into October, when an eventual unraveling occurred in parallel to an overhaul of the ruling government that enacted the fiscal package in the first place.

The Inflation Multiplier: I take the response after the Treasury reaffirmed implementation of the fiscal package as the headline estimate, since this window more closely reflects a credible commitment to the announced measures. The resulting one-year ahead inflation-compensation multiplier, which captures the change in one-year inflation compensation associated with a 1% deficit-to-annual-GDP shock, is $0.704/1.8365 \approx 0.383$ percentage points. The earlier September 26 response gives $0.323/1.8365 \approx 0.176$ percentage points, slightly below the two-year inflation multiplier of 0.19 reported by Hazell and Hobler (2024), although the horizons and inflation objects are not identical. The headline estimate also aligns more closely in timing with the forecast revision to borrowing introduced in the last subsection. The response of inflation compensation was persistent, as indicated by swaps for longer horizons depicted in figure 4.

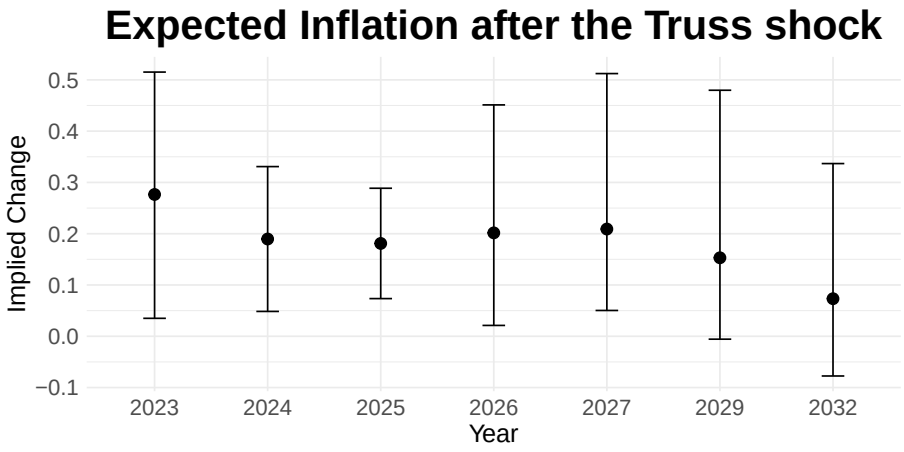


Figure 4: Change in inflation compensation at various horizons, as implied by GBP inflation swaps on September 26, 2022, at 12:00pm relative to the pre-shock period. The bins show 95% pointwise confidence intervals.

Appendix section A.2 provides further evidence on the nature of the 'mini-budget shock episode', which confirms an element of surprise in relation to the size of the unveiled fiscal package that contributed to the turmoil on financial markets reflected in the pricing of inflation swaps.

2.2 Evidence on the ex-post inflationary effect of inflation-indexed debt

A limitation of the previous narrative analysis is that the inflation measure is one of *expected* inflation recovered through the pricing of inflation swaps. I now address this, providing evidence on the effect of inflation-indexed debt on *realized* inflation. To do so, I leverage the time series of exogenous news about the present value of future U.S. federal deficits ε_t^F provided by Fritsch and Mazlish (2026), which separates changes in projected deficits on days of fiscal events from revisions driven by the macroeconomic outlook,[15] and combine it with a time series of the face value of inflation-indexed debt (TIPS) as a share of the overall face value of marketable U.S. sovereign debt, which I label θ_t. I take this to be the main indicator for the prevalence of inflation-indexed debt. Equipped with these time series, I estimate the following local projection to measure the dynamic impact of inflation-indexed debt on the price level:

$$100 \times (\log P_{t+h} - \log P_{t-1}) = \alpha_h + \beta_h \Delta\theta_t \varepsilon_t^F + \delta_{1h}\Delta\theta_t + \delta_{2h}\varepsilon_t^F + \Gamma_h Z_{t-1} + e_{t+h}, \tag{2}$$

where $h \geq 0$ indexes the forecast horizon considered and Z_{t-1} is a vector of control variables specified below. Of particular interest is the coefficient β_h, which captures the cross-effect of the identified fiscal shock ε_t^F and the concurrent change in the share of inflation-indexed debt $\Delta\theta_t$ (in percentage points) present in the economy at time t.[16]

Figure 5 depicts the impulse-responses estimated through the local projection (2). The crucial observation is that the interaction effect between the change in the share of inflation-indexed debt and the deficit news is significantly positive at all horizons until five years after the shock. In economic terms, the coefficients imply that 1% of GDP of news about future deficits, accompanied by a concurrent increase in the share of inflation-indexed debt by one percentage point, leads to an additional increase of the price level of almost 0.9% in the two years after the shock. Deficit news itself, in contrast, have only a small and mostly insignificant effect on the price level when the share of indexed debt does not change, and deficit news accompanied by a falling share of indexed debt is not inflationary. This evidence clearly links the share of inflation-indexed debt to inflation observed in response to expansionary fiscal shocks: unfunded deficits are more inflationary when their financing tilts towards inflation-indexed debt. Further details related to this analysis are provided in appendix A.3, which shows that the result is robust to year fixed effects, alternative measures of indexed debt, additional controls, sample splits, and other identified fiscal shocks.

[15] I aggregate the daily fiscal-event changes in the present value of projected federal deficits to quarters and express them as a percentage of annual nominal GDP, such that positive values denote news about higher deficits. Appendix A.3 provides further details.

[16] I work with the change in the share of inflation-indexed debt in the total debt portfolio as the share of inflation-indexed debt θ_t is trending, as shown in appendix figure A.7. Using the level of inflation-indexed debt can then induce a spurious state-dependence. More generally, since θ_t is endogenous to fiscal policy decisions, using the change in the share of indexed debt $\Delta\theta_t$ can reduce the magnitude of the bias when the level of θ_t is strongly correlated with other low-frequency components of fiscal policy that affect the outcome variable (Gonçalves et al., 2024). Effectively, using the change $\Delta\theta_t$ here captures the local derivative of inflation with respect to ε_t^F when the policy variable θ_t itself also changes contemporaneously, i.e., when the deficit news coincides with a change in the composition of government financing.

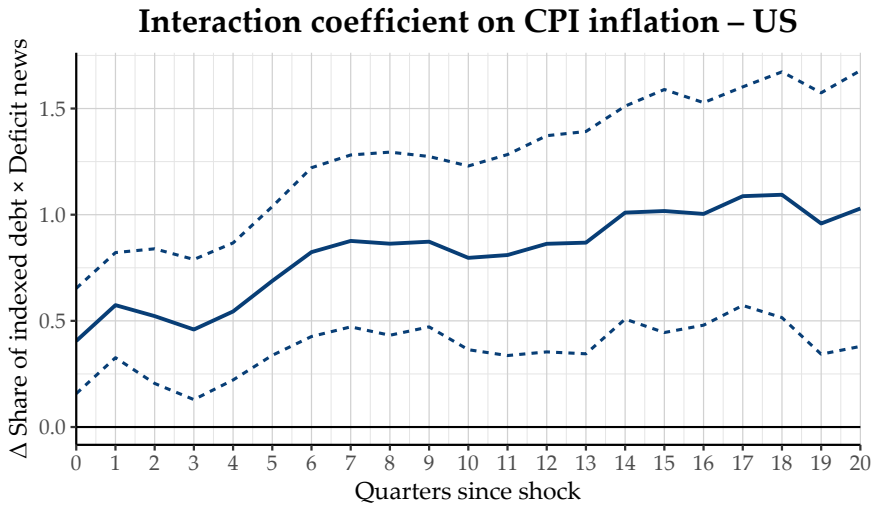


Figure 5: IRF implied by the local projection (2) through the coefficients β_h, measured in percentage points of the price level per 1% of GDP of deficit news and a concurrent increase in the share of inflation-indexed debt by one percentage point. The control vector Z contains the first four lags of the real GDP growth rate, the short-run nominal interest rate, the change in the weighted real exchange rate, a recession indicator, and the first lag of the price level difference. Standard errors are robust to heteroskedasticity and autocorrelation (Newey-West correction). Confidence intervals are provided at the 90% level. Sample length: 1980 Q3 - 2019 Q4.

3 Intuition from a one-equation price level determination model

I now introduce inflation-indexed debt in the government budget constraint and the resulting debt valuation equation. The analysis herein can be thought of as a partial equilibrium analysis that isolates the effect of indexed debt through the government budget *only*. I derive the novel result that the price level itself becomes partially backwards-looking in the intertemporal government budget equilibrium, i.e., today's price level becomes a function of the past price level. This observation holds true despite the lack of other inertia and the fact that the price level itself remains a jump variable.

The per-period government budget constraint in a world with indexed debt is given by

$$B_{t-1} + \frac{P_t}{P_{t-1}} b_{t-1} = P_t s_t + Q_t B_t + q_t b_t,$$

where B_t denotes the face value of non-indexed government debt issued at time t at price Q_t, b_t denotes the issuance value of indexed-government debt issued at time t at price q_t, lowercase letters correspond to the values for inflation-indexed debt, s_t are primary real surpluses raised (i.e., the inverse of deficits), and P_t denotes the price level. The cost of maturing inflation-indexed debt b_{t-1} (where b_{t-1} is issuance-date nominal base principal) is scaled by the gross inflation rate, P_t/P_{t-1}.[17]

To close this model as simply as possible, let $Q_t = \frac{1}{1+i_t}$ and $q_t = \frac{1}{1+r_t}$, i.e., the prices of each type

[17] Appendix F details why this formulation is the correct theoretical way to model inflation-indexed government debt. As for the empirical side of the argument, see Hall and Sargent (2011) for a verification that this is also the correct specification for indexed debt in line with how its face value adjusts empirically, absent indexation lags that I sideline here for simplicity.

of bonds equal the inverse of the respective relevant gross interest rate, where i_t is the net nominal interest rate and r_t is the net real interest rate.[18] By letting the price of indexed debt be equal to the inverse of the real interest rate, expectations of the face value adjustment of indexed debt through inflation are taken care of. Iterating this equation forward, dividing both sides by P_t, making use of the Fisher equation, and imposing a suitable no-Ponzi condition yields:

$$\frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}} = \sum_{j=0}^{\infty} \left(\prod_{m=0}^{j-1} \frac{1}{1 + r_{t+m}} \right) s_{t+j}, \tag{3}$$

This equation is the baseline government debt valuation equation without accounting for the differences in the insurance properties borne by the two types of debt, which made it possible to use the simplified bond pricing kernels Q_t and q_t as defined in the last paragraph. To the extent that equation (3) informs the evolution of the price level, *the price level itself becomes partially backwards-looking*: the real value of maturing inflation-indexed bonds depends on the past price level, not on today's price level. Intuitively, the real value of inflation-indexed bonds depends on the past price level because the face value payment of that bond is unity *at yesterday's prices*.

The dynamics of the government budget with indexed debt

I now explore the relationship between indexed debt and the price level within the government debt valuation equation through changes to fiscal surpluses. The goal is to explore how indexed debt changes the mechanisms inducing fiscal inflation in the clearest possible way.

The model is set up (in terms of outstanding bonds and expected surpluses), such that $P_{-1} = 1$. Therefore, the initial present discounted value (PDV) of surpluses is equal to the real value of the stock of debt. The economy has a finite horizon of 11 periods $t \in \{-1, 0, 1, ..., 9\}$, such that all debt has to be repaid by the government in period 9 using appropriate surpluses. All these assumptions jointly ensure a price level of $P_t = 1 \ \forall t$ in the absence of shocks. The impulse to the system is a one-period decrease of surpluses by 10% in period 0, announced at the same time. After period 0, the PDV of surpluses returns to its pre-shock value.

Figure 6 shows the reaction of the price level in response to a one-period decrease in surpluses, either in period 0 or in period 4 (while still being announced in period 0). Both panels also illustrate the response induced by the government debt valuation equation in an alternative world without indexed debt. In the case without indexed debt, the decrease in real surpluses simply induces a temporary upwards adjustment of the price level proportional to the decrease in surpluses, which returns back to its initial state subsequently, since the PDV of surpluses is equal to the pre-shock value after the shock. The difference in the shock timing merely matters by postponing the period of adjustment to the time after the shock actually materialized.

[18] This equals the fair price of either type of debt for a household maximizing its cumulative discounted utility with an exogenous endowment and perfect foresight.

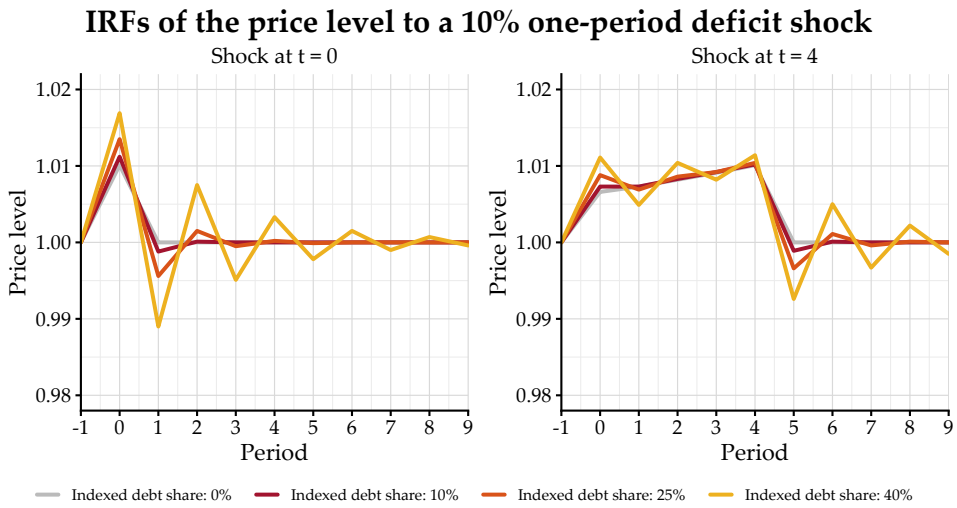


Figure 6: IRFs to a 10% decrease in the surplus in $t = 0$ for various levels of indexed debt.

However, when the share of inflation-indexed debt is strictly positive, the impact response is exacerbated. Focus first on the left panel, where the shock occurs in period 0. Given that the initial price level P_{-1} is fixed in the moment of the shock at time 0 (for the left panel), it is not possible to devalue the stock of inflation-indexed debt when the shock occurs. Therefore, the devaluation of the remaining (non-indexed) stock of bonds must be *larger* relative to the case without inflation-indexed debt: the price level must go up by a larger amount in the shock period because of the existence of inflation-indexed debt.

Once the shock vanishes, the price level oscillates when indexed debt is present instead of returning directly to steady-state. How can this be? Since the PDV of surpluses returns to its pre-shock level in $t = 1$, the stock of debt is suddenly worth *too little*: inflation-indexed debt is not worth much in $t = 1$ due to the high price level at $t = 0$, which is the factor by which b_t is normalized to 'real' terms in period 1. Since the funding shortfall caused by the deficit shock is now gone, the real value of non-indexed debt (B_0/P_1) must actually *increase* to make up the 'under-valuation' of indexed debt: therefore, P_1 must *decrease* (increasing the real value of non-indexed debt) to let the debt valuation equation hold. In the subsequent period, the price level from the previous period is now *too low*, increasing the value of indexed debt and pushing down the real value of non-indexed debt through a higher price level. This mechanism repeats itself until convergence to the initial equilibrium.[19]

In the right panel, with the shock being announced in period 0 but only materializing in period 4, the PDV of surpluses already decreases in the moment of announcement, remaining below its initial value until period 4. The oscillations induced by inflation-indexed debt decrease in size until period 4, and subsequently pick up after period 4 in line with the mechanism described above. The fact that the oscillations are decreasing in magnitude leading up to the shock is caused by the PDV of surpluses *not* being constant between periods 0 and 4 in this example: the closer period 4

[19] Cochrane (2001) explores a similar result in figure 4 of his paper, driven by a non-geometric maturity structure and the presence of long-term debt.

becomes, the more the PDV of surpluses actually decreases as the discounting of the (future) shock period reduces. This buffers the price level oscillations on the way to the period of the shock.

4 A heterogeneous-agent general equilibrium model with indexed debt

Having studied the relevance of indexed debt in a simplified framework that isolates its effect on the debt valuation equation, I now introduce inflation-indexed debt in a rich state-of-the-art macroeconomic model. Given that inflation-indexed debt delivers desirable insurance features to households by providing an income smoothing source that yields certain real returns, the model must necessarily bear relevance to imperfect consumption smoothing, borrowing constraints, and market imperfections precluding perfect risk-sharing across households. Otherwise, households would be indifferent between the two types of debt up to first-order. I work with a heterogeneous-agent model in the spirit of Kaplan et al. (2018), utilizing the efficient algorithms for solving such models provided by Auclert et al. (2021) and paying close attention to the peculiarities of determinacy in incomplete-market models exposed by Brunnermeier et al. (2024), which are intimately related to the government debt valuation equation.

Households: Heterogeneous households are indexed by i. Each individual household chooses consumption c_{it} and asset holdings B_{it} and b_{it} while supplying union-intermediated labor N_{it} to maximize their cumulative discounted utility

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(u(c_{it}) - v(N_{it}) \right) \right],$$

subject to two budget constraints, one for the liquid household budget and one for the separate evolution of indexed debt in the household savings portfolio:

$$\begin{aligned} P_t c_{it} + Q_t B_{it} &= \ell_{it}(1-\tau_t) W_t N_{it} + P_t \mathcal{D}_{it} + B_{i,t-1} - d_{it} a_{it}, \\ q_t b_{it} &= \Pi_t b_{i,t-1} + d_{it} a_{it}, \end{aligned}$$

where $a_{it} \in \{0,1\}$ is the portfolio-adjustment state and $\ell_{it} \equiv \frac{e_{it}^{1-\vartheta_\tau}}{\int e_{it}^{1-\vartheta_\tau} di}$ denotes the household's productivity, denoted in normalized efficiency units. The idiosyncratic productivity state e_{it} follows the Markov process described below, while ϑ_τ governs the progressivity of labor income. $W_t \equiv w_t P_t$ denotes the nominal wage per efficiency unit. Firm profits are distributed to households through $\mathcal{D}_{it}$, where $\int \mathcal{D}_{it} di = \mathcal{D}_t$, and profits are distributed in proportion to productivity, $\mathcal{D}_{it} = \ell_{it} \mathcal{D}_t$. Q_t and q_t are the nominal prices of one unit of non-indexed debt B_{it} and indexed debt b_{it}, respectively.

Through the two separate budget constraints, I posit that consumption is only possible from the non-indexed savings portfolio. In effect, indexed debt cannot be transformed into consumption as easily as non-indexed debt. This assumption reflects the significantly smaller liquidity of inflation-

indexed bond markets, even relative to their market size (Andreasen et al., 2021; Fleming and Krishnan, 2012), and gives rise to an ex-ante yield differential. d_{it} captures household-specific transfers between the liquid budget and the less accessible portfolio of indexed bond holdings, which can only occur when $a_{it} = 1$, which in turn happens with probability ν_b. This is a Calvo-type portfolio adjustment friction, as in Graham and Wright (2007) and Auclert et al. (2024). Finally, households are subject to standard borrowing constraints $B_{it} \geq -\underline{B}$ and $b_{it} \geq -\underline{b}$.

To solve the household block, it is necessary to distinguish whether a household is able to adjust its holdings of indexed debt in a given period. The state variables are the household-specific past asset holdings (B_-, b_-), the Markov state e_i, and the adjustment state a_i. We suppress the subscript i where possible for notational simplicity. For households with $a = 1$, the value function is

$$V_t(1, e; B_-, b_-) = \max_{c,B,b} u(c) - v(N) + \beta \mathbb{E}_t \left[V_{t+1}(a', e'; B, b) \right], \tag{4}$$

subject to

$$Pc + QB + qb = \ell(e)(1 - \tau)WN + P\mathcal{D} + B_- + \Pi b_-,$$

and $B \geq -\underline{B}$, $b \geq -\underline{b}$. The evolution of the Calvo-type portfolio adjustment friction is i.i.d., with $\mathbb{P}(a' = 1) = \nu_b$. For households with $a = 0$, indexed holdings cannot be chosen and the indexed payoff is automatically rolled into the indexed account. Their value function is

$$V_t(0, e; B_-, b_-) = \max_{c,B} u(c) - v(N) + \beta \mathbb{E}_t \left[V_{t+1} \left(a', e'; B, \frac{\Pi}{q} b_- \right) \right], \tag{5}$$

subject to $B \geq -\underline{B}$ and the 'main' budget constraint

$$Pc + QB = \ell(e)(1 - \tau)WN + P\mathcal{D} + B_-.$$

The goal is to recover policy functions $c(\cdot)$, $B(\cdot)$, and $b(\cdot)$ that solve the household problem under both adjustment states.[20] Denote by λ_{it}, μ^B_{it}, and μ^b_{it} the state-dependent budget and borrowing-constraint multipliers. The relevant first-order conditions for a household with $a_{it} = 1$ are

$$\begin{aligned} u'(c_{it}) &= P_t \lambda_{it}, \\ Q_t \lambda_{it} &= \beta \mathbb{E}_t \left[V_{B,i,t+1} \right] + \mu^B_{it}, \\ q_t \lambda_{it} &= \beta \mathbb{E}_t \left[V_{b,i,t+1} \right] + \mu^b_{it}, \end{aligned}$$

together with the complementary-slackness conditions $\mu^B_{it}(B_{it} + \underline{B}) = 0$ and $\mu^b_{it}(b_{it} + \underline{b}) = 0$. The resulting envelope conditions are

$$V_{B,i,t} = \frac{u'(c_{it})}{P_t},$$

[20] $N(\cdot)$ is effectively chosen by an intermediary labor union as explained below.

$$V_{b,i,t} = \begin{cases} \frac{u'(c_{it})}{P_t}\Pi_t = \frac{u'(c_{it})}{P_{t-1}} & \text{if } a_{it} = 1, \\ \beta \frac{\Pi_t}{q_t} \mathbb{E}_t \left[V_{b,i,t+1} \right] & \text{if } a_{it} = 0. \end{cases}$$

Thus, the equilibrium conditions give rise to fairly standard Euler equations pricing the two bonds:

$$\frac{Q_t}{P_t} u'(c_{it}) \geq \beta \mathbb{E}_t \left[V_{B,i,t+1} \right],$$

$$\frac{q_t}{P_t} u'(c_{it}) \geq \beta \mathbb{E}_t \left[V_{b,i,t+1} \right],$$

where any inequality is strict when the corresponding borrowing constraint binds. For an unconstrained holder of non-indexed debt, the first Euler equation yields

$$Q_t = \beta \mathbb{E}_t \left[\frac{u'(c_{i,t+1})}{u'(c_{it})} \frac{P_t}{P_{t+1}} \right]. \tag{6}$$

Indexed debt, in turn, is only priced by households that can adjust at time t and are unconstrained in b_{it}.

$$\frac{q_t}{P_t} u'(c_{it}) = \beta \mathbb{E}_t \left[V_{b,i,t+1} \right]. \tag{7}$$

Equation (7) does not generally collapse to a one-period consumption-SDF equation because $V_{b,i,t+1}$ remains recursive when the household cannot adjust at $t+1$. To make the retrading friction explicit, I therefore define the *next adjustment time* as

$$\tau_i(t) \equiv \inf \left\{ j \geq 1 : a_{i,t+j} = 1 \right\}.$$

For an unconstrained adjusting household, recursive substitution of the envelope condition on indexed debt gives

$$q_t = \mathbb{E}_t \left[\beta^{\tau_i(t)} \frac{u'(c_{i,t+\tau_i(t)})}{u'(c_{it})} \prod_{s=1}^{\tau_i(t)-1} \frac{1}{q_{t+s}} \right]. \tag{8}$$

The cumulative inflation terms cancel because the indexed account delivers a real payoff. The random adjustment date and the intervening forced rollovers remain in the pricing kernel. The standard bond pricing kernel $q_t = \beta \mathbb{E}_t[u'(c_{i,t+1})/u'(c_{it})]$ is nested by equation (8) when $\nu_b = 1$.

As discussed previously, B_t and b_t denote face quantities issued at time t. A unit of nominal debt pays one unit of currency at $t+1$, while a unit of indexed debt pays Π_{t+1} units of currency and therefore $1/P_t$ unit of the consumption good measured at its issuance-date price. The nominal market values at time t are $Q_t B_t$ and $q_t b_t$, and the corresponding real market values are $Q_t B_t / P_t$ and $q_t b_t / P_t$.

For the following, it will be useful to define the face-value and the market-value shares of indexed

debt:

$$\theta_t^{face} \equiv \frac{b_t}{B_t + b_t}, \qquad \theta_t^{mkt} \equiv \frac{q_t b_t}{Q_t B_t + q_t b_t}.$$

The calibration fixes the market-value composition in steady-state, while the market-value share can change away from steady-state as quantities and Q_t/q_t adjust.

Firms and production: To focus on the effects of indexed debt and its interaction with households facing uninsurable idiosyncratic income risk, I model the production block in a parsimonious way. A final-good producer combines a continuum of intermediate varieties with elasticity of substitution $\epsilon_p > 1$. Demand for variety k is

$$Y_{kt} = \left(\frac{P_{kt}}{P_t}\right)^{-\epsilon_p} Y_t.$$

Intermediate firm k produces with the linear technology $Y_{kt} = Z_t N_{kt}$, where aggregate productivity follows the process specified below. The real marginal cost is equal to $mc_t = w_t/Z_t$, and firms face Rotemberg price-adjustment costs. Real profits of firm k are thus

$$\mathcal{D}_{kt} = \frac{P_{kt}}{P_t} Y_{kt} - w_t N_{kt} - \frac{\epsilon_p - 1}{2\kappa}\left(\log \frac{P_{kt}}{P_{k,t-1}}\right)^2 Y_t.$$

Aggregate profits $\mathcal{D}_t \equiv \int \mathcal{D}_{kt} dk$ are distributed to households. The coefficient on the adjustment cost is normalized so that κ is the slope of the first-order Phillips Curve with respect to the log deviation of real marginal cost from its flexible-price value.

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \widehat{mc}_t + \mu_t^p, \tag{9}$$

where $\widehat{mc}_t \equiv \log(mc_t/\overline{mc})$ and μ_t^p is an exogenous markup shock. Forward iteration under a suitable no-Ponzi condition allows me to express the NKPC as $\hat{\pi}_t = \kappa \mathbb{E}_t \sum_{l=0}^{\infty} \beta^l \widehat{mc}_{t+l} + \mathbb{E}_t \sum_{l=0}^{\infty} \beta^l \mu_{t+l}^p$.

I close the labor market using the union-based wage-setting block following Auclert et al. (2021) (see also Auclert et al. (2025)). Labor unions represent a cross-section of households, allocate the same hours $N_{it} = N_t$ to each household and set the nominal wage subject to quadratic adjustment costs. Defining $\mathcal{U}_t \equiv \int \ell_{it} u'(c_{it})\, di$, $1 + \pi_t^w \equiv (1 + \pi_t)\frac{w_t}{w_{t-1}}$, the resulting log-linear wage Phillips curve is

$$\log(1 + \pi_t^w) = \kappa_w \left[\chi_N N_t^{1+\varphi} - \frac{(1 - \tau_t) w_t N_t}{\mu_w} \mathcal{U}_t\right] + \beta \log(1 + \pi_{t+1}^w), \tag{10}$$

where χ_N scales the disutility of labor and is calibrated so that the wage-setting condition holds in steady-state. This condition determines the real wage w_t, while firms' real marginal cost $mc_t = w_t/Z_t$ enters the price Phillips curve in equation (9).

Fiscal policy: Let T_t denote real tax revenue, G_t real government purchases, and $S_t \equiv T_t - G_t$ the real primary surplus. Consistent with the debt-unit convention above, the per-period government

budget constraint is

$$B_{t-1} + \Pi_t b_{t-1} = P_t S_t + Q_t B_t + q_t b_t, \tag{11}$$

and the aggregate tax revenue is linked to the household tax schedule by

$$T_t = \int \tau_t \ell_{it} w_t N_{it} di.$$

As pointed out by Brunnermeier et al. (2024), individual household transversality conditions do not directly imply an aggregate transversality condition for government debt under incomplete markets. This point is particularly important here because households hold the two assets in heterogeneous proportions and only a subset of households can adjust indexed holdings at a given date. For each *household*, therefore, I impose a fairly standard transversality condition

$$\lim_{T\to\infty} \beta^{T-t+1} \mathbb{E}_t \left[V_{B,i,T+1} B_{iT} + V_{b,i,T+1} b_{iT} \right] = 0. \tag{12}$$

Now, let $\mathcal{I}_t^N$ denote the set of unconstrained marginal holders of nominal debt and let $\mathcal{I}_t^I$ denote the set of unconstrained households with $a_{it} = 1$ that price indexed debt. Let ω_{it}^N and ω_{it}^I be arbitrary non-negative weights over these sets that integrate to one. It will be convenient to define the real market value of end-of-period debt and the real value of the claims inherited from the previous period in

$$V_t^d \equiv \frac{Q_t B_t}{P_t} + \frac{q_t b_t}{P_t}, \qquad W_t^d \equiv \frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}},$$

so that the government flow constraint (11) can be expressed as $W_t^d = S_t + V_t^d$. I can then analytically characterize a dynamic debt valuation equation under incomplete markets with the two types of government debt, following (Brunnermeier et al., 2024), and provide a standard present-value debt valuation equation under some additional assumptions.

Proposition 1 (i) *The flow government budget constraint under the appropriate bond pricing kernels is given by*

$$\frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}} = S_t + \mathbb{E}_t \left[m_{t,t+1}^N \frac{B_t}{P_{t+1}} \right] + \mathcal{L}_t^I(b_t), \tag{13}$$

where

$$m_{t,t+1}^N \equiv \int_{\mathcal{I}_t^N} \omega_{it}^N \beta \frac{u'(c_{i,t+1})}{u'(c_{it})} di$$

and

$$\mathcal{L}_t^I(b_t) \equiv \mathbb{E}_t \left[\int_{\mathcal{I}_t^I} \omega_{it}^I \beta^{\tau_i(t)} \frac{u'(c_{i,t+\tau_i(t)})}{u'(c_{it})} \frac{b_t}{P_t} \prod_{s=1}^{\tau_i(t)-1} \frac{1}{q_{t+s}} di \right].$$

(ii) *Suppose in addition that the set of unconstrained nominal marginal holders $\mathcal{I}_t^N$ is non-empty at every date and in every state, making the marginal-holder discount factor $M_{t,t} \equiv 1$ and $M_{t,t+j} \equiv M_{t,t+j-1} m_{t+j-1,t+j}^N$ for $j \geq 1$ well-defined. Suppose also that the aggregate transversality condition $\lim_{J\to\infty} \mathbb{E}_t \left[M_{t,t+J} V_{t+J}^d \right] = 0$*

holds and that there exists a finite constant K such that $\left|\frac{\mathbb{E}_s\left[m^N_{s,s+1}\right]}{q_s} - 1\right| \leq K$ *at every date and in every state. Then, the government debt valuation equation can be expressed in present-value terms as*

$$\frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}} = \mathbb{E}_t\left[\sum_{j=0}^{\infty} M_{t,t+j}\left(S_{t+j} - \xi_{t+j}\right)\right], \tag{14}$$

where the indexation wedge is defined as

$$\xi_t \equiv \left(\mathbb{E}_t\left[m^N_{t,t+1}\right] - q_t\right)\frac{b_t}{P_t} = \theta^{mkt}_t V^d_t\left(\frac{\mathbb{E}_t\left[m^N_{t,t+1}\right]}{q_t} - 1\right). \tag{15}$$

Proof. See appendix B.2. ■

The first valuation term in part (i) of the proposition prices the next-period real payoff of nominal debt. The second is the continuation value of the real indexed account at the next date at which its marginal holder can retrade. The indexed payoff is protected against cumulative inflation, while the covariance with the marginal utility of the pricing household and the retrading friction remain present in $\mathcal{L}^I_t(b_t)$. Household heterogeneity enters through the marginal-holder kernels and through the distribution of next adjustment dates.

Part (ii) of proposition 1 shows what is required to move from the recursive restriction to a present-value statement. The non-emptiness assumption permits the marginal holder to differ across dates and states, so that $M_{t,t+j}$ is a chain of one-period marginal-holder kernels rather than the intertemporal marginal rate of substitution of any single household. The aggregate transversality condition and the uniform bound on the indexed spread jointly eliminate the full terminal residual. Neither restriction is implied by the household-level condition (12), since adjusters and nonadjusters hold different ratios of indexed to non-indexed debt by construction.

Through the indexation wedge ξ_t in equation (15), the retrading friction enters the valuation equation. Whenever $\nu_b < 1$, the real return on indexed debt, $1/q_t$, is not priced by the one-period stochastic discount factor of any household.[21] Writing $1/q_t \equiv 1 + r^I_t$ and $1/\mathbb{E}_t[m^N_{t,t+1}] \equiv 1 + r^*_t$, I can also express the wedge approximately as

$$\xi_t \approx \theta^{mkt}_t V^d_t\left(r^I_t - r^*_t\right),$$

so that the wedge is the product of the market-value share of indexed debt and the liquidity spread that the government pays on it relative to the shadow real rate of the marginal holder of nominal debt. The wedge is non-negative whenever indexed debt trades at an illiquidity discount, in which case the debt stock that the government can support falls short of the present value of primary surpluses by exactly the discounted liquidity cost of its indexed issuance.

[21] Non-adjusters have no first-order condition in b_{it} at all, while adjusters price the continuation value $V_{b,i,t+1}$ rather than a one-period payoff.

I close the fiscal block by a standard rule for the primary-surplus-to-output ratio $s_t^p \equiv S_t/Y_t$:

$$\frac{s_t^p}{s^p} = \left(\frac{d_{t-1}^N}{d^N}\right)^{\gamma_B(\theta)} \left(\frac{d_{t-1}^I}{d^I}\right)^{\gamma_b(\theta)} e^{\varepsilon_t^F}, \tag{16}$$

where $d_t^N \equiv \frac{Q_t B_t}{P_t Y_t}$, $d_t^I \equiv \frac{q_t b_t}{P_t Y_t}$ are the real market values of non-indexed and indexed debt relative to output. ε_t^F is an AR(1) fiscal shock, which will be specified when discussing the results. Importantly, the fiscal rule coefficients γ_B and γ_b will be dependent on the split between indexed and non-indexed debt securities issued by each government θ, such that the scale of the taxation response to deviations of the market value of each debt security from steady-state takes into account what debt has been issued. I will detail this further when discussing the calibration in section 5. Given government purchases, the rule determines tax revenue through $T_t = G_t + s_t^p Y_t$, and the household tax schedule is adjusted to raise this amount. In calibrations without indexed debt ($b_{ss} = 0$), the factor involving d_{t-1}^I/d^I is omitted. In log-linearized terms, for an economy with positive steady-state supplies of both debt instruments,

$$\hat{s}_t^p = \gamma_B(\theta)\hat{d}_{t-1}^N + \gamma_b(\theta)\hat{d}_{t-1}^I + \varepsilon_t^F.$$

Monetary policy: Monetary policy follows an inertial Taylor rule with weights on inflation and output deviations from steady-state:

$$\frac{R_t^n}{R^n} = \left(\frac{R_{t-1}^n}{R^n}\right)^{\rho_M} \left[\left(\frac{\Pi_t}{\Pi}\right)^{\phi_\pi} \left(\frac{Y_t}{Y}\right)^{\phi_y}\right]^{1-\rho_M} e^{\varepsilon_t^M}. \tag{17}$$

ε_t^M is an AR(1) shock to monetary policy. In exact log-linearized terms,

$$\hat{r}_t^n = \rho_M \hat{r}_{t-1}^n + (1-\rho_M)\left(\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t\right) + \varepsilon_t^M.$$

The one-period nominal bond return is linked to the monetary policy rate. For computational reasons, I assume a small return discount, where $Q_t = \frac{e^{\chi_q}}{R_t^n}$, $\chi_q \geq 0$, so that $1/Q_t = R_t^n e^{-\chi_q}$.

Market clearing: Market clearing on the goods, labor, and asset markets is defined as follows. On the goods market,

$$C_t + G_t + \frac{\epsilon_p - 1}{2\kappa} (\log \Pi_t)^2 Y_t = Y_t. \tag{18}$$

On the labor market, supply and demand clear in efficiency units:

$$\int \ell_{it} N_{it} di = \int N_{kt} dk \equiv L_t. \tag{19}$$

For each asset class, government debt supply equals cumulative household debt demand:

$$B_t = \int B_{it} di, \qquad b_t = \int b_{it} di. \tag{20}$$

Finally, I characterize the law of motion for the distribution of households. Let $g_t^B(x)$ and $g_t^b(x)$ denote the asset policy functions at state $x = (B_-, b_-, e, a)$, let $p_e(\mathcal{E} \mid e)$ denote the productivity transition, and let $p_a(1) = \nu_b$. The distribution evolves according to

$$\Gamma_{t+1}(\mathcal{B}, \mathcal{b}, \mathcal{E}, a') = \int \mathbf{1}_{\{g_t^B(x) \in \mathcal{B},\, g_t^b(x) \in \mathcal{b}\}} p_e(\mathcal{E} \mid e) p_a(a') d\Gamma_t(x). \tag{21}$$

For a nonadjuster, the indexed policy appearing in (21) is $g_t^b(x) = (\Pi_t / q_t) b_-$.

I close the model by defining $u(c) = (c^{1-\sigma} - 1)/(1-\sigma)$ and $v(N) = \chi_N N^{1+\varphi}/(1+\varphi)$. Under this convention, σ is the inverse intertemporal elasticity of substitution and φ is the inverse Frisch elasticity of labor supply.

Equilibrium: I now define the competitive equilibrium in this economy.

Definition 1 (Competitive Equilibrium) *Given initial face quantities of government debt and an initial distribution Γ_0, a* competitive equilibrium *consists of household allocations $\{c_{it}, N_{it}, B_{it}, b_{it}, d_{it}\}_{i,t}$, firm allocations $\{Y_{kt}, N_{kt}\}_{k,t}$, fiscal allocations $\{T_t, G_t, B_t, b_t\}_t$, prices $\{P_t, P_{kt}, w_t, Q_t, q_t, R_t^n\}_t$, and a distribution Γ_t, such that households solve (4) and (5), labor unions adhere to (10), firms maximize their discounted profits subject to Rotemberg adjustment costs, profits are distributed to households, the government satisfies (11) and (16), monetary policy satisfies (17), all markets clear according to (18)–(20), and the household distribution evolves by the law of motion (21).*

Steady-state: When looking at the dynamic economy, I consider a first-order approximation around a stationary equilibrium with zero aggregate inflation, $\Pi_{ss} = 1$. The stationary distribution continues to embody idiosyncratic risk, so bond prices must be obtained jointly from household optimality and asset-market clearing. In appendix B.3, I characterize in greater detail under which additional assumptions a unique price level at the steady-state can be proven analytically.

5 Calibration and computational approach

Table 1 gives the overall model parametrization, while table 2 shows the endogenously calibrated parameters. I follow overall the approach of Auclert et al. (2021), as I apply a conceptually similar algorithm. In the preferred calibration, I vary government spending G and the household discount factor β to ensure that the goods and labor markets debt clear. The market for inflation-indexed debt is targeted with the help of ν_b, the probability of being able to access the portfolio of indexed debt actively. The market for non-indexed debt clears with a tolerance of 10^{-5}, while targeted market clearing conditions clear with close to machine precision (10^{-15}). To compare various policy combinations, I here consider policy coefficients determining whether a given policy mix is fiscally-

led (effectively yielding unfunded deficits) or monetary-led, reminiscent of Bianchi et al. (2023). The policy coefficients $\{\phi_\pi, \phi_y, \gamma_B, \gamma_b\}$ should thus be taken as indicative and related to suitable active/passive policy combinations in the sense of Leeper (1991), but not as calibrated feedback rules.[22] When deviating from the baseline parameterizations summarized in tables 1 and 2, I will explicitly introduce novel parameters as suitable.

Parameter	Description	Value	Source/Target
Firms			
Y	Steady-state output	1	Normalization
ϵ_p	Elasticity of substitution between product varieties	10.09	Firm mark-up of 11% (Auclert et al., 2025); $\epsilon_p = \mu_p/(\mu_p - 1)$, $\mu_p = 1.11$
κ	Slope of price Phillips Curve	0.007	Hazell et al. (2022)
κ_w	Slope of wage Phillips Curve	0.1	Auclert et al. (2025)
μ_w	Steady-state wage mark-up	1	No wage mark-up
Households			
σ	Inverse of intertemporal elasticity of substitution	1	Standard value
φ	Inverse of Frisch elasticity of labor supply	1	Standard value
$\underline{B}$	Lower bound of non-indexed debt holdings	0	
$\underline{b}$	Lower bound of indexed debt holdings	0	
ρ_e	Persistence of AR(1) shocks to HH productivity	0.966	Auclert et al. (2021)
σ_e	S.d. of AR(1) shocks to HH productivity	0.92	Auclert et al. (2021)
Government			
T/G	tax-to-government purchases ratio	1.03	See explanation below
r^*	Natural rate of interest	0.0125 (annual)	Benigno et al. (2024)
χ_q	Instrument-specific return wedge	1.00 bp (annual)	Nominal-indexed return spread
ρ_M	Inertia in Taylor-type interest rate rule	0	Simplification
ϕ_π	Monetary policy reaction to inflation deviations from steady-state	$\{0.25,\ 1.5\}$	For fiscally-led/monetary-led policy mix (Bianchi et al., 2023)
ϕ_y	Monetary policy reaction to output deviations from steady-state	0.075	
$\bar{\gamma}_B$	Reference fiscal policy reaction to deviations of market value of non-indexed debt from steady-state	$\{0.05, 8\}$	See text
$\bar{\gamma}_b$	Reference fiscal policy reaction to deviations of market value of indexed debt from steady-state	1.00	See text
η_I	Elasticity of γ_b to the indexed share of debt	0.50	Share-weighted fiscal rule, see text
η_N	Elasticity of γ_B to the non-indexed share of debt	1.00	Share-weighted fiscal rule, see text
Computational parameters			
n_z	Number of points in asset grid for household productivity shock	11	
n_b	Number of points in asset grid for indexed debt	70	Auclert et al. (2021)
n_B	Number of points in asset grid for non-indexed debt	50	Auclert et al. (2021)
$\bar{B}$	Maximum holdings of non-indexed debt in asset grid	150	
$\bar{b}$	Maximum holdings of indexed debt in asset grid	150	Value at which numerical stability achieved
T	Number of periods used in simulations of Jacobians	500	

Table 1: Baseline parametrization for the quantitative estimation

[22] Macroeconomists tended to focus on calibrations in which ϕ_π, the parameter capturing the reaction of monetary policy to deviations of the inflation rate from steady-state, is larger than one, commonly known as the 'Taylor Principle'. As Nakamura et al. (2025) show, this notion is rejected in the data. Instead, the Taylor principle can be reinterpreted as *prescriptive*, that is, as a policy prescription according to which the central bank ought to act, but not as *descriptive*. This supports the idea of calibrating models with $\phi_\pi < 1$. Kaplan (2025a) explains further why policy rules not adhering to the Taylor principle deserve consideration in monetary macroeconomics.

The model period is one quarter, and quarterly steady-state output is normalized to $\bar{Y} = 1$. All flow variables, including G and T, are consequently measured per quarter. Debt is a stock, and the empirical debt ratios reported in table 2 use the usual denominator of annual GDP. Writing annualized steady-state output as $\bar{Y}^{ann} = 4\bar{Y}$, I define the reported debt ratios as $\bar{d}^j = D^j/\bar{Y}^{ann}$ for $j \in \{N, I\}$. The debt ratios entering the quarterly model and the fiscal rule are therefore $d^j = \frac{D^j}{\bar{Y}} = 4\bar{d}^j$. Hence, the U.K. targets $\bar{d}^N = 0.8176$ and $\bar{d}^I = 0.3024$ correspond in model units to $d^N = 3.2704$ and $d^I = 1.2096$. Total debt is thus 4.48 times quarterly output, or 1.12 times annual GDP. The annual natural real rate $r^{*,ann} = 0.0125$ is converted to the quarterly rate used in the model, $r^{*,q} = (1 + r^{*,ann})^{1/4} - 1 = 0.00311$. Allowing for the calibrated instrument spread, the government budget yields in steady-state then implies $S = r^N D^N + r^I D^I = 0.013853$. Finally, imposing $T/G = 1.03$ determines $G = S/(1.03 - 1) = 0.461770$ and $T = 1.03G = 0.475623$. Finally, by imposing zero steady-state inflation, I nullify the possibility of distortionary inflation that could induce a wedge to the long-run natural rate.

The calibration delivers overall reasonable estimates of the endogenous parameters that are in line with the parametrization of Auclert et al. (2021). The level of government spending is not targeted to its empirical counterpart, yet the estimated government spending share of GDP is close to the share of government spending in GDP in the U.K. since 2021, which hovers at around 45%. The calibration delivers overall reasonable estimates of the endogenous parameters. I additionally set the instrument-specific return wedge χ_q to target the annual spread between nominal and indexed debt. This separates the spread target from the elasticity of substitution between product varieties and from the fiscal shock.

The fiscal policy parameters, γ_B and γ_b, have the specific in-built feature that they will vary in proportion to the share of debt that is actually inflation-indexed. Note that by the fiscal rule (16), and denoting the total stock of debt through $d_t = d_t^N + d_t^I$, a first-order expansion of that rule shows that $\log(d_t/d) \simeq (1-\theta)\log(d_t^N/d_N) + \theta\log(d_t^I/d^I)$, so stabilization of the total debt burden with a hypothesized elasticity γ would give the same results as the posted fiscal rule (16) with the suitable coefficients $\gamma_B = \gamma(1-\theta)$ and $\gamma_b = \gamma\theta$. Because of that observation, I postulate that the fiscal rule reaction parameters are actually functions of the relative shares of indexed/non-indexed debt, such that:

$$\gamma_b(\theta) = \bar{\gamma}_b \left(\frac{\theta}{\theta^{UK}}\right)^{\eta_I}, \qquad \gamma_B(\theta) = \bar{\gamma}_B \left(\frac{1-\theta}{1-\theta^{UK}}\right)^{\eta_N}. \tag{22}$$

The U.K. portfolio split, according to which I postulate 27% of all debt to be inflation-indexed, is acting as the normalization of the coefficients. Under $\eta = 0$, the reaction coefficients would be common across all debt-splits, while under $\eta = 1$, the coefficients move in exact proportion to the debt shares. I chose η_N as a simple normalization, while the fact that $\eta_I = 0.5$ indicates that governments only imperfectly internalize the specifics of the return structure of indexed debt relative to simple nominal debt bills. At $\eta_I = 0.5$, the indexed debt coefficients become 1.000 and

0.583 for the U.K. and U:S. calibrations, respectively. Note that because $\gamma_b(\theta) \to 0$ as $\theta \to 0$, equation (22) implies that the no-indexed-debt economy administers a fiscal rule by which the indexed debt term simply vanishes. For the reference reaction levels, I set $\bar{\gamma}_b$ to 1 in *both* the fiscally-led and monetary-led policy specification, which allows me to ensure that *all* variation on the fiscal side of things is tied to γ_B, reminiscent of standard models without indexed debt. I set $\bar{\gamma}_B = 0.05$ in the fiscally-led policy mix (which is standard), and $\bar{\gamma}_B = 8$ in the monetary-led policy mix (which is relatively high, but ensures a sufficient distance to explosive solutions). In sum, therefore, all results must be seen as being a consequence of the debt composition *and* its effects on the behavior of the fiscal policy authority jointly.

Finally, to pin down the tax rate in steady-state, I exogenously fix the tax-to-government-spending ratio at $T/G = 1.03$. This assumption runs counter to currently observed budget deficits, but solving the model under steady-state deficits is hardly feasible under positive steady-state real interest rates.[23] However, the assumption of positive surpluses in steady-state remains qualitatively and quantitatively in line with recent long-run forecasts of the current budget deficit for the United Kingdom, provided by the Office for Budget Responsibility (2026) in their historical official forecasts database (table CB).[24]

Real Market value of debt/Annual GDP	HH discount factor	$\mathbb{P}$(adjustment)	Govt. spending
	Main calibration: U.K. debt portfolio		
$D^N/(4\bar{Y})$: 0.8176, $D^I/(4\bar{Y})$: 0.3024	$\beta = 0.9833$	$\nu_b = 0.2054$	$G = 0.4618$
	Counterfactual: U.S. debt shares		
$D^N/(4\bar{Y})$: 1.0171, $D^I/(4\bar{Y})$: 0.1029	$\beta = 0.9833$	$\nu_b = 0.1245$	$G = 0.4611$
	Counterfactual: no indexed debt		
$D^N/(4\bar{Y})$: 1.12, $D^I/(4\bar{Y})$: 0.	$\beta = 0.9832$	$\nu_b = 0.0347$	$G = 0.4608$

Table 2: Calibrated parameters across different debt scenarios

In terms of government debt, I mainly compare three different steady-state calibrations: one which follows the observed modal split of sovereign debt into non-indexed and indexed debt in the United Kingdom (which is the G7 country with the highest share of indexed debt, such that $D^N/(4\bar{Y}) = 0.8176$ and $D^I/(4\bar{Y}) = 0.3024$), and two counterfactual calibrations with either a split between indexed and non-indexed debt in accordance to the U.S. sovereign debt portfolio (i.e., $D^N/(4\bar{Y}) = 1.0171$ and $D^I/(4\bar{Y}) = 0.1029$), or the complete absence of any indexed debt (i.e., $D^N/(4\bar{Y}) = 1.12$ and $D^I/(4\bar{Y}) = 0$).[25] I therefore exogenously postulate the same steady-state gross bond supply across the calibrations, given that bond supply as a share of GDP is a relatively low-frequency variable, and vary the shares of the two types of bonds. Many of the exercises will resolve around the differences between these calibrations, as I will mainly focus on the effect that indexed government debt has on economic aggregates.

Computational details - using the Sequence-Space Jacobian

[23] Kaplan et al. (2026) solve a model with negative surpluses and a negative steady-state real interest rate.

[24] Conditional on a $\sim 40\%$ share of government spending in GDP, the projected 1% budget surplus in the long-run as a share of GDP is equivalent to a ratio of sovereign income to spending of 1.025.

[25] Appendix figure C.1 shows the distribution of household assets in steady-state.

The simulation of the model is derived using the Sequence-Space Jacobian method developed in Auclert et al. (2021), which itself constitutes an evolution of the methods pioneered by Reiter (2009). The computational method I employ generates perfect-foresight solutions in aggregates in response to time-zero perturbations of exogenous disturbances, but it maintains the non-linearity underlying the responses of heterogeneous households.

First, the heterogeneous household block is solved, taking aggregate prices as given, for both the policy functions (through backwards iteration) and the distribution of asset holdings (through forwards iteration). Both solve with a numerical tolerance of up to 10^{-14}, and are subsequently used to inform other blocks of the model (such as firm optimality, government policies, and market clearing) and to generate updates of aggregates where necessary. The two components (heterogeneous households and aggregate) interact and iterate until convergence, which is reached with a numerical threshold of 10^{-9}, which is reasonable given the high degree of complexity underlying household behavior in the presence of two types of assets.

6 Quantitative insights: the joint role of indexed debt and policy rules

With the computational algorithm at hand, I solve and estimate the model's aggregate impulse-responses for a number of shocks, using the parametrization from table 1, but varying the calibration of the debt shares in line with table 2. Here, I will mostly focus on the effects of unanticipated disturbances to *government spending* G_t, which directly influence the surpluses raised by the government in any given period.[26] Unless stated otherwise, impulse period $t = 0$ is the first quarter in which the shock is realized. Quarterly inflation is $100 \log(P_t/P_{t-1})$, annualized quarterly inflation is $400 \log(P_t/P_{t-1})$, and the cumulative log price change at horizon h is $100[\log P_{t+h} - \log P_{t-1}]$. The cumulative first-year price change is $100[\log P_3 - \log P_{-1}]$, equivalently one quarter of the sum of the first four annualized quarterly inflation responses. Tables and figure captions state which of these objects is reported.

Throughout this section, the fiscal innovation that I consider is a *jointly unfunded government-purchases innovation*, rather than a government-purchases shock or a surplus-rule shock in isolation. Let $g_t \equiv \Delta G_t/\bar{Y}$, where $g_t = \rho_G^t g_0$ and $g_0 = 0.01$. I feed in the two perfectly correlated exogenous paths $\Delta G_t = \bar{Y} g_t$, $\varepsilon_t^F = -\frac{g_t}{\bar{s}^p}$ into the model. The first path raises real government purchases in the goods resource constraint and therefore represents government absorption rather than a household transfer. The second path accommodates the spending increase through the primary-surplus rule. To first order, its direct contribution to the primary surplus is $\bar{S}\varepsilon_t^F = -\Delta G_t$. Hence, $\Delta S_t = -\Delta G_t + \bar{s}^p \Delta Y_t + \bar{S}\left[\gamma_B(\theta)\hat{d}_{t-1}^N + \gamma_b(\theta)\hat{d}_{t-1}^I\right]$, and therefore $\Delta T_t = \bar{s}^p \Delta Y_t + \bar{S}\left[\gamma_B(\theta)\hat{d}_{t-1}^N + \gamma_b(\theta)\hat{d}_{t-1}^I\right]$. Thus, the spending is not accompanied by a discretionary one-for-one increase in contemporaneous tax revenue, although revenue and the proportional labor tax rate continue to respond endogenously through output, the tax base, and debt demand. A shock to G_t alone would instead be contemporaneously tax-financed, while a shock to ε_t^F alone would lower the primary surplus

[26] Appendix C provides an overview of the dynamic responses to expansionary monetary policy shocks.

without increasing government purchases.

I first look at the role that inflation-indexed debt plays for the amplification of shocks as evidenced through simulated moments, in line with the principal focus of the paper. To get a detailed grasp of the effect of inflation-indexed debt for aggregate variables, I compare the simulated volatility of a number of macroeconomic aggregates across all calibrations (U.K. calibration, counterfactual U.S. distribution of debt across the two types, and issuance of non-indexed debt only) and across both 'common' policy combinations (passive monetary/active fiscal (fiscally-led) and active monetary/passive fiscal (monetary-led)). The results of this exercise are presented in table 3.

	Normalized standard deviations across policy scenarios					
	PM/AF-U.K.	PM/AF-U.S. split	PM/AF-NoIndex	AM/PF-U.K.	AM/PF-U.S. split	AM/PF-NoIndex
G	1.000	1.000	1.000	1.000	1.000	1.000
Y	3.261	2.237	1.790	0.921	0.940	0.943
C	2.781	1.677	1.159	0.151	0.130	0.129
π	0.494	0.302	0.199	0.081	0.090	0.093
r^I	0.332	0.139	0.051	0.170	0.119	0.065
r^*	0.149	0.098	0.073	0.088	0.092	0.093
N	3.261	2.237	1.790	0.921	0.940	0.943

Table 3: Normalized standard deviations of aggregate variables in response to fiscal shocks with $\rho_G = 0.5$. r^I is the ex-ante real yield on indexed debt, r^* the ex-ante real return on non-indexed debt.

The three left-hand columns yield one of the major quantitative insights of the paper: the volatility of consumption and inflation strictly increase in the presence of inflation-indexed debt, conditional on being in the fiscally-led policy case (i.e., conditional on deficits being effectively unfunded). Of particular interest in that regard is the fourth row of table 3, which captures the volatility of inflation in response to unfunded increases in government purchases. Conditional on being in the fiscally-led policy scenario, the unweighted volatility of inflation is around 50% of the volatility induced by the government spending increase, compared to only 20% in the counterfactual without any inflation-indexed debt. With a calibrated share of indexed sovereign debt of about 27%, on average, a one percentage point increase in the share of inflation-indexed debt more or less then corresponds to a 1.09 percentage point increase in the volatility of inflation (relative to the normalized government spending volatility) in response to unfunded increases in government purchases.

The effect of indexed debt increases in the share of inflation-indexed debt when the policy calibration follows the fiscally-led policy mix, in line with the prescription from section 3. To the best of my knowledge, this paper is among the first to quantitatively evaluate the impact that inflation-

indexed debt can have on the volatility of inflation, and how such changes in volatility are directly related to the monetary-fiscal policy nexus. Under the monetary-led policy scenario, no such increase in the volatility of inflation is visible. In fact, higher shares of debt being indexed to inflation here contribute to *decreasing* inflation volatility, while consumption volatility increases only slightly, which we will explore in greater detail now.

Impulse-responses to expansionary fiscal spending shocks

I will now look in more detail at the impulse-responses to unfunded increases in government purchases and the role borne by inflation-indexed debt when an unexpected spending increase occurs. As the persistence of any shock is relevant, the model will be simulated under different possible autocorrelations of the fiscal shock to highlight the role of persistence and the forward-looking nature of the intertemporal government budget constraint. The initial focus rests on the case of a *fiscally-led policy mix* in line with the first parametrization introduced in table 2, i.e., the baseline calibration to the U.K. economy. Figure 7 plots IRFs of aggregate variables in response to a 100bp expansionary fiscal shock that increases the need for fiscal spending when the shock has a standard persistence of $\rho_G = 0.5$.

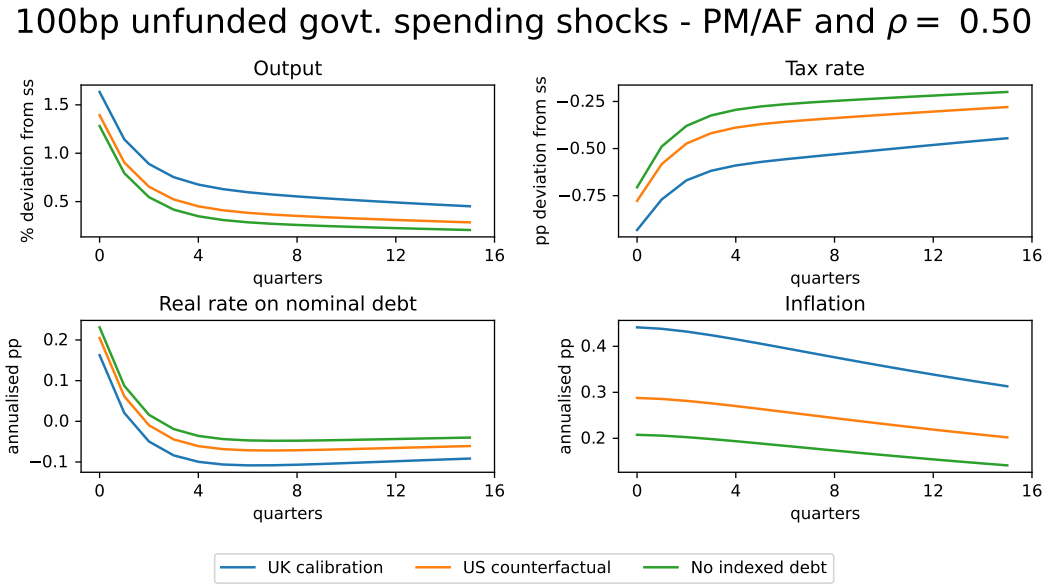


Figure 7: IRFs to an unfunded increase in government purchases of 1% of quarterly steady-state GDP on impact, $\rho_G = 0.5$, under a fiscally-led policy mix. Inflation and the ex-ante real return on nominal debt are annualized.

A number of observations is worth highlighting: the responses of output and tax rates are in line with canonical macroeconomic models and the expected reactions in response to the fiscal expansion: there is an instantaneous increase in output that persists alongside the expenditure increase. The increase in government spending here generates a slight crowding-in due to expected positive wealth effects on behalf of the output, leading to a positive (but limited) fiscal multiplier in line with some of the modern estimates of the impact of expansionary fiscal policy (Ramey and

Zubairy, 2018; Ramey, 2019). The output increase is largest for high levels of debt indexation, and it generally persists at high and positive levels under precisely this calibration.

As for the change to the marginal tax rate, the fiscal expansion is accompanied by a tax-based easing relative to steady-state levels, in line with the specification of active fiscal policy. In particular, that easing is strongest for higher levels of indexed debt under the chosen policy calibration, which beyond the impact period is directly related to the adjustment of the fiscal rule to the indexed debt term: since each fiscal authority only imperfectly internalizes the consequences of indexed debt issuance in their tax adjustment, higher shares of indexed debt correspond to less stringent taxation changes overall. The fact that the tax *rate* changes are overall negative comes, on impact, from the expansion of the tax base $w_t N_t$ at nearly unchanged tax revenue ($\Delta T_0 = \bar{s}^p \Delta Y_0$, as the lagged debt terms of the fiscal rule have not yet moved), and thereafter from the fact that the real value of government bonds relative to output, in particular of indexed bonds, decreases below steady-state (reflecting bond-price changes and household portfolio adjustment).

The evolution of the real interest rate on non-indexed debt, r_t^*, in response to the fiscal impulse is also tightly linked to the share of inflation-indexed debt in each calibration. In all calibrations, the real interest rate appreciates on impact of the shock, as the monetary authority in each economy is not fully passive, but instead *partially* tightens the screws through higher nominal rates, although they do so insufficiently to curb the full nature of the demand-driven expansion induced by fiscal spending. With positive levels of inflation-indexed debt, however, the impact change of the real interest rate is dampened: nominal rates rise the most in the U.K. calibration, but since $\phi_\pi < 1$ and inflation is persistent, the real rate rises the least there (by 0.163 p.p. annualized, against 0.205 p.p. with U.S. debt shares and 0.231 p.p. without indexed debt). The ex-ante real yield on indexed debt, $r_t^I = 1/q_t - 1$, instead rises by more on impact, by 0.393 p.p. annualized in the U.K. calibration. The difference is the liquidity spread $r_t^I - r_t^*$ that enters the indexation wedge ξ_t in equation (15), which widens by about 23 annualized basis points on impact in the U.K. calibration and turns negative after about three quarters.

Finally, the panel on the bottom right quantifies the focal point of this paper - the change in inflation in response to the fiscal expansion. Without inflation-indexed debt, annualized quarterly inflation peaks at about 0.21 p.p., compared with 0.44 p.p. in the U.K. calibration and 0.29 p.p. in the counterfactual with U.S. debt shares. Over the first year, the corresponding cumulative price increases are about 0.434%, 0.283%, and 0.204% for the U.K., U.S.-share, and no-index calibrations, respectively. The U.K. first-year price response is therefore about 0.23 percentage points larger than without indexed debt.[27] These responses are qualitatively in line with the empirical evidence presented in section 2.

Summarizing, I therefore find that turning on the debt indexation channel of government debt (i.e., setting inflation-indexed debt to positive values) significantly amplifies the nominal effects

[27] Appendix table C.1 reports the cumulative first-year price changes and the annualized impact inflation responses under both policy mixes.

of fiscal expansions, with amplification of output and consumption responses as well. The model nests the expected reaction to a fiscal expansion under a fiscally-led policy mix with non-Ricardian households: output and inflation co-move positively, and once inflation-indexed debt is present, inflationary pressure becomes more pronounced and persistent.[28]

—

I now highlight what changes in the simulations when a *monetary-led policy mix* is considered instead, corresponding to fiscal policy turning 'passive' in the language of Leeper (1991). The calibration of the policy parameters in this case follows from table 1, with $\phi_\pi = 1.5$ and $\bar{\gamma}_B = 8$. Figure 8 summarizes the results from this exercise for persistent fiscal shocks, $\rho_G = 0.5$.

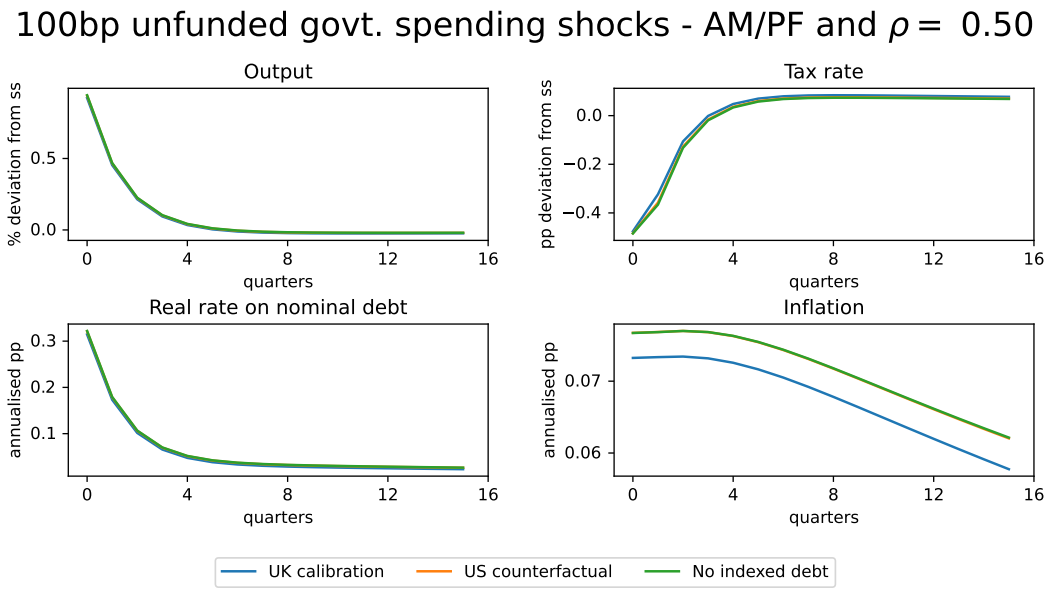


Figure 8: IRFs to an initially unfunded increase in government purchases of 1% of quarterly steady-state GDP on impact, $\rho_G = 0.5$, under a monetary-led policy mix. Inflation and the ex-ante real return on nominal debt are annualized.

The response of output turns out to be smaller relative to the fiscally-led policy mix for the calibration without indexed debt, indicating less of a crowding in of output across the board. Crucially, indexed debt alters the output response only slightly in this calibration, although differences in consumption and asset returns remain. The amplification of output and inflation by indexed debt is therefore much weaker than under the fiscally-led policy mix.

The fact that there is little quantitative difference between the U.K., U.S., and no-indexed-debt calibrations holds across the board, including for tax rates. The real rate tightening is principally steeper, in line with the specification of monetary policy, and is nearly identical across the three calibrations. Since this specification follows a conventional monetary-led policy mix, fiscal policy, as exemplified through the tax rate, passively adjusts to ensure that the government budget con-

[28] Complementary impulse-response functions of bond prices and quantities, as well as of the price level itself, are provided in appendix C.

straint holds. The tax rate initially decreases, but subsequently rises slightly above steady-state as fiscal policy provides backing for the spending increase.

Most importantly, under the monetary-led policy mix, the reaction of inflation is muted, with substantially smaller deficit-inflation multipliers than under the fiscally-led policy mix. Nonetheless, some inflationary pressure exists, and it is somewhat larger in the calibrations with no or little indexed debt. The reason for that lies in the absence of risk-sharing among households and the imperfect insurance properties borne by normal (non-indexed) bonds: in the light of such incomplete markets, the government spending measure exhibits a greater degree of Ricardian dis-equivalence, impacting households through a wealth effect. This wealth effect contributes to a reduction in household demand, but slight inflationary pressure as the unfunded increase in government purchases is not fully crowded out.

Appendix C presents further simulation results, in particular related to the IRFs of bond prices and interest rates, household policy functions and monetary policy shocks. In particular the revaluation of the bonds as expressed through their prices are of interest, as they confirm the above arguments that the revaluation of the intertemporal government debt valuation equation belongs to the main determinants of the inflationary response.

Decomposing the price level response under the fiscally-led policy mix

The government debt valuation equation, as shown through Proposition 1, allows an accounting-based decomposition of the channels of inflation in the model by postulating that this equation is informing the rate of inflation under the fiscally-led policy mix (following Bianchi et al. (2023) and Kaplan et al. (2026)), decomposing the various drivers of inflation in general equilibrium across the debt share calibrations. To that goal, we substitute in the first-order equations on indexed and non-indexed debt from the unconstrained households into (13), using $D_t^N \equiv Q_t B_t / P_t$ and $D_t^I \equiv q_t b_t / P_t$ for the real market values of the two debt instruments, and then perform a first-order approximation, which amounts to an equation decomposing the drivers of inflation:

$$\hat{\pi}_t \approx \underbrace{\widehat{R}_{t-1}^N}_{\text{lagged policy rate}} + \frac{1}{(1+r^N)D^N}\left[\underbrace{D^I \Delta r_{t-1}^I}_{\text{cost of serving indexed debt}} \underbrace{-\Delta S_t}_{\text{primary surplus}} + \underbrace{\underset{\sim}{Q}_t}_{\text{rollover and mkt-value adjustment}}\right], \tag{23}$$

where $\underset{\sim}{Q}_t \equiv (1+r^N)\Delta D_{t-1}^N + (1+r^I)\Delta D_{t-1}^I - \Delta D_t^N - \Delta D_t^I$ summarizes debt absorption by households. Figure 9 shows the results of this decomposition of inflation in all three debt-share calibrations under a conventionally fiscally-driven policy mix, as defined in table 1.

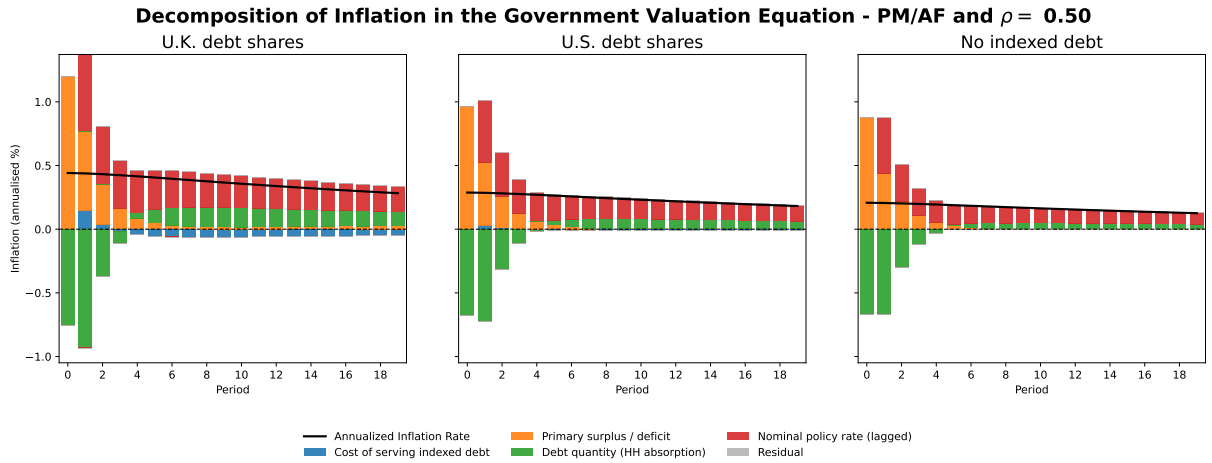


Figure 9: Decomposition of inflation in response to a 1% unfunded increase in government purchases under a fiscally-led policy mix.

The cost of serving maturing inflation-indexed debt indeed can contribute to inflation in the U.K. calibration, but compared to section 3, this richer general equilibrium model clearly highlights the direct power of the deficit-driven revaluation and of additional interest expense as primary drivers, although these are notably scaled up even in general equilibrium through the presence of indexed debt. Thus, indexed debt primarily works as an *indirect* mechanism scaling up the nominal revaluation driven through surpluses and an interest-expense effect, and not directly through higher face values of indexed debt. Economically, this is mirrored by wealth effects on the households which are net-positive. As the inflation rate remains elevated, there is a premium that the government can exert from issuing indexed debt, creating slight deflationary pressure in the U.K. calibration.

The interaction between monetary policy and the share of indexed debt

I now zoom into the joint role borne by inflation-indexed debt and the monetary policy rule. To that goal, I fix the monetary policy coefficients at the levels summarized by table 1 for the fiscally-led policy mix and vary the share of inflation-indexed debt in the government debt portfolio, $\theta_t^{mkt} = \frac{q_t b_t}{Q_t B_t + q_t b_t}$, between $[0, 0.27)$, capturing the three calibrations analyzed above, while also varying the monetary policy reaction coefficient to inflation between $[0, 1]$. Under these parametrizations, monetary policy is conventionally considered "passive". The reaction of the price level in the first year after the fiscal impulse across the monetary policy specifications and various shares of inflation-indexed debt in the sovereign debt portfolio is depicted in figure 10.



Figure 10: Cumulative one-year reaction of prices in response to fiscal spending shocks under a fiscally-led policy mix.

On the x-axis, I vary the share of indexed debt in the total debt portfolio (while maintaining a constant overall relation between the gross stock of debt and GDP), while the colors indicate the chosen monetary reaction coefficient ϕ_π. Thus, orange and especially brown colors reflect 'less passive' monetary policy in the conventional sense (as the monetary authority acts closer to the standard proposition of inducing a real rate tightening in response to inflationary pressure), while greener colors reflect 'more passive' monetary policy.

Generally, the inflationary pressure is increasing in the share of inflation-indexed debt. The magnitude of the consequences increasing indexed debt, however, is striking: increasing the share of indexed debt by 5 percentage points can boost the one-year deficit-inflation multiplier by 0.02 percentage points when monetary policy is fully passive, but a similar increase in the indexed debt share can boost the deficit-inflation multiplier by 0.1 percentage points for $\phi_\pi \in (0.5, 1)$. The marginal effect of an increase in the share of inflation-indexed debt is smaller for more permissive monetary policy, whereas it is larger when monetary policy tightens the nominal rate somewhat.

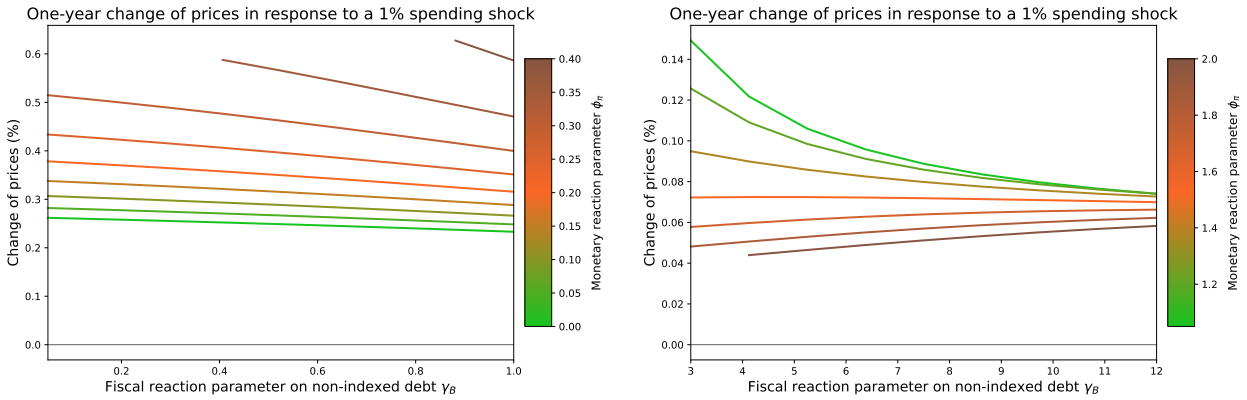
Fiscal-monetary policy combinations, inflation, and determinacy

The functional form of the fiscal reaction rule has been kept constant, in line with specification (22), in the previous analysis. However, fiscal and monetary policy do not operate in a vacuum from one another, and it makes sense to consider in greater detail how monetary and fiscal policy jointly matter for inflationary dynamics in a model with indexed debt. I now focus more directly on the link between the fiscal and monetary reaction rules, keeping inflation-indexed debt at constant and elevated levels corresponding to the share of inflation-indexed debt present in the U.K.

To keep the results simple, I focus on appropriate splits into policy areas that are considered conventionally fiscally-led and monetary-led. The fiscally-led policy mix is conventionally character-

ized by $\bar{\gamma}_B \in [0, 1)$ and $\phi_\pi \in [0, 1)$, while the monetary-led policy mix is conventionally characterized by $\bar{\gamma}_B >> 1$[29] and $\phi_\pi \geq 1$, leaving aside in either case the possibility of negative policy parameters. In this exercise, I compare one-year inflation in response to a 100bp expansionary fiscal shock across various values of the fiscal anchor $\bar{\gamma}_B$ and the monetary reaction parameter ϕ_π. The results for the fiscally-led policy mix are described by panel a of figure 11 and the results for the monetary-led policy mix are described by panel b.

Under the fiscally-led policy mix in panel a, the deficit-inflation multipliers are principally larger than under the monetary-led policy mix in panel b. This holds across all particular calibrations. Relative monetary passivity (embedded by $\phi_\pi \to 0$, i.e., by the green lines in panel a) induces smaller inflation multipliers, as do larger values of the fiscal policy parameter $\bar{\gamma}_B$. Note that, interestingly, the distance between the lines is larger than the slope of each line. Therefore, under such fiscally-led policy mixes, the deficit-inflation multiplier is more sensitive to the monetary reaction parameter, similar to what was explored when varying the share of inflation-indexed debt. Fiscal decisions, such as the exact sensitivity of tax rates to deviations of debt from steady-state or the exact share of debt indexed to inflation, are clearly anchoring the extent of the inflationary effect of expansionary fiscal policy, but the monetary authority's behavior clearly shapes the ultimate response to a significant degree as well.



(**a**) Fiscally-led policy mix

(**b**) Monetary-led policy mix

Figure 11: The one-year deficit-inflation multiplier under fiscally-led and monetary-led policy mixes. Missing line segments correspond to areas where a unique saddle path-stable equilibrium is not attained.

Within the confines of the monetary-led policy mix explored in panel b, more restrictive monetary policy (increasing ϕ_π) always decreases the size of the deficit-inflation multiplier. A more restrictive fiscal policy (increasing $\bar{\gamma}_B$), however, does only locally decrease the inflationary pressure in response to the fiscal impulse when monetary policy is close to a real rate peg. Flipping around the result of the fiscally-led policy mix, the effect of marginal changes to the fiscal policy parameter here generally outweigh the effects induced by changes to monetary policy. Under the monetary-led policy mix, therefore, the fiscal policy anchor is particularly informative about the

[29] See the discussion in section 5 for the reasoning behind considering values of $\bar{\gamma}_B$ that are significantly above 1.

deficit-inflation multiplier. In sum, the policy authority that is usually considered 'passive' has still quite significant influence on the exact size of the deficit-inflation multiplier within its constraint set.

Equilibrium stability of the numerical simulations

As a final exercise, I consider for which values of the fiscal and monetary policy parameters the linearized system implies a unique, saddle path-stable equilibrium. In doing so, I exploit the 'winding number criterion' developed in Auclert et al. (2025), which is consistent with the use of the associated sequence-space Jacobian methodology to solve the dynamic model.[30]

Figure 12 summarizes the determinacy properties of the model on a grid of the fiscal policy anchor $\bar{\gamma}_B$ and the main monetary policy reaction parameter ϕ_π, with the remainder of the calibration being unchanged, and separately for a lower value of $\bar{\gamma}_b = 1$ and a higher value of $\bar{\gamma}_b = 2$.

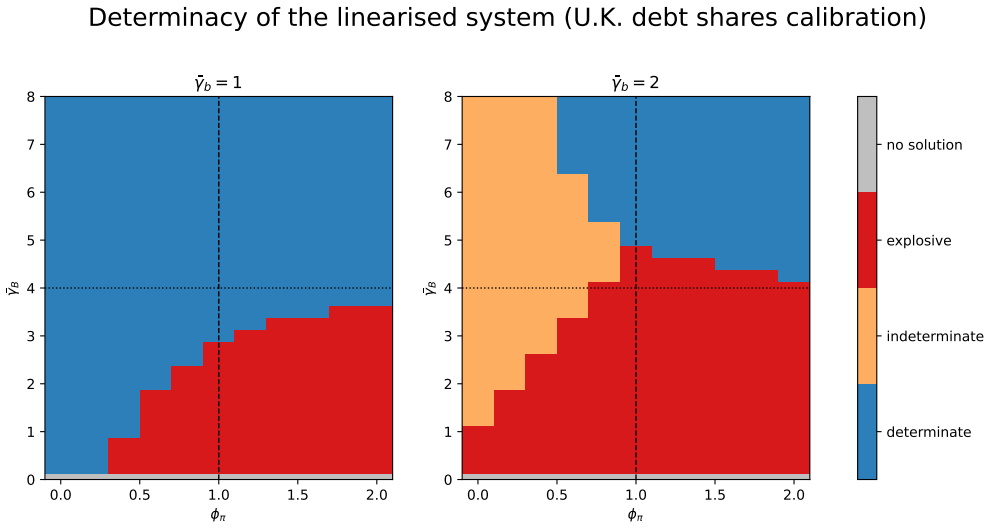


Figure 12: Determinacy of the generalized Jacobian in relation to choices for the fiscal and monetary policy reaction coefficients under two different values of $\bar{\gamma}_b$. The black lines mark the reference values $\phi_\pi = 1$ and $\bar{\gamma}_B = 4$.

In line with recent evidence on determinacy properties in non-Ricardian models (Kaplan, 2025a; Rachel and Ravn, 2025), the standard notions of determinate policy spaces do not apply one-for-one when fiscal policy is non-Ricardian. In the left panel, where $\bar{\gamma}_b = 1$ makes indexed debt almost neutral for determinacy properties, we actually nest the result of Rachel and Ravn (2025), by which the saddle path-stable area becomes continuous, overcoming the distinction between conventional active fiscal and monetary policy. The 'explosive' area characterizes what would conventionally be considered an 'active/active' policy mix, and is fairly large due to the coupling of the debt valuation

[30] A detailed exposition of the 'winding number criterion' can be found in Auclert et al. (2025). Intuitively, it is related to the Blanchard and Kahn (1980)-conditions, which are cast in state-space. The winding number criterion provides a generalizable mapping of the Blanchard-Kahn conditions onto the sequence-space, i.e., allowing for infinitely many 'quasi-roots' of the linearized system. The prerequisites to apply the winding number criterion, such as the quasi-Toeplitz property of the generalized Jacobian, are not violated (the corresponding results are available upon request).

block to the output-inflation nexus.

Changing the fiscal anchor for indexed debt in the tax rule to $\bar{\gamma}_b = 2$ allows for the emergence of sunspot equilibria under monetary policy that would conventionally be considered passive, as shown in the right-hand panel of figure 12. Intuitively, when fiscal policy reacts very strongly to deviations of indexed debt from steady-state in its taxation rule (i.e., under high $\bar{\gamma}_b$), the government debt valuation equation *necessarily* has a stable root as the tax adjustment covers significantly for any additional face value increase coming from an inflation-driven increase in the face value of indexed debt. Pairing this with a monetary rule that induces a stable root in the output-inflation nexus, problems of indeterminacy can emerge. In sum, therefore, excessive reactiveness of the fiscal authority to deviations of the real value of *indexed* debt to steady-state can contribute to system instability in the form of the possible emergence of sunspot shocks, despite the possible good intentions underlying such an excessively reactive fiscal rule.

The distributional consequences of indexed debt

As the model features a complete distribution of households with wealth percentile-specific asset holdings, I can also speak to the distributional effects induced by the presence of inflation-indexed government debt. In the following, I will focus only on the *differences* in the model outcomes for the calibrations to the U.K. portfolio split relative to the counterfactual without indexed debt, while appendix D shows the corresponding level effects in response to an expansionary fiscal policy shock for both of the calibrations considered. Here, I principally consider only the calibration to the fiscally-led policy mix, which has proven itself to be the case under which indexed debt has significant quantitative effects.

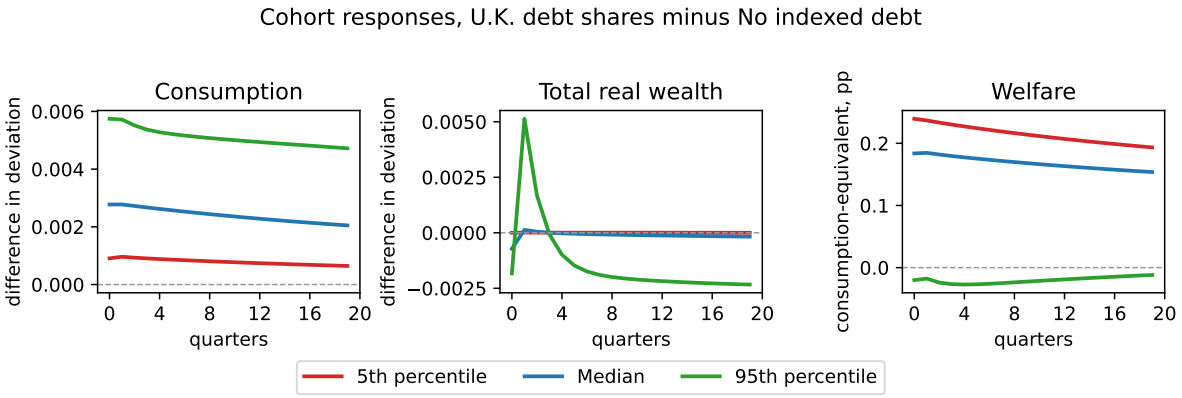


Figure 13: Differences in wealth-percentile specific impulse-responses to an expansionary government spending shock of 1% of quarterly GDP between the U.K. and no-indexed-debt calibrations. Consumption and wealth are in model units, with quarterly steady-state output normalized to one. Welfare is in consumption-equivalent percentage points.

Figure 13 highlights how a 1% government spending/GDP impulse propagates to household-specific variables in the model, contrasting the bottom of the wealth distribution (red line) with the middle (blue line) and the top (green line), where I difference the responses of each given

wealth percentile for the U.K. calibration with the counterfactual calibration to a model with the same gross debt quantity, but less indexed debt.

The presence of indexed debt generally delivers larger consumption gains to households at the top of the wealth distribution under the fiscal expansion, who enjoy a consumption increase of around 0.6% of steady-state output that is *fully* attributed to the presence of inflation-indexed debt in the model, with the same increase being only about 0.1% of steady-state output at the bottom of the wealth distribution. Relative to each group's own steady-state consumption, however, the increase is larger at the bottom (0.68%) than at the top (0.43%). This is mostly driven by higher labor income in the presence of indexed debt, as aggregate hours, which the labor union allocates equally to all households, rise by more, as seen in figure 14. In welfare terms, the gains from indexed debt are also largest at the bottom of the wealth distribution (0.24 p.p. in consumption-equivalent terms at the 5th percentile and 0.18 p.p. at the median), while households at the 95th percentile are slightly worse off (-0.02 p.p.), as their larger nominal portfolios are more exposed in absolute terms to the devaluation induced by the stronger inflation response.

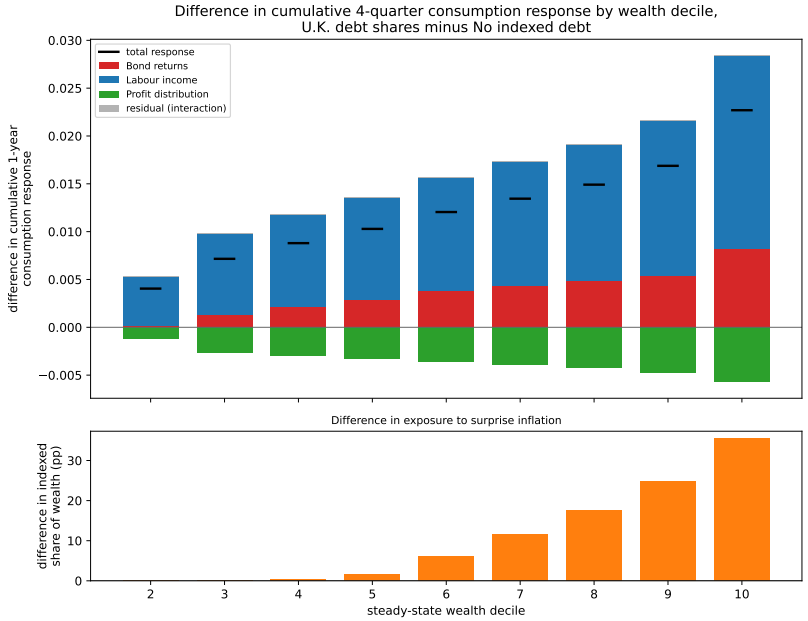


Figure 14: Top panel: differences in the sum of the first four quarterly consumption responses between the U.K. and no-indexed-debt calibrations, in model units with quarterly steady-state output normalized to one. Deciles 2–10 are displayed. Bottom panel: decile-specific differences in steady-state holdings of indexed debt for each wealth decile (expressed in percentage points of total assets).

Figure 14 zooms into the channels underlying the consumption differences across the displayed deciles of the wealth distribution in the first year after the impulse, decomposing the exact channels leading to the differences in real consumption. It turns out that a quite significant part of the differences in the consumption response coming from the change in debt indexation can be attributed to differences in labor income earned by households at the top of the wealth distribution, even after accounting for the opposing effect of the uniform redistribution of firm profits.

Note that these effects are still driven by the presence of indexed debt, since the plot shows the *differences* between the U.K. and the no-index-debt calibrations, so the changes to labor income and profit distribution here amount to *indirect* effects of the presence of indexed debt. Portfolio-driven differences in returns earned on government debt held across the distribution, which can be seen as the *direct* effect of indexed debt, account for a difference of about 0.2% of annual steady-state output between the highest and lowest displayed wealth deciles over the first year, which highlights the significance of the debt indexation-insurance mechanism enjoyed by households at the top of the wealth distribution.

7 Long-term indexed debt in a tractable general equilibrium framework

Section 6 explored in a rich quantitative model how inflation-indexed debt and policy rules interact to give rise to inflationary pressure in response to unfunded increases in government purchases. In that model, however, characterizing analytically how the composition of the sovereign debt portfolio shapes the inflation response is not possible, the use of spending shocks (as opposed to government transfer shocks) has direct and complicated consequences for aggregate market clearing, and the possible inclusion of long-term debt would make the framework even more complicated. I therefore revisit the mechanism in the tractable New Keynesian model with mortality risk developed by Angeletos et al. (2024), which builds on Blanchard (1985).[31] The mortality friction serves as a parsimonious proxy for short effective planning horizons and the elevated marginal propensities to consume that are central to heterogeneous-agent models, while retaining a higher degree of analytical tractability. I also make use of this model to discuss briefly the consequences of increasing the duration of inflation-indexed debt beyond one period.

Inflation-indexed debt will matter significantly for inflation in this class of model as well. Angeletos et al. (2024) show that with finite household horizons, an expansionary deficit shock is partially self-financing, and that the self-financed share converges to unity as the direct response of taxes to the outstanding debt stock is made arbitrarily weak. In that limit, the finite-horizon economy delivers an impact inflation response that an economy with infinitely-lived households only delivers under a fiscally-led policy mix, and this observational equivalence has been read as making the fiscally-led policy mix dispensable for the analysis of deficit-driven inflation. I here show that the introduction of a long-term indexed instrument whose required real return responds to the deficit-induced boom allows this type of observational equivalence to fail for any strictly positive share of *long-term* inflation-indexed debt.

My organizational device is the comparison of two different economies, which I index by $j \in \{A, B\}$. The two economies share the same debt portfolio, the same monetary rule and the same fiscal rule, and they differ only in the survival probability, with $\omega_A = \omega \in (0, 1)$ and $\omega_B = 1$. What separates them is therefore not a difference in policy but a difference in what closes the model. In

[31] In Kawalec (2026), I consider a related model in richer detail, focusing on the abilities of monetary and fiscal policymakers to undo the distortionary effects of indexed debt.

economy A, the outstanding debt stock enters aggregate demand directly, because finite horizons break Ricardian Equivalence, and the equilibrium is pinned down by saddle path stability of a dynamic system in debt and output. In economy B, the demand block does not respond to the debt stock at all, and the equilibrium is pinned down by the government debt valuation equation, which is the sense in which economy B is fiscally-led. Following Angeletos et al. (2024), I let the direct response of taxes to the outstanding debt stock, denoted by τ_d, become arbitrarily weak, taking $\tau_d \to 0^+$ in economy A and setting $\tau_d = 0$ in economy B. As shown below, this is also the only region in which the fiscally-led benchmark exists, so it is the region in which the two economies can be compared. In the following, uppercase variables denote levels and lowercase variables denote deviations from the deterministic steady-state, while debt, taxes and household wealth are measured in absolute deviations from steady-state and are scaled by Y^{SS}.

The debt portfolio and the pricing of long-term indexed debt

In this model, a share $\theta \in [0, 1)$ of the steady-state real market value of privately held government debt is inflation-indexed and the remaining share $1 - \theta$ is one-period nominal debt. The indexed component has a geometric maturity structure governed by $\delta \in [0, 1)$, such that a unit outstanding at the end of period t pays one unit of the consumption good at $t + 1$ and δ^k units at $t + 1 + k$ for every $k \geq 1$. If $\delta = 0$, indexed debt becomes one-period only, while larger values of δ correspond to longer duration.

Let Q_t^j denote the real price of the indexed security in economy j without the coupon paid, and let q_t^j denote its log-deviation from steady-state. Denote by r_t^j the proportional deviation of the expected gross real return on the one-period nominal instrument from its steady-state value β^{-1} and by ψy_t^j, with $\psi \geq 0$, the expected proportional excess real return that the indexed instrument must offer over the nominal one. Then, the indexed security satisfies

$$Q_t^j = \frac{\beta \mathbb{E}_t \left[1 + \delta Q_{t+1}^j\right]}{1 + r_t^j + \psi y_t^j}, \tag{24}$$

so that in the deterministic steady-state $Q^{SS} = \beta/(1 - \beta\delta)$, and a first-order approximation of equation (24) around that point delivers

$$q_t^j = -\left(r_t^j + \psi y_t^j\right) + \beta\delta \mathbb{E}_t q_{t+1}^j. \tag{25}$$

The indexed claim cannot be revalued by an inflation surprise, which is the mechanism isolated in section 3, but its real market price *can* be revalued through movements in the real return that the market requires on it. That second margin exists only when the required return itself moves, and only when the claim has positive duration, since $\delta = 0$ collapses equation (25) to $q_t^j = -(r_t^j + \psi y_t^j)$ with no consequence for the value of the outstanding stock.

Policy rules and the government budget

Monetary policy sets the proportional deviation of the expected gross real return on the one-period nominal instrument according to

$$r_t^j = \phi y_t^j, \tag{26}$$

as in Angeletos et al. (2024). The proportional deviation of the expected gross real return on the government debt portfolio is then the market-value weighted average of the two instrument-return deviations,

$$(1-\theta)\,\phi y_t^j + \theta\,(\phi+\psi)\,y_t^j = (\phi+\theta\psi)\,y_t^j. \tag{27}$$

Fiscal policy follows the rule of Angeletos et al. (2024) without modification,

$$t_t^j = -\varepsilon_t + \tau_d\left(d_t^j + \varepsilon_t\right) + \tau_y y_t^j, \tag{28}$$

where ε_t is the deficit shock, $\tau_d \in [0,1)$, and $\tau_y \in (0,1)$. No indexation adjustment enters equation (28). Expected indexation payments are compensated ex ante through the price at which the indexed instrument is issued, so they impose no additional financing requirement on the fiscal authority, and only the loss of the surprise inflation margin and the revaluation of the outstanding indexed stock do.

The consolidated government flow budget constraint in real market value terms is

$$d_{t+1}^j = \frac{1}{\beta}\left(d_t^j - t_t^j\right) + \frac{1}{\beta}\frac{D^{SS}}{Y^{SS}}\,(\phi+\theta\psi)\,y_t^j - \frac{1}{\beta}\,(1-\theta)\,\frac{D^{SS}}{Y^{SS}}\left(\pi_{t+1}^j - \mathbb{E}_t\pi_{t+1}^j\right) + \theta\delta\frac{D^{SS}}{Y^{SS}}\left(q_{t+1}^j - \mathbb{E}_t q_{t+1}^j\right). \tag{29}$$

The third term on the right-hand side is the erosion of the one-period nominal component by surprise inflation, attenuated by the factor $1-\theta$ because the indexed component cannot be eroded. The fourth term is the revaluation of the long-term indexed component, which carries no inflation surprise because the claim is real, and which is weighted by δ. Although the price surprise enters the proportional holding return with coefficient $\beta\delta$, converting that proportional deviation into a change in the gross return introduces the steady-state gross return β^{-1}. Starting from steady-state before the date zero shock is observed, equation (29) implies the following necessary revaluation of the stock of debt on impact of a shock:

$$d_0^j = -\frac{1}{\beta}\,(1-\theta)\,\frac{D^{SS}}{Y^{SS}}\pi_0^j + \theta\delta\frac{D^{SS}}{Y^{SS}}q_0^j. \tag{30}$$

Aggregate supply is described by the standard New Keynesian Phillips Curve, $\pi_t^j = \kappa y_t^j + \beta\mathbb{E}_t\pi_{t+1}^j$, which for a bounded equilibrium can be iterated forward to $\pi_t^j = \kappa\mathbb{E}_t\sum_{k=0}^{\infty}\beta^k y_{t+k}^j$.

Impact inflation and the failure of the observational equivalence

The equilibrium of either economy is characterized by a first-order difference system in the outstanding debt stock and output, which is derived in appendix E. The persistence of the equilibrium path is governed by the smaller root of the characteristic equation of that system, which I denote by $\rho_A(\tau_d)$ in economy A and by ρ_B in economy B, and which solves

$$\rho^2 - \left[\frac{1-\tau_d}{\beta} + 1 + \sigma(\phi+\theta\psi) + \frac{\left(1-\beta\omega_j\right)\left(1-\omega_j\right)}{\beta\omega_j}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)\right)\right]\rho + \frac{(1-\tau_d)\left[1+\sigma(\phi+\theta\psi)\right]}{\beta} = 0, \tag{31}$$

where $\omega_A = \omega \in (0,1)$ and $\omega_B = 1$. I write $\bar{\rho}_A \equiv \lim_{\tau_d \to 0^+} \rho_A(\tau_d) = \rho_A(0)$. Setting $\omega_j = 1$ removes the third term in the bracket, upon which equation (31) factorizes into $\left(\rho - \frac{1-\tau_d}{\beta}\right)(\rho - 1 - \sigma(\phi+\theta\psi))$ for every τ_d, so that the root governing the equilibrium path of economy B is exactly $\rho_B = 1 + \sigma(\phi+\theta\psi)$. The entire content of the result below is that this root differs from its limiting counterpart in the economy with mortality risk, and that inflation-indexed debt of positive duration is the instrument through which that difference reaches the price level.

Proposition 2 *Suppose that*

$$-\frac{1}{\sigma} < \phi + \theta\psi \leq 0, \qquad \tau_d \in [0, 1-\beta), \qquad \tau_y > 0. \tag{32}$$

Then, economy A admits a unique bounded equilibrium for every $\tau_d \in [0,1-\beta)$, while at $\tau_d = 0$ economy B admits a unique fiscally-led equilibrium selected by the intertemporal government budget condition. The relevant limiting roots satisfy $0 < \bar{\rho}_A < \rho_B = 1 + \sigma(\phi+\theta\psi) \leq 1$, and impact inflation in response to a one-off positive deficit shock $\varepsilon_0 > 0$ is, in the limit $\tau_d \to 0^+$ for economy A and at $\tau_d = 0$ for economy B,

$$\pi^j_{\varepsilon,0} = \frac{\kappa\,\varepsilon_0}{\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi) + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa + \theta\delta\frac{D^{SS}}{Y^{SS}}(\phi+\psi)\frac{1-\beta\rho_j}{1-\beta\delta\rho_j}}, \qquad j \in \{A,B\}, \tag{33}$$

where $\rho_j = \bar{\rho}_A$ for $j = A$ and $\rho_j = \rho_B$ for $j = B$, and where both denominators are weakly larger than $\tau_y + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa$ and hence strictly positive. The two denominators differ by exactly

$$\Delta \equiv \frac{\theta\beta\delta(1-\delta)\frac{D^{SS}}{Y^{SS}}(\phi+\psi)\left[1+\sigma(\phi+\theta\psi)-\bar{\rho}_A\right]}{(1-\beta\delta\bar{\rho}_A)\left(1-\beta\delta\left[1+\sigma(\phi+\theta\psi)\right]\right)}, \tag{34}$$

so that $\pi^A_{\varepsilon,0} > \pi^B_{\varepsilon,0}$ if and only if $\Delta < 0$. Consequently, for every $\omega \in (0,1)$, every $\delta \in (0,1)$ and every $\phi + \psi \neq 0$,

$$\pi^A_{\varepsilon,0} = \pi^B_{\varepsilon,0} \quad \Longleftrightarrow \quad \theta = 0. \tag{35}$$

Proof. See appendix E. ■

The restriction on τ_d in (32) is imposed for economy B and not for economy A, following in a sense Angeletos et al. (2024). Economy B admits a unique equilibrium only while the tax response to the debt stock is too weak to stabilize that stock on its own, that is, while $\tau_d < 1 - \beta$, and for larger values of τ_d the representative-agent economy is indeterminate for the posted small value of ϕ, with the Ricardian allocation that leaves impact inflation at zero being one element of a continuum. Economy A is determinate on the strictly larger range established in part two of lemma 5, and on the whole of $[0, 1)$ when the required real return on the debt portfolio is pegged. The presence of mortality risk therefore delivers a determinate equilibrium in a policy region in which the representative-agent economy has none, which is the sense in which the fiscally-led policy mix has been argued to be dispensable.

Equation (34), however, qualifies that equivalence. First, finite household horizons make the deficit-induced boom less persistent in economy A, since lemma 5 establishes that $\bar{\rho}_A < \rho_B$. Second, because $\beta\delta < \beta$ for indexed debt of positive but finite duration, the indexed-bond price and inflation discount this common output path at different rates. The resulting ratio $(1 - \beta\rho) / (1 - \beta\delta\rho)$ is strictly decreasing in ρ, so that for each unit of impact inflation the less persistent path in economy A generates a larger revaluation of the outstanding indexed stock. With $\phi + \psi < 0$ that revaluation is a capital gain for bondholders, it raises the real market value of government liabilities, and it therefore requires a larger inflation response to validate the same deficit. With $\phi + \psi > 0$, the indexed claim instead incurs a capital loss and the direction of the discrepancy is reversed. In the economy without indexed debt or any other debt of positive duration, the difference in front-loading has nothing to attach itself to, and the two economies deliver the same impact inflation.

The condition $\phi + \theta\psi \leq 0$ in (32) expresses the known idea that monetary policy cannot adhere to the Taylor principle for the representative-agent economy to admit a fiscally-led equilibrium. The common peg $\phi = \psi = 0$ is one knife-edge configuration at which the observational equivalence survives partial indexation. More generally, for a strictly positive indexed-debt share and strictly positive duration, equivalence survives if and only if $\phi + \psi = 0$, while every admissible configuration with $\theta > 0$, $\delta \in (0, 1)$ and $\phi + \psi \neq 0$ breaks it. Equivalence also survives when $\theta = 0$ or $\delta = 0$, because in either case no outstanding indexed instrument can transmit differences in the timing of the output response to impact inflation. A second observation follows immediately from equation (34). For $\theta \in (0, 1)$ and $\delta \in (0, 1)$, pegging the expected real return on the debt *portfolio*, that is, setting $\phi + \theta\psi = 0$ with $\psi > 0$, does not restore the equivalence, because the indexed claim is priced by the return required on that instrument and not by the return earned on the portfolio, and $\phi + \psi = (1 - \theta)\psi > 0$ in that case.

A useful benchmark is thus the peg itself at which equation (33) collapses to a single closed form common to both economies.

Corollary 1 *Under the conditions of proposition 2 and at $\phi = \psi = 0$, impact inflation is the same in both*

economies and equals

$$\pi_{\varepsilon,0} = \frac{\kappa}{\tau_y + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa}\,\varepsilon_0, \qquad \frac{\partial \pi_{\varepsilon,0}}{\partial \theta} = \frac{\frac{1}{\beta}\kappa^2 \frac{D^{SS}}{Y^{SS}}}{\left(\tau_y + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa\right)^2}\,\varepsilon_0 > 0. \tag{36}$$

Corollary 1 is the analytical counterpart of the quantitative finding of section 6 and of the empirical estimates of section 2. The deficit-inflation multiplier is strictly increasing in the share of inflation-indexed debt, and the mechanism is exactly the one isolated in the one-equation model of section 3, namely that only the fraction $1-\theta$ of the inherited stock can be devalued by an inflation surprise, so that a given funding shortfall requires a larger price level adjustment the larger is the indexed share. The magnitude implied by equation (36) is smaller than the one obtained in the quantitative model, which is to be expected, since the tractable framework holds the real interest rate exogenous, abstains from household heterogeneity, and captures only the impact revaluation rather than the full dynamic feedback of the price level on itself. Away from the peg, an additional revaluation channel arises whose direction depends on the response of the required return on the indexed instrument. When $\phi + \psi < 0$, the capital gain on the outstanding indexed stock reinforces the loss of the erosion margin, while when $\phi + \psi > 0$, the indexed claim instead incurs a capital loss and the revaluation channel operates in the opposite direction. The total effect of the indexed share away from the peg additionally depends on the response of the portfolio return and on the endogenous persistence of the equilibrium path. Under the calibration $\beta = 0.99$, $\sigma = 1$, $\kappa = 0.055$, $\tau_y = 0.25$, $D^{SS}/Y^{SS} = 1.12$, $\omega = 0.8$, $\delta = 0.95$ and $\psi = 0$, used below in figures 15 and 16, moving the indexed share from the value observed in the United States to the value observed in the United Kingdom raises the deficit-inflation multiplier by four percent at the peg.

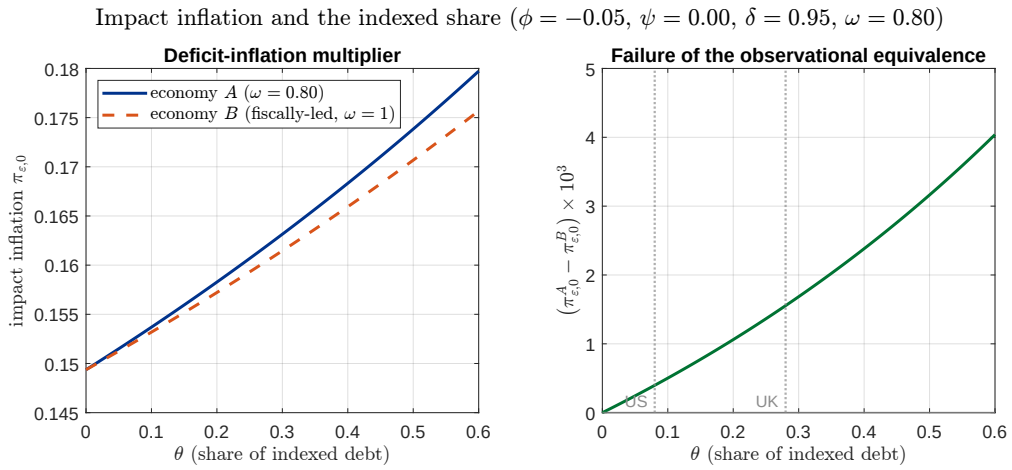


Figure 15: A visualization of proposition 2 and the associated difference in impact inflation between economies A and B.

The result qualifies the reading of Angeletos et al. (2024) under which the fiscally-led policy mix can be dispensed with in the analysis of deficit-driven inflation. That intuition is exact in an econ-

omy whose entire liability structure is nominal, and it also survives with indexed debt when the required real return on the indexed instrument is pegged or when the indexed instrument has zero duration. Once a government issues indexed liabilities of positive duration and the required return on that instrument responds to the deficit-induced boom, the two classes of models no longer deliver the same inflation response to the same deficit, because they differ in the timing of the boom that the deficit sets off and the indexed instrument prices that timing.

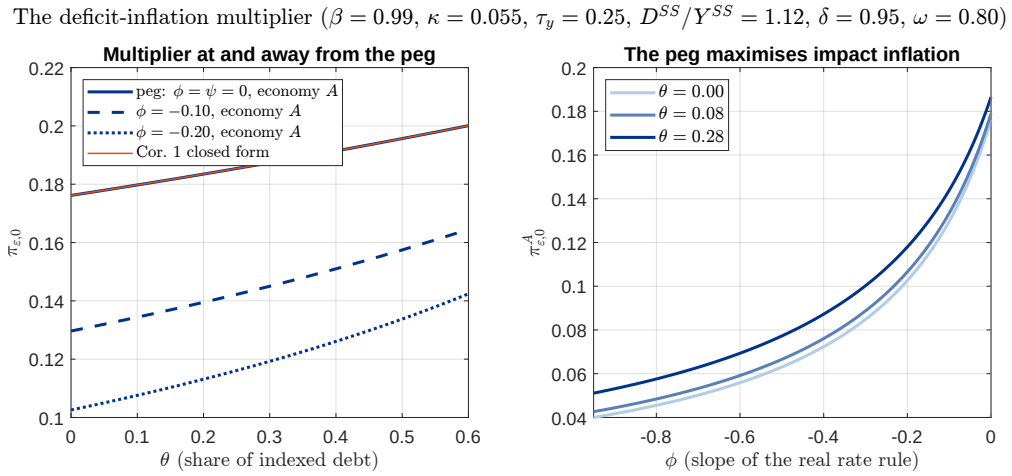


Figure 16: The effect of the monetary rule and the share of inflation-indexed debt on impact inflation in economy A.

8 Conclusion

This paper introduced inflation-indexed debt into non-Ricardian general equilibrium models. I first provided empirical evidence on the role of inflation-indexed debt as a major determinant of inflationary dynamics with the help of local projections applied to U.S. data, as well as with a specific large fiscal shock in the U.K. in September 2022. Next, I established in a simplified model that such debt itself suffices to make the price level backward-looking: the previous price level therefore matters for the determination of today's price level. I then introduced inflation-indexed debt in a state-of-the-art macroeconomic model with imperfect markets and household heterogeneity, characterizing conditions under which the debt valuation equation holds analytically. In the U.K. calibration, the cumulative first-year price increase after an unfunded government spending shock of 1% of quarterly GDP is about 0.23 percentage points larger than without indexed debt. Then, I provided model-driven evidence that inflation-indexed debt can indeed exacerbate the inflationary response to unfunded increases in government purchases, in particular when fiscal policy is considered conventionally 'active' in the sense of Leeper (1991). Finally, I contrasted fiscally-led policy mixes and other mechanisms inducing non-Ricardian household behavior in a tractable model to complement the quantitative analysis, showing that long-term indexed debt can break the observational equivalence between fiscally-led mechanisms and finite household horizons when the required real return on the indexed instrument responds to the boom.

Both the empirical and theoretical results derived in this paper thus tarnish the classic notion that inflation-indexed bonds always limit inflation in a given country by offering governments a commitment device to 'not inflate the debt away', as argued by Campbell and Shiller (1996). While this notion can remain true absent incompletely funded government deficit shocks, my results point out that once the government budget is ex-post (after debt issuance) in disarray, the inflationary consequences of funding shortfalls can increase in the share of inflation-indexed debt. Issuance of indexed debt can therefore backfire despite its great ability to serve as an ex ante commitment device, as argued rigorously by Schmid et al. (2026).

Despite these conclusions, more can be done to emphasize the interaction between inflation-indexed debt and inflation. A complete estimation of the model based off long-running samples of U.K. and U.S. data with a particular focus to the fiscal and monetary rules would further strengthen the conclusions of this paper. Another refinement can be the inclusion of long-term government debt in the quantitative model: as Cochrane (2001) and Barro and Bianchi (2026) show, the maturity structure of government debt matters for the trade-off between front-loaded and delayed inflation responses to deficit shocks.

Finally, inflation-indexed debt can inform recent policy debates on the possible regressivity or progressivity of inflation as implicit taxation. As evidenced by figure A.1 in the appendix and partially explored in section 6, inflation-indexed debt, which serves as an insurance device against unexpected inflation, seems to be particularly skewed in household portfolios towards the highest decile of the income distribution. The quantitative exercise shows that larger absolute consumption gains at the top need not imply larger welfare gains: the consumption-equivalent gains from indexed debt are largest at the bottom. A more thorough analysis of the welfare effects of unexpected inflation to households at varying income levels, and a resulting characterization as optimal policy, would make the case for issuing inflation-indexed government debt clearer.

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Appendix

A Additional empirical evidence

A.1 The distribution of government debt holdings across households

Publicly available microdata reinforces the idea of an unequal distribution of indexed debt in household portfolios. This brief section focuses on the U.S. due to the superior availability of household-level data on asset holdings.

Figure A.1 plots the real (2017) Dollar value of nonindexed and indexed government debt holdings of households questioned in the U.S. Survey of Consumer Finances (SCF), separated by income deciles.[32] The left-hand panel of figure A.1 reflects the well-known left tail of bond holdings of households in the income distribution, by which households at the upper end of the income distribution hold a significantly larger share of sovereign bonds. The right-hand panel of figure A.1 reflects a less well-known observation: this left tail is *vastly* more pronounced for indexed sovereign bonds, with the top income decile holding almost 40% of such bonds in the sample.

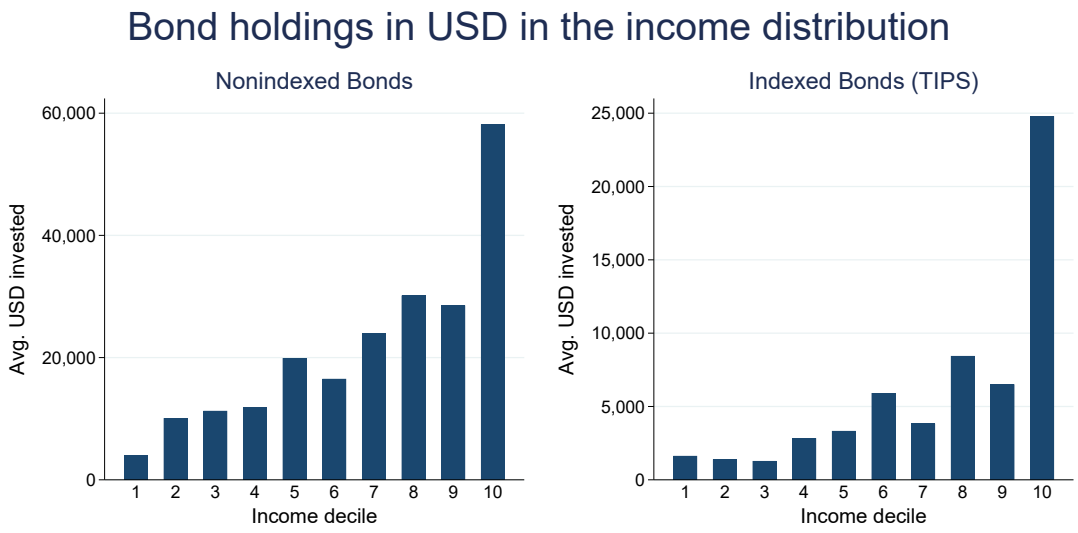


Figure A.1: Distribution of indexed and non-indexed debt holdings across household income deciles, denoted in real (2017) USD. Data source: Survey of Consumer Finances (U.S.). Sample period: 2014-2019.

Figure A.2 provides further evidence in favor of the distribution implied by the model, which has not been a targeted moment. Just as in the model (figure C.1), the density of the asset distribution of both indexed and non-indexed bonds exhibits a significant skew, which is more pronounced overall for inflation-indexed debt. In particular, the size of the bins, even if not exactly matched, broadly reflects the distribution of the model very well.

[32] I chose income deciles due to their clear definition in the survey with a single question. Constructing individual wealth variables is possible with the survey data, albeit this process is subject to particular choices about what to consider as household wealth. For most definitions of wealth, the results continue to hold qualitatively.

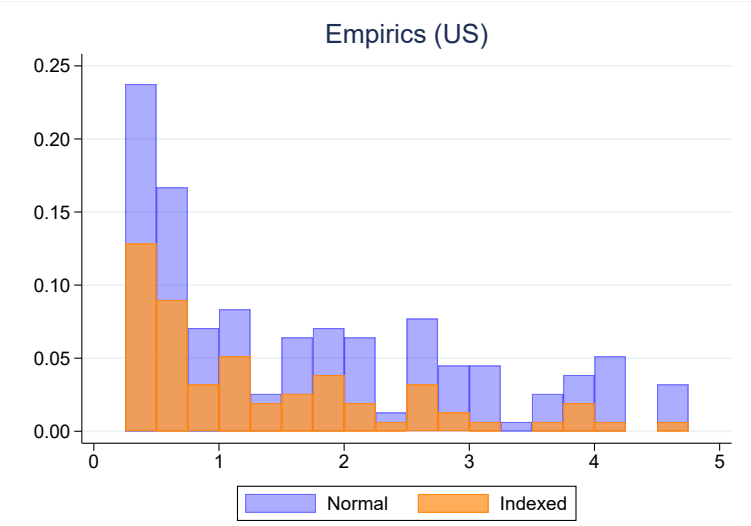


Figure A.2: Density of indexed and non-indexed debt holdings in the U.S. Survey of Consumer Finances, snapshot from October 2019.

A.2 Additional evidence on the September 2022 'mini-budget' shock

In this part of the appendix, I provide additional details on the fiscal policy measure dubbed "The Growth Plan" in September 2022 in the United Kingdom, commonly known as the 'mini-budget', and the reaction on financial markets to its announcement. To do so, I utilize ticker-frequency data on market prices (and thus yields), debt quantities, and risk perceptions. The data comes from the Bank of England (BoE), the U.K. Treasury, and Bloomberg Financial Services.

I begin by examining the degree to which the policy announcement can be informative about the propensity of a type of 'fiscally-led policy mix' in a wider sense, i.e., whether the policy measures around this particular fiscal shock can be placed in a context at which monetary policy passively adjusts to the fiscal policy measure, taking the fiscal announcement as given.[33] A measure that is plausibly related to debt sustainability concerns introduced by the budget announcement as well as to the prospective monetary reaction are expected overnight interest rates. These are the interest rates used for overnight bank lending activities on financial markets, instrumented using swaps on overnight lending between the day at question and the day of the next monetary policy meeting. Normally, these swaps follow the prevailing nominal interest rate closely (with a spread of a couple of basis points), as any other rate would induce arbitrage by the possibility of a risk-free hedge using the current overnight nominal interest rate, unless there are rate change expectations in the near future that were previously not priced in. As figure A.3 shows, however, the turmoil introduced by the 'mini-budget' caused a remarkable wedge between the two rates:

[33] Determining uniquely whether a given policy announcement, or a given time period, clearly relates to a monetary-led or a fiscally-led policy mix in a narrow sense, i.e., in relation to the respective policy rules and how they inform the stability of the underlying economic system, is generally not possible purely based off time-series data. Simply put, the "Taylor Principle" cannot be tested as its impact on the uniqueness properties surrounding macroeconomic models depends on off-equilibrium threats that cannot be observed under the condition of the Taylor Principle itself holding (Cochrane, 2011; Neumeyer and Nicolini, 2025).

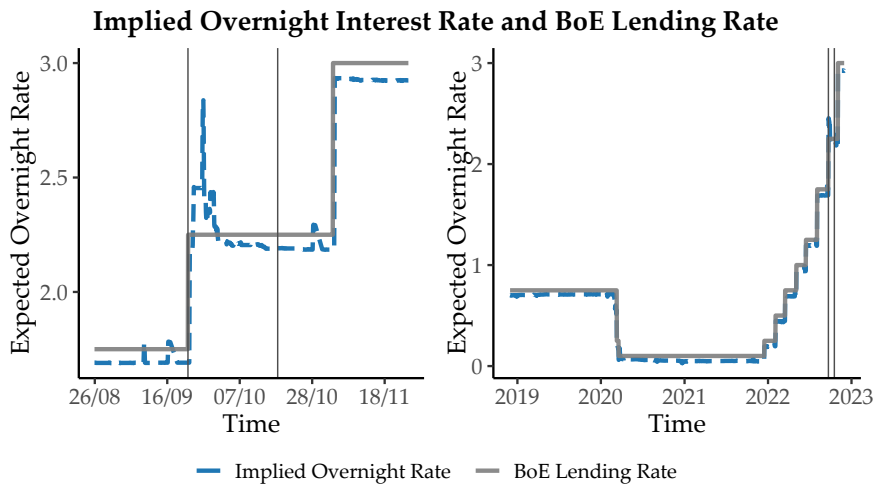


Figure A.3: The prevailing Bank of England Lending Rate and the Implied Overnight Interest Rate, derived by instrumenting overnight interest swaps from today to the expected next meeting of the BoE Monetary Policy Committee.

As can be inferred from the right-hand side panel for a period of five years, and on the left-hand side in more detail for the period of interest, the implied overnight interest rate follows the BoE lending rate closely with a roughly constant spread, exhibiting jumps around meeting dates of the BoE Monetary Policy Committee in alignment with monetary policy decisions.

The period of the mini-budget, which commenced one day after a Monetary Policy Committee (MPC) meeting (September 23 and September 22, respectively), induced movements in the expected overnight rates that were not observed at any other point in time - despite no MPC meeting in near sight.[34] Expected overnight rates shot up far beyond the then-prevailing BoE bank lending rate by up to 50 basis points. Such movements can be caused by an array of different possibilities: it could be either that fiscal policy caused a shift in market expectations of monetary policy in the short-term, thus implying that monetary policy was considered to be 'reactive' to the fiscal policy announcement, or that the mini-budget was expected to have such detrimental consequences on inflation that the BoE was required to react immediately, or it might be reflective of liquidity issues in the swap market in the same period.

Correspondingly, and as can be inferred from figure A.4, prices of Gilt futures dropped sharply during the period in which the U.K. mini-budget was expected to be put in place.

[34]On September 27, then-BoE chief economist Huw Pill stated that the proposed U.K. government budget might require a "significant monetary response", indicating readiness on behalf of the BoE to adjust the monetary stance, but no concrete emergency meeting date had been proposed at that point.

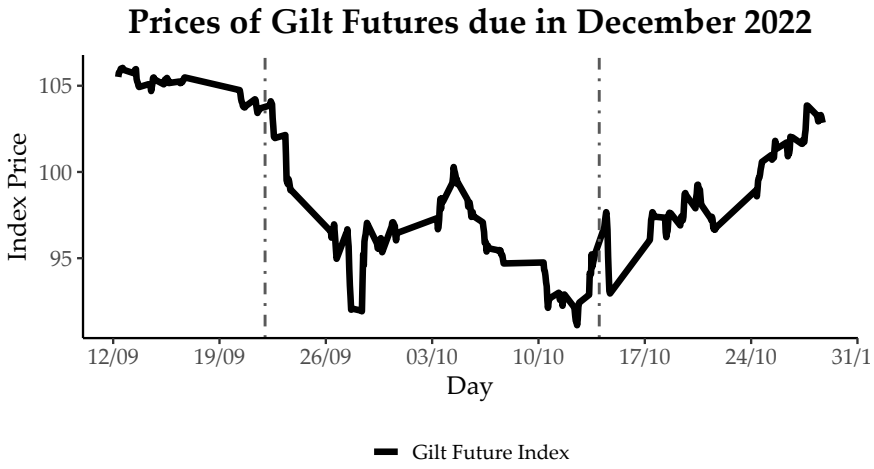


Figure A.4: Evolution of the weighted Gilt Future Index for futures due in December 2022, weighting Gilt prices based on a normalized face value of 100, after adjusting for expected inflation. Data source: Bloomberg.

An important caveat is that reducing the observed dynamics to expected revaluations of bonds and prospective interest rate movements does not capture all aspects related to this fiscal policy announcement. Uncertainty surrounding the proposed policy measures might have also been an important contributor to market reactions. Figure A.5 plots the FTSE 100 IVI Index that can plausibly serve as a proxy for uncertainty.

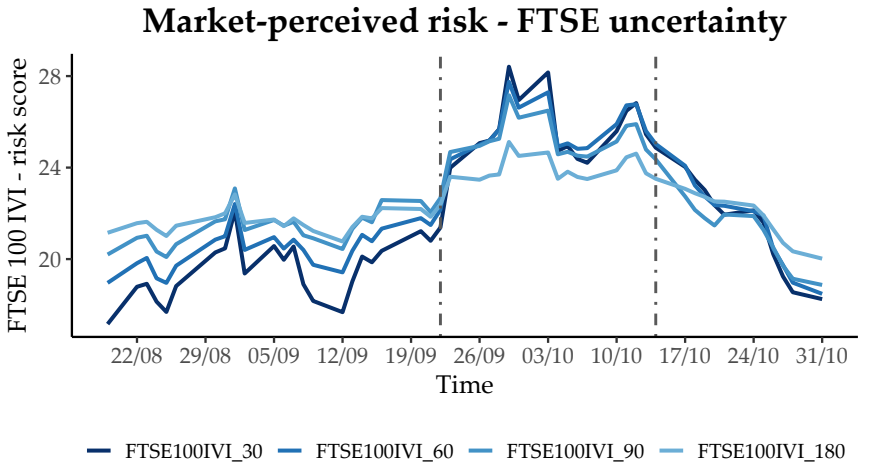


Figure A.5: The uncertainty index over equity of the largest publicly traded British companies. The lines measure implied uncertainty in 30-day, 60-day, 90-day, and 180-day forward-looking windows.

The mini-budget episode coincided with an inversion of the market-perceived risk relative to the forecast horizon: whereas in the periods before and after the mini-budget announcement market risk was perceived to be higher in the medium-term (180 days) than in the short-term (30 or 60 days), the opposite has been the case during this short-lived fiscal episode.

A final and related aspect is the possibility of elevated default risk, which has been omitted from the

model presented in this paper. In figure A.6, I present the Credit Default Swap spread alongside the quantity of Gilts maturing in the near-term around September 2022, which plausibly indicate near-term fiscal refinancing pressures.

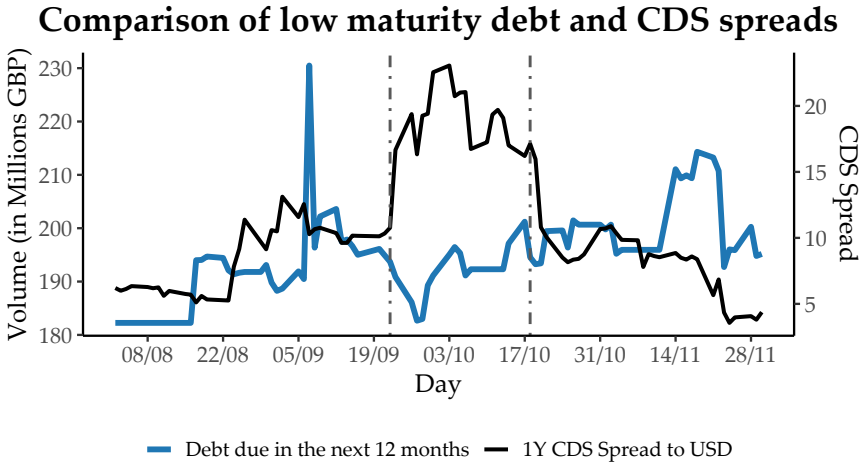


Figure A.6: Government debt maturing over the course of the next 12 months, plotted against the 1-year spread of Credit Default Swaps on U.K. Gilts over USD futures. Data source: Bloomberg.

During the period of the 'mini-budget' announcement, the amount of maturing government debt was at rather low levels, i.e., there was no inherent pressure to refinance a large quantity of maturing obligations around the time of the 'mini-budget' announcement. Yet the black line, which plots short-term Credit Default Swaps based off U.K. Gilts, was elevated during the 'mini-budget' episode, confirming a disconnect of the total amount of maturing debt from perceived market risk. While the two measures have a correlation of 0.49 in the period leading up to the announcement of the mini-budget, that correlation drops to -0.51 between the middle of September and the middle of October 2022.

All these data points reinforce the idea that the 'mini-budget' announced by the U.K. Treasury in September 2022 was indeed an unexpected fiscal measure that significantly subverted perceived fiscal sustainability, with wide-spread ramifications for expected real returns on government bonds, expected inflation and interest rates, and elevated risk levels.

FAQ: the 'mini-budget shock' episode

In addition to the evidence derived using market data, I here present narrative viewpoints complementing the understanding of the 'mini-budget' episode.

Why did markets possibly reverse the uptick in inflation expectations initially?

- On September 26, around 4.00pm, Kwasi Kwarteng announced to publish a 'medium-term fiscal plan', which possibly indicated greater restraint in fiscal policy: (Bloomberg).
- On September 28, a further plausible shock to perceptions of fiscal sustainability occurred: Moody's explicitly deemed the mini-budget to put U.K. debt sustainability in danger, fol-

lowed by a same-day increase of inflation expectations: (Reuters).

- Likewise, on September 28, in a reversal of expectations caused by the September 26 statement, the Treasury explicitly rejected for the first time since the initial announcement any idea of reneging on the additional budget shortfall, thereby re-affirming expectations about the fiscal policy measure actually being pushed through. See: (BBC). On the very same day, the Bank of England intervened in bond markets by re-starting long-dated government bond purchases, which, again, plausibly re-affirmed the idea that the Treasury will not back down. This was announced at around 11.20am - see: (X). Swap breakeven rates shot up by 60bps in the three hours after the statement made by the Bank of England.

Why were inflation swaps priced much higher in August 2022 compared to the dynamics occurring in September and October 2022?

- On August 17, 2022, the ONS released a report of CPI inflation being 10.1%, breaking the 10% barrier for the first time in 40 years, also beating the private sector forecasts decisively (Bloomberg). This occurred alongside a significant depreciation of the British Pound (FT). Likewise, implied interest rate raises corresponding to expectations of vastly more aggressive monetary tightening from that period onward alongside a yield curve inversion appeared around August 15 (FT).
- Implied one-year ahead inflation expectations peaked at around 8% in August. This is still *vastly* below the forecasts released in August 2022 by major financial market actors: the Goldman Sachs forecast of one-year ahead inflation amounted to 14.8%, with a 'negative' scenario of 22.1% annual inflation for the U.K. implied in their August 2022 briefings (FT). Relative to that forecast, the change in inflation swaps implied in that policy uncertainty episode was relatively benign. This period of increased inflation expectations also coincided with record prices on natural gas spot markets in the U.K.

A.3 Further details on the Local Projection exercise

To shed further light on the evidence presented in section 2.2, I here provide the additional information related to the local projection exercise on U.S. data presented in figure 5, a battery of robustness exercises, and additional evidence using other identified fiscal shocks as well as U.K. data.

Data: The deficit news ε_t^F of Fritsch and Mazlish (2026) measures daily changes in the present value of projected federal deficits over the following eleven fiscal years, and classifies these changes into changes on days of fiscal events (such as the passage of legislation or the release of budget projections) and endogenous revisions driven by the macroeconomic outlook. I sum the fiscal-event changes within each quarter and divide them by annual nominal GDP in the preceding quarter, such that positive values denote news about higher deficits. Over the period 1980-2019, the series has a standard deviation of 2.5% of GDP, and 118 out of 158 quarters contain deficit news. The share of inflation-indexed debt θ_t is the face value of TIPS (excluding the accrued inflation

adjustment) as a share of the face value of marketable Treasury debt at the end of each quarter, constructed from the security-level data of the Monthly Statement of the Public Debt and from the Economic Report of the President.[35] The price level is the U.S. consumer price index (all urban consumers, not seasonally adjusted, quarterly average). Figure A.7 plots the time series of the level and the quarterly change of the share of indexed debt in the U.S., showing the secular increase in the share of indexed debt until 2008, its sharp fall during the largely bill-financed crisis response of 2008-2010, and its gradual recovery thereafter.

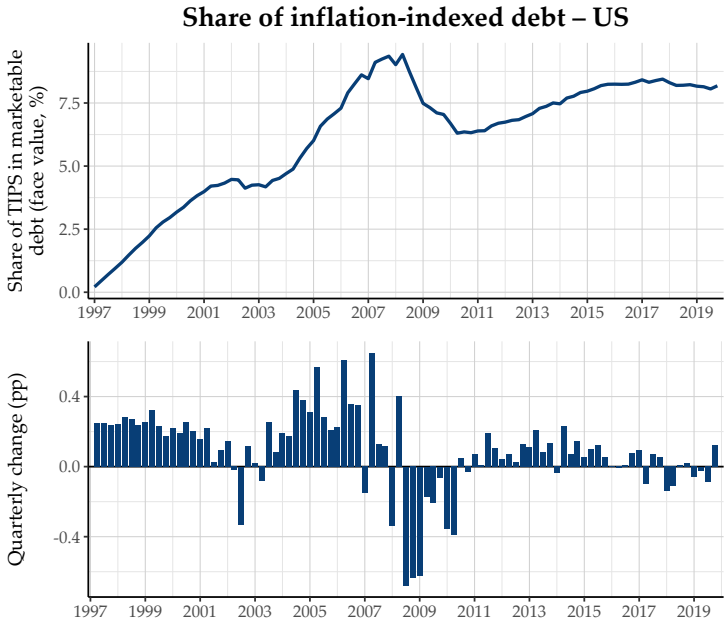


Figure A.7: Time series of the level of the share of inflation-indexed debt (TIPS, face value) in marketable U.S. Treasury debt as well as the quarterly change thereof.

Table A.1 gives the concrete numbers corresponding to figure 5, specifying the exact coefficients of the interaction effect of $\Delta\theta_t\,\varepsilon_t^F$ and the individual effects of the change in the indexed debt share $\Delta\theta_t$ and the deficit news ε_t^F on the cumulative price level change from the pre-shock period -1 until the period specified.

While neither the deficit news itself nor the change in the share of indexed debt itself significantly impacts the price level at these horizons, the interaction effect of the change in the share of indexed debt with the deficit news follows the pattern given in figure 5. Figure A.8 translates the coefficients into responses of the price level to 1% of GDP of deficit news for representative concurrent changes in the share of indexed debt. When the share rises by 0.28 percentage points in the quarter of the news (the 90th percentile across quarters with deficit news since 1997), the price level is 0.33% higher after two years, while the response is insignificant, and if anything negative, when the share falls by 0.24 percentage points (the 10th percentile).

[35] The security-level inflation adjustment of TIPS is reported from June 2004 onwards. For earlier periods, I reconstruct it from the reference CPI of the outstanding securities. The share is zero before the first TIPS auction in January 1997.

Dependent variable: $100 \times (\log P_{t+h} - \log P_{t-1})$								
Horizon h:	0	1	2	3	4	5	6	7
Deficit News	-0.01	-0.01	0.02	0.06	0.06	0.06	0.07	0.08
	(0.03)	(0.04)	(0.05)	(0.06)	(0.06)	(0.06)	(0.06)	(0.07)
Δ Index Share	0.16	0.46	0.21	0.58	0.92	0.73	0.08	0.15
	(0.33)	(0.62)	(0.70)	(0.72)	(0.81)	(0.88)	(1.00)	(1.03)
Deficit News × Δ Index Share	0.41***	0.57***	0.52***	0.46**	0.54***	0.69***	0.82***	0.88***
	(0.15)	(0.15)	(0.19)	(0.20)	(0.20)	(0.21)	(0.24)	(0.25)
Additional Controls	Y	Y	Y	Y	Y	Y	Y	Y
Year Fixed Effects	N	N	N	N	N	N	N	N
Observations	158	157	156	155	154	153	152	151
R^2	0.425	0.468	0.518	0.580	0.590	0.588	0.603	0.605

Table A.1: Local Projection results for the U.S. The controls include the first four lags of the real GDP growth rate, the short-run nominal interest rate, the change in the weighted real exchange rate, a recession indicator, and the first lag of the price level difference. Standard errors are robust to heteroskedasticity and autocorrelation (Newey-West correction).

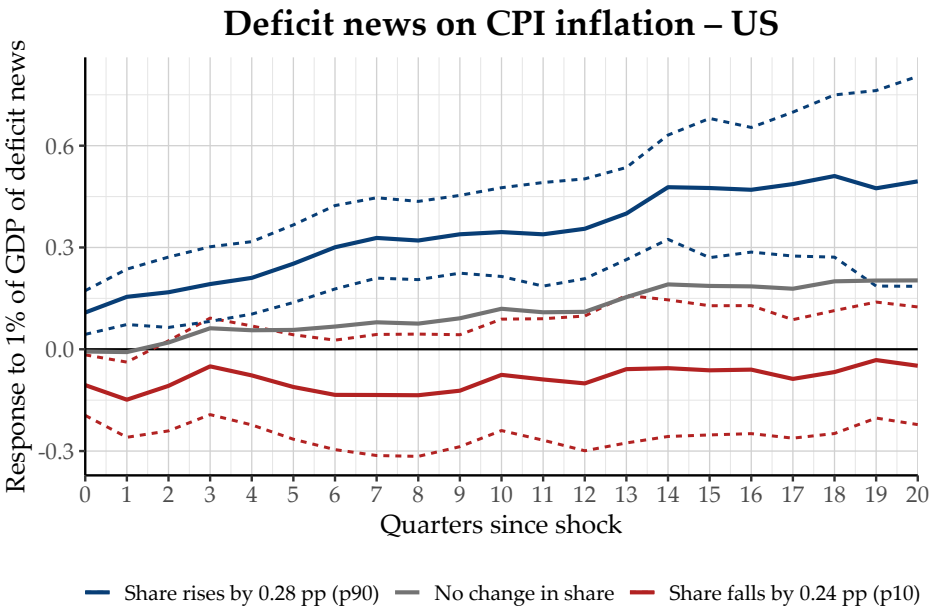


Figure A.8: Cumulative response of the price level (in percentage points) to 1% of GDP of deficit news, implied by the local projection (2), for a concurrent increase in the share of inflation-indexed debt by 0.28 percentage points (90th percentile), no change, and a decrease by 0.24 percentage points (10th percentile). Confidence intervals are provided at the 90% level.

Year fixed effects: Adding year fixed effects to the local projection (2) identifies the interaction effect only from the timing of deficit news within a given year. Since deficit news is concentrated in a few years, the year fixed effects absorb a large part of its variation. Figure A.9 and table A.2 show that the interaction effect remains positive at all horizons, and significant at the 5% level at all horizons except two, with a smaller magnitude of about 0.5 at a horizon of seven quarters.

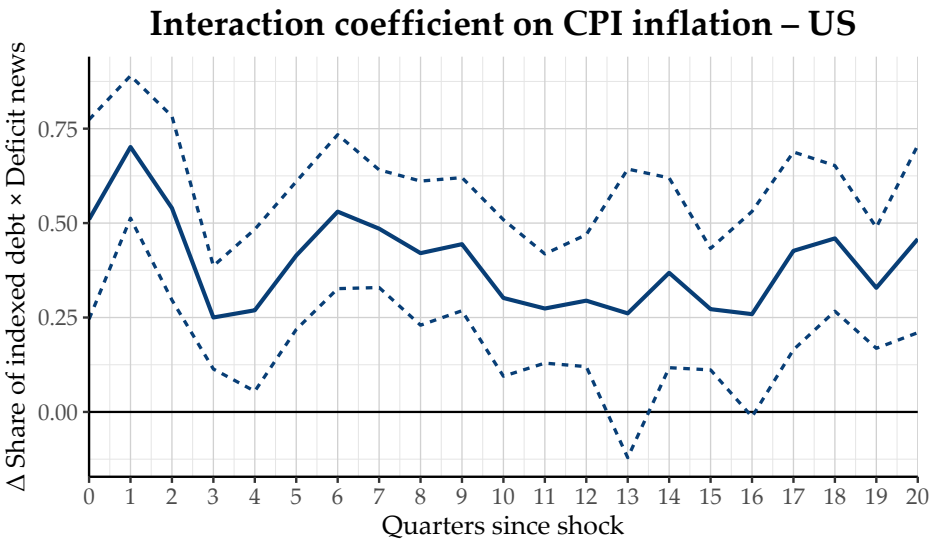


Figure A.9: IRF implied by the local projection (2) through the coefficients β_h, with year fixed effects added to the control vector Z. Standard errors are robust to heteroskedasticity and autocorrelation (Newey-West correction). Confidence intervals are provided at the 90% level. Sample length: 1980 Q3 - 2019 Q4.

Dependent variable: $100 \times (\log P_{t+h} - \log P_{t-1})$								
Horizon h:	0	1	2	3	4	5	6	7
Deficit News	-0.02	-0.07***	-0.07**	-0.02	-0.05**	-0.08**	-0.08*	-0.07**
	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)	(0.04)	(0.05)	(0.03)
Δ Index Share	0.66	1.17	0.11	0.53	1.15*	0.88	-0.29	0.20
	(0.48)	(0.76)	(0.74)	(0.50)	(0.63)	(0.74)	(0.70)	(0.45)
Deficit News × Δ Index Share	0.51***	0.70***	0.54***	0.25***	0.27**	0.41***	0.53***	0.49***
	(0.16)	(0.11)	(0.15)	(0.08)	(0.13)	(0.12)	(0.12)	(0.09)
Additional Controls	Y	Y	Y	Y	Y	Y	Y	Y
Year Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	158	157	156	155	154	153	152	151
R^2	0.605	0.784	0.846	0.925	0.918	0.912	0.940	0.963

Table A.2: Local Projection results for the U.S. with year fixed effects. The controls are as in table A.1. Standard errors are robust to heteroskedasticity and autocorrelation (Newey-West correction).

Robustness: Table A.3 reports the interaction coefficient β_h at horizons of three, seven, and eleven quarters for a battery of alternative specifications, both without and with year fixed effects, and figures A.10 and A.11 show the full impulse responses for 16 of them. The checks vary the deficit news (restricting it to days that are classified exclusively as fiscal events, or adding the endogenous revisions), the timing and the measurement of the share of indexed debt, the set of controls (adding the monetary policy surprises of Bauer and Swanson (2023), oil prices and foreign output growth, global real factors such as commodity prices, the oil supply news of Känzig (2021) and world industrial production, global financial factors such as the VIX, the shadow policy rate and geopolitical risk, narrative tax shocks, as well as the debt-to-GDP ratio and past inflation together

with their interaction with the deficit news), the sample, the price index, and the inference. Across the 32 specifications in table A.3 (excluding the placebo and the reference row), the interaction coefficient at a horizon of seven quarters is positive in all cases, and significant at the 5% level in 27 (without year fixed effects) and 26 (with year fixed effects) cases.

	(1) No fixed effects			(2) Year fixed effects		
Horizon h:	3	7	11	3	7	11
Baseline, equation (2)	0.46** (0.20)	0.88*** (0.25)	0.81*** (0.29)	0.25*** (0.08)	0.49*** (0.09)	0.27*** (0.09)
Shock: fiscal-event days only	0.93** (0.39)	1.74*** (0.44)	1.67*** (0.51)	0.51*** (0.16)	0.85*** (0.14)	0.58*** (0.10)
Shock: total news (fiscal events and endogenous revisions)	0.18** (0.08)	0.30*** (0.11)	0.29** (0.12)	0.07 (0.04)	0.11* (0.06)	0.04 (0.05)
Placebo: endogenous (macroeconomic) revisions only	0.21 (0.20)	0.28 (0.29)	0.29 (0.29)	0.09 (0.09)	0.07 (0.10)	-0.00 (0.09)
Shock: narrative tax shocks (Mierzwa, 2024), reference	0.52 (1.12)	1.89 (1.71)	11.66* (6.92)	-0.45 (0.85)	-4.85** (2.28)	-2.22 (2.94)
Timing: lagged change $\Delta\theta_{t-1}$	0.49** (0.23)	0.93*** (0.29)	0.89*** (0.31)	0.24** (0.10)	0.45*** (0.13)	0.28*** (0.09)
Timing: lead change $\Delta\theta_{t+1}$	0.88*** (0.17)	1.39*** (0.21)	1.10*** (0.24)	0.41*** (0.13)	0.66*** (0.12)	0.35*** (0.10)
Timing: $\Delta\theta_t$, controlling for $\Delta\theta_{t-1}$ and $\varepsilon_t^F\Delta\theta_{t-1}$	0.56 (0.40)	1.02* (0.56)	0.96* (0.53)	0.32* (0.17)	0.53*** (0.20)	0.35*** (0.11)
Measure: positive changes only, $\max(\Delta\theta_t, 0)$	0.51 (0.48)	1.42** (0.63)	1.45** (0.63)	0.66*** (0.19)	0.89*** (0.34)	0.72*** (0.20)
Measure: change of the inflation-uplifted share	0.31** (0.15)	0.62*** (0.19)	0.54** (0.21)	0.16** (0.07)	0.33*** (0.09)	0.17** (0.08)
Measure: uplifted share, net of mechanical indexation	0.44*** (0.15)	0.78*** (0.19)	0.67*** (0.22)	0.20** (0.08)	0.39*** (0.10)	0.20** (0.09)
Measure: growth rate $\Delta \log \theta_t$ (per 10 log points)	0.17 (0.11)	0.28* (0.17)	0.23 (0.15)	0.07 (0.06)	0.13 (0.10)	0.10 (0.06)
Controls: + θ_{t-1} and $\varepsilon_t^F\theta_{t-1}$	0.39** (0.17)	0.68*** (0.22)	0.49** (0.23)	0.27*** (0.08)	0.51*** (0.09)	0.30*** (0.09)
Controls: + monetary policy surprises (Bauer and Swanson, 2023)	0.65*** (0.18)	1.16*** (0.22)	1.01*** (0.28)	0.21** (0.09)	0.48*** (0.09)	0.32*** (0.12)
Controls: + oil price and foreign GDP growth	0.43** (0.17)	0.95*** (0.27)	0.66** (0.28)	0.15 (0.10)	0.44*** (0.12)	0.22* (0.13)
Controls: + global real factors	0.51*** (0.18)	1.03*** (0.25)	1.00** (0.40)	0.27* (0.15)	0.35* (0.20)	0.34 (0.27)
Controls: + global financial factors	0.62*** (0.23)	0.85*** (0.17)	0.60** (0.27)	0.36*** (0.13)	0.46*** (0.06)	0.26** (0.11)
Controls: + narrative tax shocks	0.50** (0.20)	0.91*** (0.25)	0.84*** (0.29)	0.26*** (0.08)	0.49*** (0.09)	0.28*** (0.09)
Controls: + debt/GDP and ε_t^F×debt/GDP	0.53*** (0.20)	0.98*** (0.25)	0.93*** (0.28)	0.23*** (0.08)	0.45*** (0.09)	0.23** (0.10)
Controls: + inflation over $t-4,\dots,t-1$ and its interaction	0.58*** (0.19)	1.02*** (0.23)	0.88*** (0.30)	0.18* (0.10)	0.34*** (0.10)	0.21* (0.13)
Controls: + inflation over $t-8,\dots,t-1$ and its interaction	0.55*** (0.18)	0.99*** (0.22)	0.87*** (0.29)	0.15 (0.10)	0.42*** (0.11)	0.22 (0.14)
Controls: + four lags of inflation	0.54*** (0.20)	0.99*** (0.24)	0.86*** (0.29)	0.18* (0.10)	0.35*** (0.10)	0.24* (0.14)
Sample: excluding 2008–2009	0.28 (0.31)	1.02*** (0.35)	1.07** (0.42)	0.47*** (0.18)	0.60** (0.27)	0.58*** (0.18)
Sample: excluding 2008	-0.04 (0.18)	0.29 (0.28)	0.23 (0.27)	0.15 (0.10)	0.39*** (0.11)	0.26** (0.11)
Sample: excluding 1997	0.40** (0.19)	0.81*** (0.24)	0.78*** (0.29)	0.25*** (0.08)	0.50*** (0.09)	0.29*** (0.09)
Sample: 1997 Q2 onwards	0.55*** (0.17)	0.81*** (0.19)	0.50** (0.23)	0.27** (0.11)	0.45*** (0.08)	0.29* (0.17)
Sample: 1980–2007	0.46 (0.53)	1.08 (0.82)	1.33* (0.79)	0.41* (0.22)	0.04 (0.25)	0.32 (0.24)
Prices: seasonally adjusted CPI	0.44** (0.20)	0.87*** (0.24)	0.81*** (0.28)	0.25*** (0.08)	0.49*** (0.10)	0.28*** (0.08)
Prices: core CPI	-0.02 (0.14)	0.27 (0.20)	0.39 (0.24)	-0.06 (0.06)	0.11 (0.08)	0.01 (0.07)
Prices: PCE	0.45*** (0.16)	0.75*** (0.20)	0.70*** (0.26)	0.22*** (0.06)	0.34*** (0.08)	0.20*** (0.08)
Prices: core PCE	0.19** (0.08)	0.38*** (0.11)	0.43*** (0.15)	0.05 (0.04)	0.10* (0.06)	0.04 (0.06)
Prices: GDP deflator	0.42*** (0.08)	0.69*** (0.15)	0.79*** (0.25)	0.17*** (0.06)	0.18** (0.08)	0.08 (0.08)
Inference: lag-augmented LP (Montiel Olea and Plagborg-Møller, 2021)	0.59** (0.28)	1.11*** (0.37)	1.16*** (0.41)	0.26** (0.12)	0.44*** (0.13)	0.30** (0.14)
Inference: standard errors clustered by year	0.46 (0.31)	0.88** (0.38)	0.81** (0.39)	0.25** (0.10)	0.49*** (0.12)	0.27*** (0.09)

Table A.3: Robustness of the interaction coefficient β_h of the local projection (2) at horizons of three, seven, and eleven quarters, without (1) and with (2) year fixed effects. Each row changes one element of the baseline specification. Coefficients are measured in percentage points of the price level per unit of the shock and per percentage point of the concurrent change in the share of inflation-indexed debt. Standard errors are robust to heteroskedasticity and autocorrelation (Newey-West correction) except where stated otherwise.

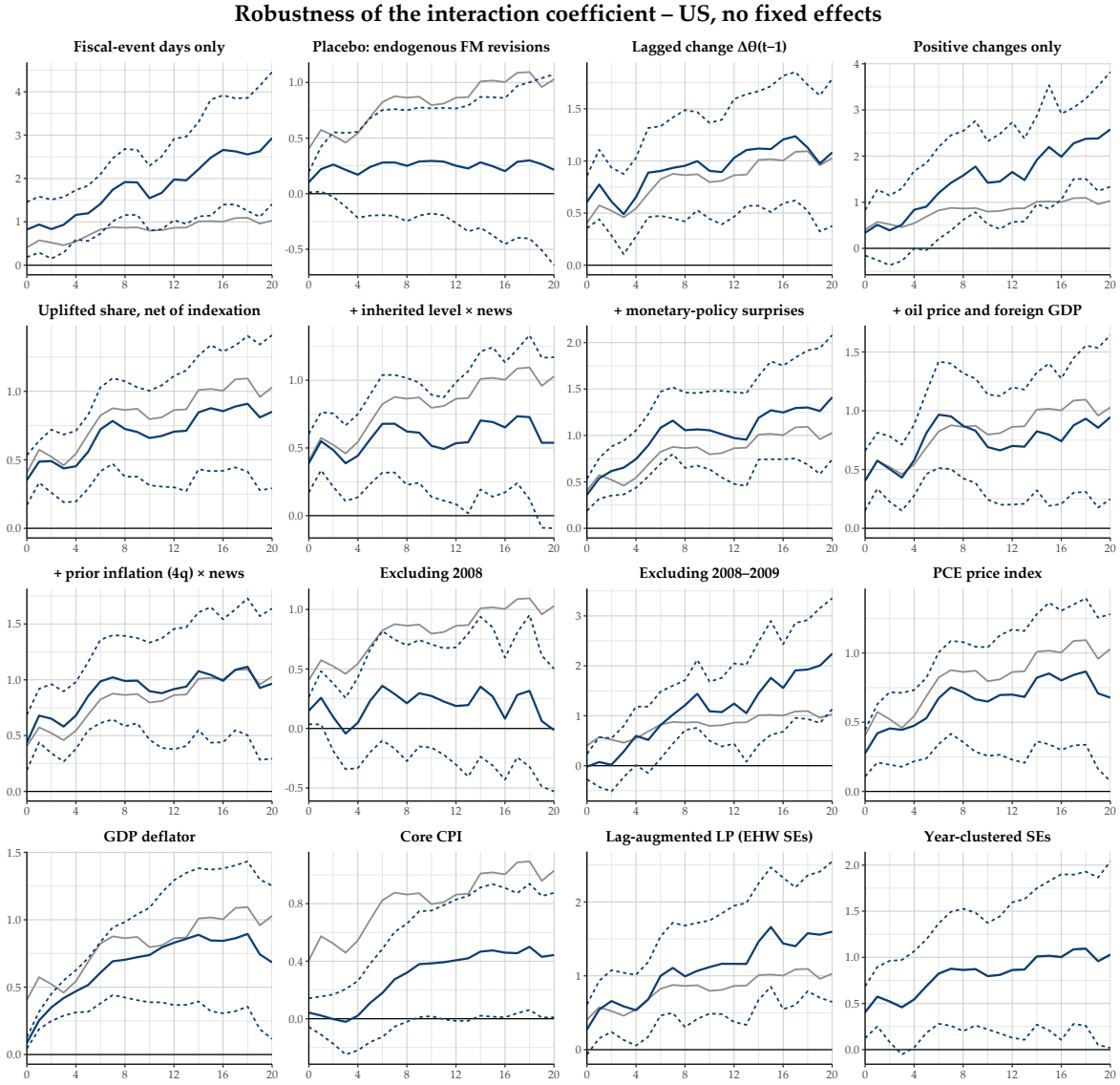


Figure A.10: IRFs of the interaction coefficient β_h of the local projection (2) for 16 of the robustness checks in table A.3, without year fixed effects. The gray line shows the baseline estimate from figure 5. Confidence intervals are provided at the 90% level.

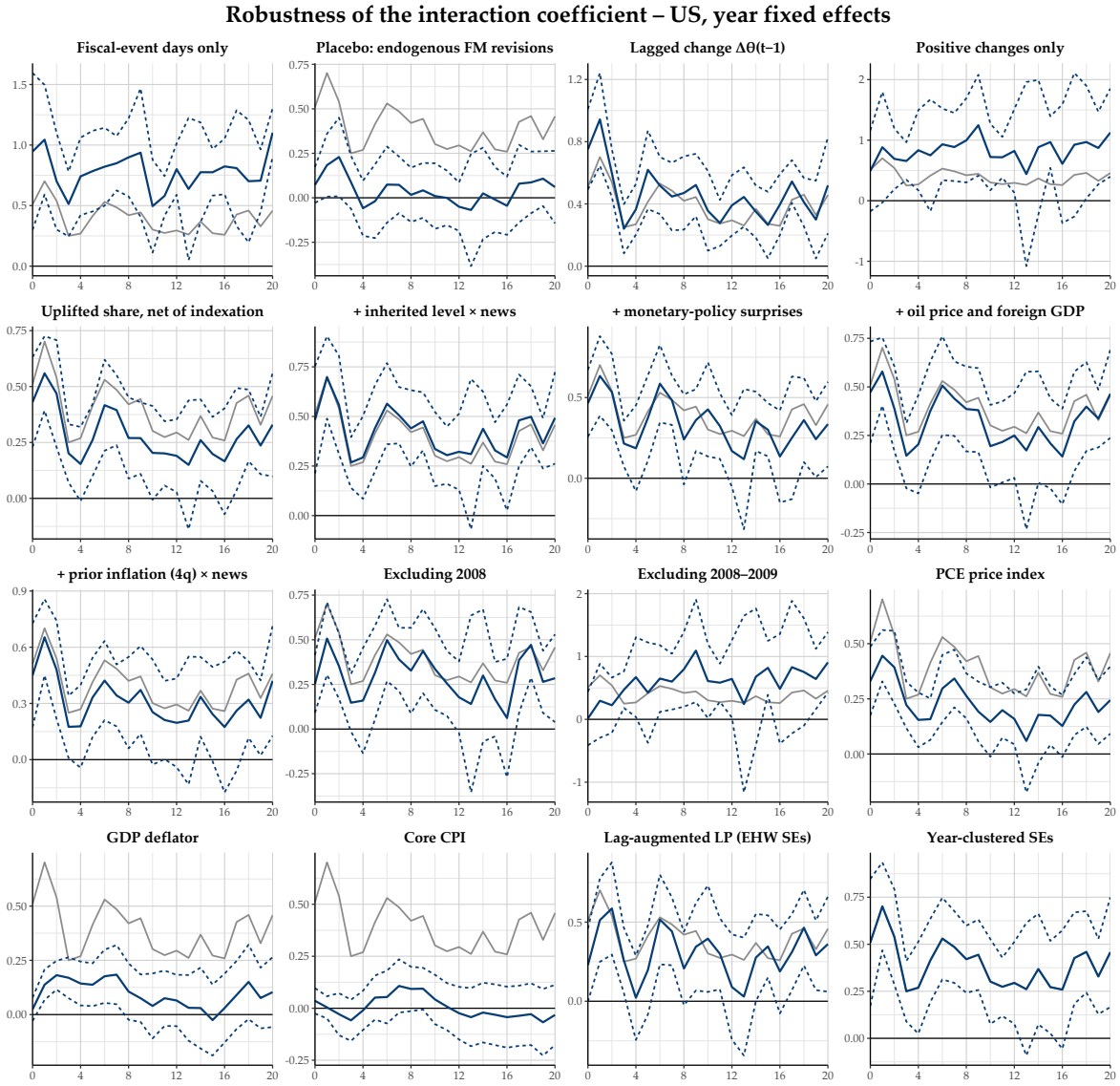


Figure A.11: IRFs of the interaction coefficient β_h of the local projection (2) for 16 of the robustness checks in table A.3, with year fixed effects. The gray line shows the baseline estimate from figure A.9. Confidence intervals are provided at the 90% level.

Notably, the placebo, which uses only the endogenous revisions of projected deficits driven by the macroeconomic outlook, yields no significant interaction. Additionally, the timing and the measurement of the share of indexed debt matter in the way implied by the government debt valuation equation: the lagged change $\Delta\theta_{t-1}$ yields a similar interaction on its own, since changes in the share are persistent, but loses its explanatory power once the concurrent change is included. Controlling for the lagged change and its interaction with the news leaves the concurrent interaction positive, at 1.02 without and 0.53 with year fixed effects at a horizon of seven quarters (table A.3). The growth rate $\Delta \log \theta_t$ is dominated by the first quarters after the introduction of TIPS, in which the share is close to zero. Positive changes of the share alone yield a larger interaction.

The role of the 2008 financial crisis: Without year fixed effects, the estimate draws in part on the fourth quarter of 2008, in which the U.S. Treasury financed the crisis response largely through bills, such that the share of indexed debt fell by 0.64 percentage points, while deficit news amounted to 6.7% of GDP and the price level fell over the subsequent two years. This quarter accounts for about 0.4 of the coefficient of 0.88 at a horizon of seven quarters, and excluding the year 2008 from the sample renders the interaction insignificant in this specification. The result survives the exclusion of 2008 with year fixed effects (0.39) and for positive changes of the share only (0.98), and excluding both 2008 and 2009 leaves the interaction at 1.02. Controlling for inflation over the four or eight quarters before the shock, together with its interaction with the deficit news, leaves the interaction coefficient at 0.34-0.42 with and at 0.99-1.02 without year fixed effects (see the corresponding rows of table A.3).

Financing composition: Since $\Delta\theta_t \approx (1-\theta_{t-1})\Delta IL_t/D_{t-1} - \theta_{t-1}\Delta N_t/D_{t-1}$, the concurrent change in the share combines new indexed borrowing ΔIL_t and new nominal borrowing ΔN_t, relative to total debt D_{t-1}. Interacting the deficit news separately with both (in percent of GDP) shows that deficit news accompanied by more nominal borrowing is significantly less inflationary in both specifications (-0.25 without and -0.07 with year fixed effects at a horizon of seven quarters), whereas deficit news accompanied by more indexed borrowing is more inflationary with year fixed effects (1.05) and imprecisely estimated without. The interaction does not reflect a larger fiscal impulse either, since the interaction of the deficit news with the change in the share has no significant effect on the cumulative primary deficit.

Other fiscal shocks: Table A.4 re-estimates the local projection (2) with other identified fiscal shocks in place of the deficit news. For the U.S., I use the legislated changes to projected deficits reported by the Congressional Budget Office in each revision of its baseline projections (Congressional Budget Office, 2026), which are independent of the deficit news used above; government spending shocks identified as in Blanchard and Perotti (2002), both from forecast errors of professional forecasters as in Ramey (2011) and from excess returns of defense contractors as in Fisher and Peters (2010); the military spending news of Ramey and Zubairy (2018) and Ben Zeev and Pappa (2017); and the narrative tax shocks of Mierzwa (2024) and Romer and Romer (2010). Figure A.12 shows the corresponding impulse responses per one-standard-deviation shock. The interaction with the concurrent change in the share of indexed debt is positive and significant for the legislated changes to projected deficits (0.27 per 1% of GDP, or about 60% of the baseline estimate per standard deviation of the shock), which is driven by legislated spending, and for all three government spending shocks identified from VAR innovations, forecast errors, and stock returns, with magnitudes per standard deviation close to the baseline estimate. With year fixed effects, the interaction remains significant for the legislated changes to projected deficits and for the spending shocks of Blanchard and Perotti (2002). The result is not confirmed by the military spending news, which contains very few events after the introduction of TIPS in 1997, while the narrative tax shocks are too rare after 1997 to identify the interaction. Dating the legislated changes one quarter before their publication by the CBO removes the effect.

Shock	Sample	Obs. with $\varepsilon_t \Delta\theta_t \neq 0$	β_7, no FE	per s.d. of ε_t	β_7, year FE
United States					
Deficit news (Fritsch and Mazlish, 2026), baseline	1980Q3–2019Q4	59	0.88*** (0.25)	2.18	0.49*** (0.09)
Deficit news, fiscal-event days only	1980Q3–2019Q4	51	1.74*** (0.44)	3.96	0.85*** (0.14)
CBO legislated changes to projected deficits (Congressional Budget Office, 2026)	1984Q3–2019Q4	46	0.27*** (0.10)	1.25	0.12* (0.07)
CBO legislated changes, first five projected years	1984Q3–2019Q4	46	0.52** (0.25)	1.02	0.30* (0.16)
CBO legislated changes, dated one quarter earlier	1984Q3–2019Q4	46	-0.23 (0.14)	-1.07	-0.03 (0.08)
CBO legislated changes, revenues	1984Q3–2019Q4	46	-0.25 (0.95)	-0.59	0.27 (0.44)
CBO legislated changes, outlays	1984Q3–2019Q4	46	0.28*** (0.11)	1.00	0.12 (0.07)
Placebo: endogenous revisions (Fritsch and Mazlish, 2026)	1980Q3–2019Q4	79	0.28 (0.29)	0.55	0.07 (0.10)
Narrative tax shocks (Mierzwa, 2024)	1980Q3–2019Q4	32	1.89 (1.71)	0.47	-4.85** (2.28)
Narrative tax shocks (Romer and Romer, 2010)	1980Q3–2007Q4	4	-2.79 (3.77)	-0.71	-3.94 (2.92)
Government spending shocks (Blanchard and Perotti, 2002)	1980Q3–2019Q4	92	14.56** (6.61)	1.84	10.29** (4.45)
Forecast errors of federal spending (Ramey, 2011)	1981Q4–2019Q4	92	10.77* (5.92)	2.26	2.50 (3.40)
Military spending news (Ramey and Zubairy, 2018)	1980Q3–2015Q4	16	-1.70 (1.49)	-1.91	-1.47*** (0.46)
Defense news (Ben Zeev and Pappa, 2017)	1980Q3–2007Q4	44	-3.45 (2.82)	-2.13	-3.53*** (1.23)
Defense excess returns (Fisher and Peters, 2010)	1980Q3–2008Q4	48	9.75*** (3.73)	2.71	6.93 (5.20)
United Kingdom					
Private sector PSNB forecast revisions, purged	1997Q4–2019Q4	88	-0.04 (0.08)	-0.03	0.00 (0.04)
Official PSNB forecast revisions, purged (Office for Budget Responsibility, 2026)	1984Q1–2019Q4	73	-0.16* (0.09)	-0.22	0.02 (0.08)
Official PSNB forecast revisions	1980Q3–2019Q4	74	-0.13 (0.29)	-0.18	0.14 (0.09)
Official cyclically-adjusted PSNB forecast revisions	2003Q4–2019Q4	34	0.05 (0.07)	0.07	-0.18*** (0.05)
OBR policy-driven PSNB revisions, two years	2010Q4–2019Q4	19	-6.46 (6.99)	-0.85	8.46*** (2.47)
OBR policy-driven PSNB revisions, five years	2010Q4–2019Q4	19	0.70 (1.77)	0.40	2.73*** (0.55)
Placebo: OBR underlying revisions	2010Q4–2019Q4	19	-0.00 (0.56)	-0.00	-2.94*** (0.79)
Narrative tax shocks (Mierzwa, 2024)	1980Q3–2019Q4	123	8.11** (3.46)	1.31	0.90 (1.31)
Narrative tax shocks (Cloyne, 2013)	1980Q3–2010Q4	96	9.26*** (3.27)	1.51	1.92 (1.46)
Government spending shocks (Blanchard and Perotti, 2002)	1980Q3–2019Q4	156	-0.13 (1.73)	-0.03	0.38 (1.01)

Table A.4: Interaction coefficient β_7 of the local projection (2) with alternative fiscal shocks in place of the deficit news, measured in percentage points of the price level per unit of the shock (in percent of GDP, except for the defense news and the defense excess returns) and per percentage point of the concurrent change in the share of inflation-indexed debt. The fifth column scales the coefficient by the standard deviation of the shock. The third column counts the quarters in which both the shock and the change in the share are non-zero. Standard errors are robust to heteroskedasticity and autocorrelation (Newey-West correction).

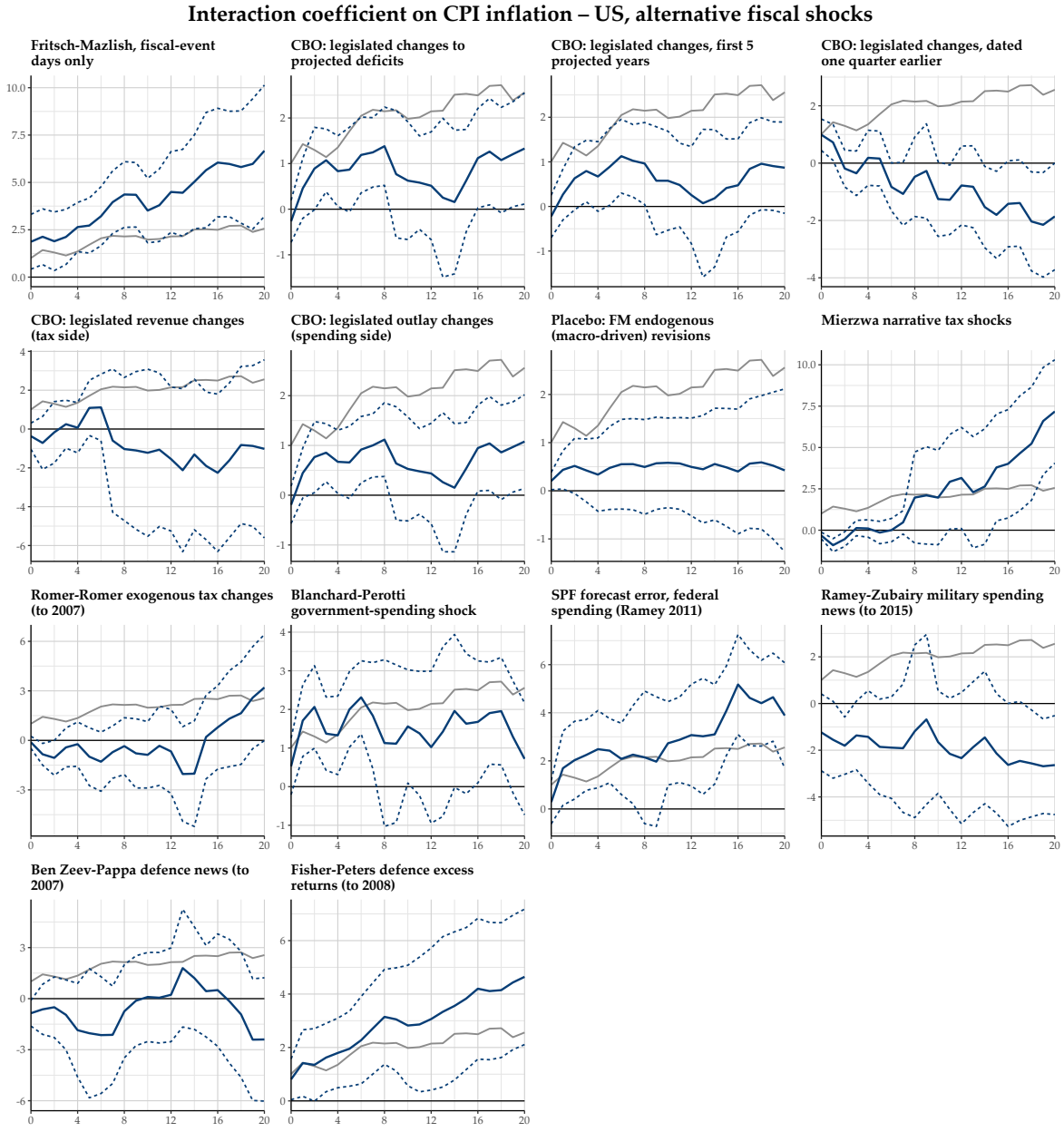


Figure A.12: IRFs of the interaction coefficient β_h of the local projection (2) with alternative U.S. fiscal shocks in place of the deficit news, per one-standard-deviation shock, without year fixed effects. The gray line shows the baseline estimate per one-standard-deviation deficit news. Confidence intervals are provided at the 90% level.

United Kingdom: The lower part of table A.4 shows the corresponding estimates for the U.K., using the concurrent change in the face-value share of index-linked gilts. Here, I measure deficit news as revisions of the private sector forecasts of Public Sector Net Borrowing for the current and the next fiscal year, as used in section 2.1, purged of contemporaneous revisions of the forecasts for GDP growth, inflation, the Bank Rate, and unemployment. Alternatively, I also measure deficit news as revisions of the official borrowing forecasts of the Treasury and the OBR between consec-

utive fiscal events since 1984 (Office for Budget Responsibility, 2026), and as the policy-driven revisions to the official forecasts reported by the OBR since 2010. None of these measures yields a robust positive interaction for the U.K.: the policy-driven revisions reported by the OBR yield a positive interaction only with year fixed effects in a short sample (2010-2019), in which the corresponding placebo is significant as well, and the narrative tax shocks yield a positive interaction only without year fixed effects. Possible reasons include the shorter horizon of the U.K. forecast revisions relative to the present value of eleven years of deficits used for the U.S., the inflation-targeting regime of the Bank of England throughout most of the sample, and the possibility that forecasters perceive revisions to borrowing as funded.

Evidence from narrative tax shocks: using instead the narratively identified tax shocks of Mierzwa (2024) with the growth rate of the share of inflation-indexed debt, $\Delta \log \theta_t$, for the U.K., the U.S., and Germany, I find that using just legislated tax changes also yields a positive and significant interaction for the U.K. (0.81 at a horizon of seven quarters) and the U.S. (0.24), which, however, does not survive the inclusion of year fixed effects (0.14 and -0.12, respectively). Narrative tax shocks are moreover rare after the introduction of indexed debt in the U.S. and in Germany, and U.K. narrative tax cuts are not followed by higher realized deficits, such that these shocks are less suited to capture unfunded deficits than the deficit news used in this paper.

A.4 Local projections conditional on policy regimes under the past level of indexed debt

As a further robustness test of the empirical exercise, I now highlight the effects of the prevalent policy mix for the local projection estimates, in line with the discussions underpinning the model in section 4. At the same time, I also link the interaction coefficient between the share of indexed debt and the identified fiscal disturbance to its *level*, examining this specification as well. Note that I restrict the analysis here to the U.S., given that most of the evidence on the possible prevalence of a fiscally-led policy mix is squarely related to U.S. policy dynamics.

Overall, the estimation follows the earlier version of the exercise with narrative tax shocks described at the end of appendix A.3. That is, I leverage the time series of exogenous fiscal policy surprises from Mierzwa (2024), and combine it with a corresponding series of inflation-indexed debt, taking the share of inflation-indexed debt in the overall sovereign debt portfolio as the main indicator for the intensity of the prevalence of inflation-indexed debt. Concretely, I estimate the following local projection here:

$$\log P_{t+h} - \log P_{t-1} = \alpha_h + \beta_h \omega_{t-1} \varepsilon_t^F + \delta_{1h} \omega_{t-1} + \delta_{2h} \varepsilon_t^F + \Gamma_h Z_{t-1} + e_{t+h}, \tag{A.1}$$

where $h \geq 0$ indexes the forecast horizon considered and Z_{t-1} is a vector of control variables specified below. Of particular interest is the coefficient β_h, which captures the cross-effect of the identified fiscal shock ε_t^F and the past level of the share of inflation-indexed debt ω_{t-1}. I estimate this equation separately for two different policy regimes, creating effectively a *state-dependent local*

projection, as described by Jordà and Taylor (2025). In detail, I estimate the specification (A.1) separately for periods of *fiscally-led policy mixes* and *monetary-led policy mixes*, as identified by Chen et al. (2022) for the United States.[36] I label the regimes here 'active fiscal policy' and 'passive fiscal policy' in line with Chen et al. (2022).

Figure A.13 summarizes this exercise, with the left-hand panel showcasing the fiscally-led policy mix and the right-hand panel showing the monetary-led policy mix. Under the monetary-led policy mix, the response of the price level on the interaction between the share of inflation-indexed debt and the identified fiscal policy shock is relatively imprecisely estimated, but - if anything - there is at least a temporary negative effect on the price level when more debt is indexed. In the case of a fiscally-led policy mix, however, medium-term price pressures arising from the identified fiscal policy disturbance are significantly increasing in the share of inflation-indexed debt. In the period 4-5 years after the initial shock, a 1% deficit-to-GDP shock accompanied by a corresponding level difference of 1% in the share of inflation-indexed debt boosts observed changes to the price level by up to 1.5% relative to a case where the share of inflation-indexed debt is lower.

These estimates should, however, be regarded as suggestive only. They are obtained without year fixed effects and with standard errors clustered by year, and with year fixed effects the interaction under the fiscally-led policy mix is no longer significant. Moreover, the fiscally-led periods with a positive stock of indexed debt contain only 15 quarters with narrative tax shocks, and the estimate is dominated by the third quarter of 2003. In line with appendix A.3, the level of indexed debt is thus a less reliable state variable than its concurrent change.

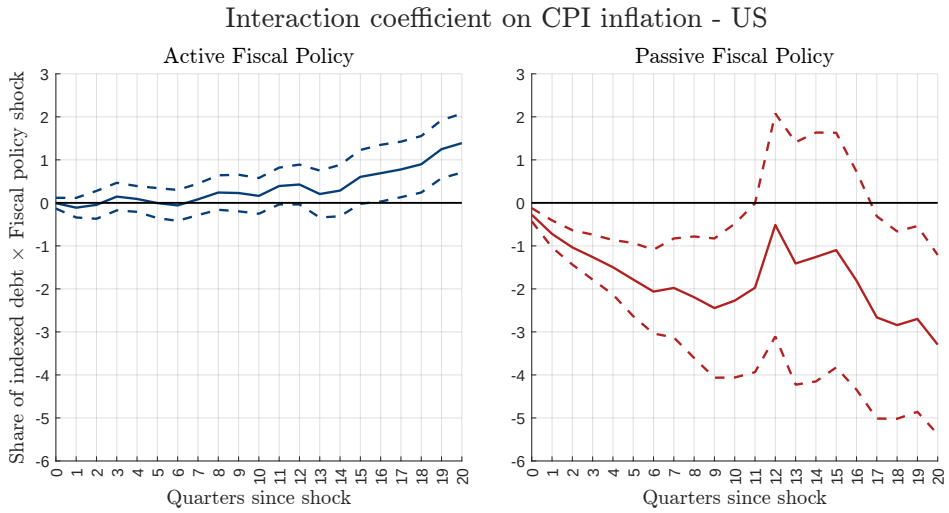


Figure A.13: IRF implied by the local projection (A.1) through the coefficients β_H. The control vector Z contains the first four lags of the real GDP growth rate, the short-run Fed Funds rate, the change in the weighted real exchange rate, and the first lag of the price level difference. The specification used here contains neither year fixed effects nor a recession indicator. Standard errors are clustered by year. Confidence intervals are provided at the 90% level. Sample length: 1980 Q3 - 2020 Q4.

[36] While the results in their paper only extend for the period until the Great Financial Crisis, I utilize their characterization to further pin down the prevalent policy mix until 2020, inclusive. The results are qualitatively robust to characterizing the prevalent fiscal-monetary regime as Bianchi and Melosi (2022) do.

B Derivations and proofs from the main text

B.1 Derivations from section 4

On the transversality condition with uninsurable idiosyncratic risk

I first clarify the sense in which a cumulative stochastic discount factor can be used to obtain a government debt valuation equation. The government flow constraint can be written as

$$W_t^d = S_t + V_t^d,$$

where W_t^d is the real value of inherited government liabilities and V_t^d is the real market value of newly issued liabilities.

Under complete markets, there exists a common one-period stochastic discount factor $m_{t,t+1}^c$ that prices the entire government debt portfolio, so that

$$V_t^d = \mathbb{E}_t \left[m_{t,t+1}^c W_{t+1}^d \right].$$

Defining the cumulative stochastic discount factor by $M_{t,t}^c \equiv 1$ and

$$M_{t,t+j}^c \equiv \prod_{s=0}^{j-1} m_{t+s,t+s+1}^c,$$

forward iteration gives

$$W_t^d = \mathbb{E}_t \left[\sum_{j=0}^{J} M_{t,t+j}^c S_{t+j} \right] + \mathbb{E}_t \left[M_{t,t+J+1}^c W_{t+J+1}^d \right].$$

Under the corresponding aggregate terminal condition, this converges to

$$W_t^d = \mathbb{E}_t \left[\sum_{j=0}^{\infty} M_{t,t+j}^c S_{t+j} \right].$$

This construction cannot be applied under incomplete markets by simply replacing the common SDF with an unweighted average of household-specific SDFs. An unweighted average one-period SDF need not price either aggregate debt position, and a ratio such as $\mathcal{M}_{t+l,t+l+1}/\mathcal{M}_{t,t+1}$ is not a cumulative discount factor. Moreover, the bond-pricing conditions are conditional-expectation restrictions and do not imply state-by-state identities linking bond prices to SDFs at consecutive dates.

An appropriate cumulative kernel can nevertheless be constructed from one-period marginal-holder kernels. In the present model, the nominal marginal-holder kernel prices nominal debt, but it does not price the indexed claim without a wedge because indexed debt is valued through the

continuation derivative of an adjusting household. Its cumulative product therefore produces the indexation wedge in proposition 1, rather than the wedge-free complete-market valuation equation. The next subsection derives this representation directly from the asset-specific household optimality conditions.

The household-level transversality condition does not remove the need for a separate aggregate terminal restriction. Although household optimality and the no-Ponzi conditions eliminate terminal wealth when valued by the relevant household-specific marginal values, these conditions cannot generally be aggregated because marginal values and portfolio positions are heterogeneous.

B.2 Derivation of equations (13) and (14) and proof of proposition 1

Proof. Here, I present the derivation of the asset-specific dynamic-trading valuation in proposition 1. The crucial feature is that the indexed bond is priced by an adjusting marginal holder through the continuation derivative $V_{b,i,t+1}$. The derivation does not substitute the contemporaneous consumption SDF for this continuation derivative when the household may be unable to adjust next period.

Part (i). For a household that can adjust indexed holdings at time t and whose indexed borrowing constraint does not bind, the first-order condition is

$$q_t \lambda_{it} = \beta \mathbb{E}_t \left[V_{b,i,t+1} \right]. \tag{B.2}$$

The envelope condition on indexed debt is given by

$$V_{b,i,t+1} = \begin{cases} \Pi_{t+1} \lambda_{i,t+1} & \text{if } a_{i,t+1} = 1, \\ \beta \frac{\Pi_{t+1}}{q_{t+1}} \mathbb{E}_{t+1} \left[V_{b,i,t+2} \right] & \text{if } a_{i,t+1} = 0. \end{cases}$$

Let $\tau_i(t)$ denote the first adjustment date after t. Repeated substitution into equation (B.2), followed by an application of the Law of Iterated Expectations, gives

$$q_t \lambda_{it} = \mathbb{E}_t \left[\beta^{\tau_i(t)} \lambda_{i,t+\tau_i(t)} \frac{\prod_{s=1}^{\tau_i(t)} \Pi_{t+s}}{\prod_{s=1}^{\tau_i(t)-1} q_{t+s}} \right]. \tag{B.3}$$

Using $\lambda_{it} = u'(c_{it})/P_t$ and $\prod_{s=1}^{j} \Pi_{t+s} = P_{t+j}/P_t$, equation (B.3) becomes equation (8). For a marginal indexed holder, multiplying that price equation by the aggregate real face quantity b_t/P_t gives

$$\frac{q_t b_t}{P_t} = \mathbb{E}_t \left[\beta^{\tau_i(t)} \frac{u'(c_{i,t+\tau_i(t)})}{u'(c_{it})} \frac{b_t}{P_t} \prod_{s=1}^{\tau_i(t)-1} \frac{1}{q_{t+s}} \right]. \tag{B.4}$$

The expression multiplying the household's marginal-utility ratio on the right-hand side of equation (B.4) is the real value of the indexed account when it can next be liquidated. Indeed, nonad-

justment at an intermediate date implies $b_{i,t+s} = (\Pi_{t+s}/q_{t+s})b_{i,t+s-1}$. Cumulative inflation cancels when the eventual nominal indexed payoff is divided by the contemporaneous price level, while the forced rollover through q_{t+s} remains. Averaging equation (B.4) over unconstrained adjusting marginal holders using the weights ϖ_{it}^I yields

$$\frac{q_t b_t}{P_t} = \mathcal{L}_t^I(b_t). \tag{B.5}$$

For non-indexed debt, the envelope condition is $V_{B,i,t+1} = \lambda_{i,t+1}$. The first-order condition of any unconstrained marginal holder therefore gives

$$Q_t = \beta \mathbb{E}_t \left[\frac{u'(c_{i,t+1})}{u'(c_{it})} \frac{P_t}{P_{t+1}} \right].$$

Multiplying by B_t/P_t and averaging over unconstrained nominal marginal holders using ϖ_{it}^N gives

$$\frac{Q_t B_t}{P_t} = \mathbb{E}_t \left[m_{t,t+1}^N \frac{B_t}{P_{t+1}} \right]. \tag{B.6}$$

Dividing the government flow constraint (11) by P_t gives

$$\frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}} = S_t + \frac{Q_t B_t}{P_t} + \frac{q_t b_t}{P_t},$$

which is the statement $W_t^d = S_t + V_t^d$ used in the main text. Substituting equations (B.6) and (B.5) proves equation (13).

The household-level terminal restriction used in this derivation and in the solution of the household problem is

$$\lim_{T \to \infty} \beta^{T-t+1} \mathbb{E}_t \left[V_{B,i,T+1} B_{iT} + V_{b,i,T+1} b_{iT} \right] = 0,$$

supplemented by no-Ponzi conditions when positions can be negative. Summing the household conditions does not create an aggregate government TVC because household marginal values and portfolio positions are heterogeneous. This is why the transversality restriction is imposed separately in part (ii) rather than derived.

Part (ii). The claims inherited by the government at $t+1$ have the real value

$$W_{t+1}^d = \frac{B_t}{P_{t+1}} + \frac{b_t}{P_t}, \tag{B.7}$$

where the second term is measurable with respect to time t information. The indexed instrument delivers its real payoff with certainty one period ahead irrespective of which households can retrade at $t+1$, since the retrading friction governs the price q_t rather than the redemption value of the account. Discounting equation (B.7) with the nominal marginal-holder kernel assumed in the

proposition and using equation (B.6) gives

$$\mathbb{E}_t\left[m^N_{t,t+1}W^d_{t+1}\right] = \mathbb{E}_t\left[m^N_{t,t+1}\frac{B_t}{P_{t+1}}\right] + \frac{b_t}{P_t}\mathbb{E}_t\left[m^N_{t,t+1}\right] = \frac{Q_tB_t}{P_t} + \frac{q_tb_t}{P_t} + \left(\mathbb{E}_t\left[m^N_{t,t+1}\right] - q_t\right)\frac{b_t}{P_t},$$

that is,

$$\mathbb{E}_t\left[m^N_{t,t+1}W^d_{t+1}\right] = V^d_t + \xi_t, \tag{B.8}$$

with ξ_t defined in equation (15). The wedge captures the retrading friction for the aggregate valuation equation, and it is non-zero because no household equates its one-period marginal rate of substitution to q_t when $\nu_b < 1$. Nonadjusters have no first-order condition in b_{it}, and the first-order condition (B.2) of an adjuster prices the continuation derivative $V_{b,i,t+1}$, which by the envelope condition coincides with $\Pi_{t+1}\lambda_{i,t+1}$ only in the event $a_{i,t+1} = 1$.

Combining equation (B.8) with the flow constraint $W^d_t = S_t + V^d_t$ delivers the recursion

$$W^d_t = S_t - \xi_t + \mathbb{E}_t\left[m^N_{t,t+1}W^d_{t+1}\right]. \tag{B.9}$$

Iterating equation (B.9) forward J times and applying the Law of Iterated Expectations gives

$$W^d_t = \mathbb{E}_t\left[\sum_{j=0}^{J} M_{t,t+j}\left(S_{t+j} - \xi_{t+j}\right)\right] + \mathbb{E}_t\left[M_{t,t+J}\left(V^d_{t+J} + \xi_{t+J}\right)\right],$$

where the residual term has been rewritten using $W^d_{t+J} = S_{t+J} + V^d_{t+J}$. By equation (15) and the uniform bound assumed in part (ii),

$$|\xi_{t+J}| \leq K\theta^{mkt}_{t+J}V^d_{t+J} \leq KV^d_{t+J}.$$

Since $M_{t,t+J} \geq 0$, it follows that

$$\left|\mathbb{E}_t\left[M_{t,t+J}\xi_{t+J}\right]\right| \leq \mathbb{E}_t\left[M_{t,t+J}\left|\xi_{t+J}\right|\right] \leq K\mathbb{E}_t\left[M_{t,t+J}V^d_{t+J}\right] \to 0.$$

The aggregate transversality condition eliminates the component involving V^d_{t+J} directly and, together with the uniform bound, also eliminates the component involving ξ_{t+J}. The full residual therefore vanishes as $J \to \infty$, which proves equation (14). ■

Once the indexation wedge (which we defined in Proposition 1) vanishes, the debt valuation equation becomes very simple.

Corollary 2 *Let the assumptions from part (ii) of proposition 1 hold. The indexation wedge satisfies $\xi_t = 0$ at all dates if either*

- *no indexed debt is outstanding, $b_t = 0$ for all t, or*
- *the retrading friction is inoperative, $\nu_b = 1$, the set $\mathcal{I}^N_t \cap \mathcal{I}^I_t$ is non-empty at every date and in every*

state, and the weights ϖ_{it}^N are supported on that set.

In either case, equation (14) collapses to an expression equating the real value of debt to thee (unweighted) present discounted value of primary surpluses,

$$\frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}} = \mathbb{E}_t\left[\sum_{j=0}^{\infty} M_{t,t+j} S_{t+j}\right].$$

Proof. In the first case, the indexed face quantity is zero at every date, so $\xi_t = 0$ follows immediately from equation (15). In the second case, $\nu_b = 1$ implies $\tau_i(t) = 1$ for every household and every date, so equation (8) collapses to

$$q_t = \beta \mathbb{E}_t\left[\frac{u'(c_{i,t+1})}{u'(c_{it})}\right] \qquad \text{for all } i \in \mathcal{I}_t^I.$$

Since the weights ϖ_{it}^N are supported on $\mathcal{I}_t^N \cap \mathcal{I}_t^I$, every household receiving positive weight in $m_{t,t+1}^N$ satisfies this equation, and therefore

$$\mathbb{E}_t\left[m_{t,t+1}^N\right] = \int_{\mathcal{I}_t^N \cap \mathcal{I}_t^I} \varpi_{it}^N \, \beta \mathbb{E}_t\left[\frac{u'(c_{i,t+1})}{u'(c_{it})}\right] di = q_t,$$

so that $\xi_t = 0$. Substituting into equation (14) yields the stated expression in both cases. ■

The support restriction in the second case cannot be sidelined. A household that is an unconstrained marginal holder of nominal debt, but is constrained in indexed debt, satisfies $q_t \lambda_{it} = \beta \mathbb{E}_t[V_{b,i,t+1}] + \mu_{it}^b$ with $\mu_{it}^b > 0$, hence $\mathbb{E}_t[m_{t,t+1}^N] \leq q_t$ for that household even when $\nu_b = 1$, and the wedge does not vanish. The derivation also clarifies the relation to the dynamic-trading perspective of Brunnermeier et al. (2024): the ability to retrade enters the indexed continuation value through $\tau_i(t)$ and the intervening rollover terms. The insurance value of the instrument enters because its eventual real payoff is protected against cumulative inflation and is valued in states with the marginal utility of the household that prices the claim.

B.3 Properties of the stationary equilibrium

The main model is approximated around a steady-state without trend inflation, $\Pi_{ss} = 1$. In this subsection only, I consider the more general auxiliary question of whether a stationary equilibrium can be supported when nominal face quantities and the price level grow at a common rate μ, so that Π_{ss} need not equal one. Stationarity in this auxiliary exercise must be understood in normalized real units rather than in the unnormalized nominal face quantities used to state the contracts in the main model.

To that goal, define issuance-date real face positions by

$$\widetilde{B}_{i,t-1} \equiv \frac{B_{i,t-1}}{P_{t-1}}, \qquad \tilde{b}_{i,t-1} \equiv \frac{b_{i,t-1}}{P_{t-1}},$$

and define current normalized choices by $\widetilde{B}_{it} \equiv B_{it}/P_t$ and $\tilde{b}_{it} \equiv b_{it}/P_t$. For this auxiliary stationary exercise, the borrowing constraints are imposed in these normalized units, $\widetilde{B}_{it} \geq -\underline{B}$ and $\tilde{b}_{it} \geq -\underline{b}$. Equivalently, the corresponding nominal limits grow with the price level. This convention does not alter the main model around $\Pi_{ss} = 1$, where the price level and nominal limits are constant.

Dividing the combined budget constraint of an adjuster by P_t gives

$$c_{it} + Q_t\widetilde{B}_{it} + q_t\tilde{b}_{it} = \ell_{it}(1-\tau_t)w_tN_{it} + \mathcal{D}_{it} + \frac{\widetilde{B}_{i,t-1}}{\Pi_t} + \tilde{b}_{i,t-1},$$

while for a nonadjuster,

$$c_{it} + Q_t\widetilde{B}_{it} = \ell_{it}(1-\tau_t)w_tN_{it} + \mathcal{D}_{it} + \frac{\widetilde{B}_{i,t-1}}{\Pi_t}, \qquad \tilde{b}_{it} = \frac{\tilde{b}_{i,t-1}}{q_t}.$$

Thus, the auxiliary stationary household problem depends on gross inflation and real returns but not on the level of nominal prices.

Let $\tilde{g}^B_t(\tilde{x})$ and $\tilde{g}^b_t(\tilde{x})$ denote the normalized asset policy functions at the state $\tilde{x} = (\widetilde{B}_-, \tilde{b}_-, e, a)$, and let $\tilde{\Gamma}_t$ denote the corresponding distribution. Its law of motion is

$$\tilde{\Gamma}_{t+1}(\widetilde{\mathcal{B}}, \tilde{\mathcal{b}}, \mathcal{E}, a') = \int \mathbf{1}_{\{\tilde{g}^B_t(\tilde{x}) \in \widetilde{\mathcal{B}},\, \tilde{g}^b_t(\tilde{x}) \in \tilde{\mathcal{b}}\}} p_e(\mathcal{E} \mid e) p_a(a') d\tilde{\Gamma}_t(\tilde{x}). \tag{B.10}$$

For a nonadjuster, $\tilde{g}^b_t(\tilde{x}) = \tilde{b}_-/q_t$. The inflation term appearing in the nominal transition cancels because both the indexed face quantity and the price level are scaled by Π_t.

It is convenient to work with the real market values of the two instruments and with their stationary real returns,

$$D^N_t \equiv \frac{Q_tB_t}{P_t}, \qquad D^I_t \equiv \frac{q_tb_t}{P_t}, \qquad R^N_{ss} \equiv \frac{1}{Q_{ss}\Pi_{ss}}, \qquad R^I_{ss} \equiv \frac{1}{q_{ss}},$$

so that $d^N_t = D^N_t/Y_t$ and $d^I_t = D^I_t/Y_t$ are the debt ratios entering the surplus rule (16). For the purposes of this subsection, a *stationary equilibrium* is a competitive equilibrium in which the normalized distribution $\tilde{\Gamma}_t$, all real quantities and returns, and the gross inflation rate are constant over time. The corresponding distribution over unnormalized nominal face positions need not be invariant when $\mu \neq 0$, since it shifts at the same rate as the price level.

Neither policy rule restricts the stationary equilibrium. Evaluating the Taylor rule (17) at constant R^n, Π, and Y with $\varepsilon^M_{ss} = 0$ gives $1 = 1^{\rho_M}[1^{\phi_\pi}1^{\phi_y}]^{1-\rho_M}$, which holds for all ρ_M, ϕ_π, and ϕ_y, and the same applies to the surplus rule (16) with $\varepsilon^F_{ss} = 0$, so both rules obviously do not matter for the stationary equilibrium. What does have to be assigned exogenously is the stationary primary surplus

ratio s^p that appears as a parameter in (16), together with a specification of nominal issuance.

I therefore specify policy in the stationary equilibrium as follows.

(P1) Fiscal policy sets the face quantities of the two instruments in nominal terms, with a common issuance growth rate μ and a fixed composition, so that $B_t = (1+\mu)B_{t-1}$ and $b_t = (1+\mu)b_{t-1}$ from given initial levels $B_{-1} \geq 0$ and $b_{-1} \geq 0$ with $\theta^{const.} = b_{-1}/(B_{-1}+b_{-1})$ and $B_{-1}+b_{-1} > 0$.

(P2) Fiscal policy sets the stationary primary surplus ratio at $s^p > 0$, and the household tax schedule adjusts to implement it at every price level.

Under (P1), the normalized aggregate face quantities B_t/P_t and b_t/P_t are stationary precisely when the price level grows at the common issuance rate. The policy specification is therefore consistent with an invariant distribution $\tilde{\Gamma}_{ss}$ even when $\mu \neq 0$.

The regularity conditions I impose are the standard ones under which stationary asset demand is a well-defined function, together with two monotonicity restrictions that deliver uniqueness conditional on the existence of a stationary real allocation.

(R1) For every admissible pair of real returns and every real tax schedule, the normalized stationary household problem and the law of motion (B.10) admit a unique invariant distribution $\tilde{\Gamma}_{ss}$, so that aggregate demand for each actively supplied instrument is a continuous function.

(R2) The joint fixed point in real debt and real taxation implied by the stationary government budget exists and is unique, following Acemoglu and Jensen (2015). Write the resulting aggregate real demands as $\mathcal{A}^N(R^N, R^I)$ and $\mathcal{A}^I(R^N, R^I)$.

(R3) For each instrument supplied in strictly positive quantity, $\beta R^k_{ss} < 1$ and aggregate real demand for that instrument is finite and strictly positive.

(R4) For an interior portfolio, relative demand $\mathcal{A}^I/\mathcal{A}^N$ is strictly increasing in R^I at given R^N, so that the two instruments are gross substitutes in household portfolios.

(R5) For an interior portfolio, total real interest expenditure $(R^N-1)\mathcal{A}^N + (R^I-1)\mathcal{A}^I$ is strictly increasing in R^N along the locus defined by portfolio composition and falls below $s^p Y_{ss}$ at the lower boundary of admissible returns.

Conditions (R4) and (R5) do not impose the conditions required for a general existence theorem. They will therefore be used only to establish uniqueness conditional on existence.

Proposition 3 *Consider the auxiliary stationary economy of this subsection under* (P1), (P2), *and* (R1) *to* (R3), *and suppose that* $\theta^{const.} \in (0,1)$. *Then, the following hold:*

(i) *Every stationary equilibrium satisfies* $\Pi_{ss} = 1+\mu$, *and the consistency condition* $g_B = g_b$ *on the issuance schedule is a property of any such equilibrium, so any stationary-equilibrium inflation is driven by fiscal factors.*

(ii) *The stationary real returns of any stationary equilibrium solve the system*

$$\frac{\mathcal{A}^I(R^N_{ss}, R^I_{ss})}{\mathcal{A}^N(R^N_{ss}, R^I_{ss})} = \frac{R^N_{ss}\Pi_{ss}}{R^I_{ss}} \cdot \frac{\theta^{const.}}{1-\theta^{const.}}, \tag{B.11}$$

$$s^p Y_{ss} = \left(R^N_{ss} - 1\right)\mathcal{A}^N(R^N_{ss}, R^I_{ss}) + \left(R^I_{ss} - 1\right)\mathcal{A}^I(R^N_{ss}, R^I_{ss}). \tag{B.12}$$

Under (R4) *and* (R5)*, this system has at most one solution.*

(iii) *Conditional on the existence of a solution to* (ii)*, the stationary price level is unique, finite, and strictly positive, and is given in closed form by*

$$P_t = \frac{Q_{ss}B_t + q_{ss}b_t}{\mathcal{A}^N + \mathcal{A}^I}, \tag{B.13}$$

so that $P_t = P_0(1+\mu)^t$.

To establish the proof of this proposition, I must define two additional lemmas. The first derives the stationary government budget in real terms (with two different debt instruments), while the second establishes that the price level enters the stationary system only through the real value of nominal issuance.

Lemma 3 *In any stationary equilibrium,*

$$S_{ss} = \left(R^N_{ss} - 1\right)D^N_{ss} + \left(R^I_{ss} - 1\right)D^I_{ss}. \tag{B.14}$$

Proof. Divide the government flow constraint (11) by P_t to obtain

$$\frac{B_{t-1}}{P_t} + \frac{b_{t-1}}{P_{t-1}} = S_t + D^N_t + D^I_t.$$

Stationarity of D^N and of Q implies $D^N_{ss} = Q_{ss}B_{t-1}/P_{t-1}$, hence $B_{t-1}/P_t = D^N_{ss}/(Q_{ss}\Pi_{ss}) = R^N_{ss}D^N_{ss}$. Stationarity of D^I and of q implies $D^I_{ss} = q_{ss}b_{t-1}/P_{t-1}$, hence $b_{t-1}/P_{t-1} = D^I_{ss}/q_{ss} = R^I_{ss}D^I_{ss}$. Substituting both and rearranging gives (B.14). ■

Lemma 4 *Collect the stationary real objects in* $\mathcal{R} \equiv (\Pi_{ss}, Q_{ss}, q_{ss}, D^N_{ss}, D^I_{ss}, S_{ss}, Y_{ss}, w_{ss}, N_{ss}, mc_{ss}, \tilde{\Gamma}_{ss})$*. Every equilibrium condition of the auxiliary stationary economy restricts* $\mathcal{R}$ *alone, with the exception of the two asset market clearing conditions in* (20)*, which take the form*

$$D^N_{ss} = \frac{Q_{ss}B_t}{P_t}, \qquad D^I_{ss} = \frac{q_{ss}b_t}{P_t}. \tag{B.15}$$

The price level therefore enters the stationary system only through the real value of nominal issuance.

Proof. The normalized stationary household problem defined in this subsection depends on aggregate prices only through the real returns R^N_{ss} and R^I_{ss}, stationary gross inflation Π_{ss}, the real wage w_{ss}, real profits, and the real after-tax income schedule, and hence through $\mathcal{R}$ alone. The

Rotemberg first-order condition underlying (9) relates mc_{ss} to Π_{ss} and involves no price level. Labor market clearing (19) and goods market clearing (18) are stated in real terms, and the latter is redundant by Walras' Law. The government flow constraint reduces to (B.14) by lemma 3, which is real. The two policy rules hold identically at any stationary equilibrium, as verified above, and therefore restrict nothing. Asset market clearing in face quantities, together with the definitions of D^N and D^I, gives (B.15). ■

With these prerequisites, I can now construct the proof of proposition 3.

Proof. By (P1), the face quantities grow at the constant nominal rate μ. Invariance of the distribution $\tilde{\Gamma}_{ss}$ implies that aggregate normalized face holdings B_t/P_t and b_t/P_t are constant. Since $\theta^{const.} \in (0,1)$, both instruments are supplied in strictly positive quantities, and therefore

$$\Pi_{ss} = \frac{P_t}{P_{t-1}} = \frac{B_t}{B_{t-1}} = \frac{b_t}{b_{t-1}} = 1 + \mu.$$

This proves part (i).

Next, fix a candidate pair of real returns. By (R1), the invariant distribution over normalized real positions is unique, and since real taxation is not exogenous but must finance G_{ss} together with the real interest payments in (B.14), aggregate demand is evaluated at the fixed point in which the real tax schedule is the one implied by the demands themselves. Condition (R2) delivers existence and uniqueness of that fixed point, following the argument in section VII of Acemoglu and Jensen (2015), and continuity of $\mathcal{A}^N$ and $\mathcal{A}^I$. By (R3), both aggregate demands are finite and strictly positive in the interior case considered here.

Asset market clearing (B.15) requires both instruments to be consistent with a common price level. Dividing the second condition by the first and using $b_t/B_t = \theta^{const.}/(1-\theta^{const.})$, which holds at every date under (P1), eliminates P_t and yields (B.11), where $q_{ss}/Q_{ss} = R^N_{ss}\Pi_{ss}/R^I_{ss}$. Fix R^N and consider (B.11) as an equation in R^I. By (R4), the left-hand side is strictly increasing in R^I, while the right-hand side is strictly decreasing. Hence, whenever a solution exists, it is unique and defines a function $R^I(R^N)$. The maintained assumptions do not by themselves imply that relative demand crosses relative supply at the boundaries of the admissible return set.

Substituting $R^I(R^N)$ into (B.12), which is lemma 3 evaluated under (P2) and market clearing, gives one equation in R^N alone. By (R5), its right-hand side is strictly increasing in R^N, so whenever a solution exists it is unique. This proves part (ii).

With the real block solved, the first condition in (B.15) gives $P_t = Q_{ss}B_t/\mathcal{A}^N$, and by construction of R^I_{ss} the second condition gives the same value, $P_t = q_{ss}b_t/\mathcal{A}^I$. Adding numerators and denominators yields (B.13). Since $\theta^{const.} \in (0,1)$, both nominal supplies and both corresponding real demands are strictly positive. The right-hand side is therefore unique, strictly positive, and finite by (P1) and (R3). Since B_t and b_t grow at rate μ while $\mathcal{A}^N + \mathcal{A}^I$, Q_{ss}, and q_{ss} are stationary, $P_t = P_0(1+\mu)^t$, consistent with part (i). This proves part (iii). ■

Proposition 3 is deliberately stated for an interior debt portfolio. At $\theta^{const.} = 0$, the relative-market-clearing equation (B.11) is unavailable and the stationary equilibrium of the zero-supply indexed market must be established separately. Conditional on such a stationary solution, the active nominal market determines the price level according to

$$P_t = \frac{Q_{ss} B_t}{\mathcal{A}^N}.$$

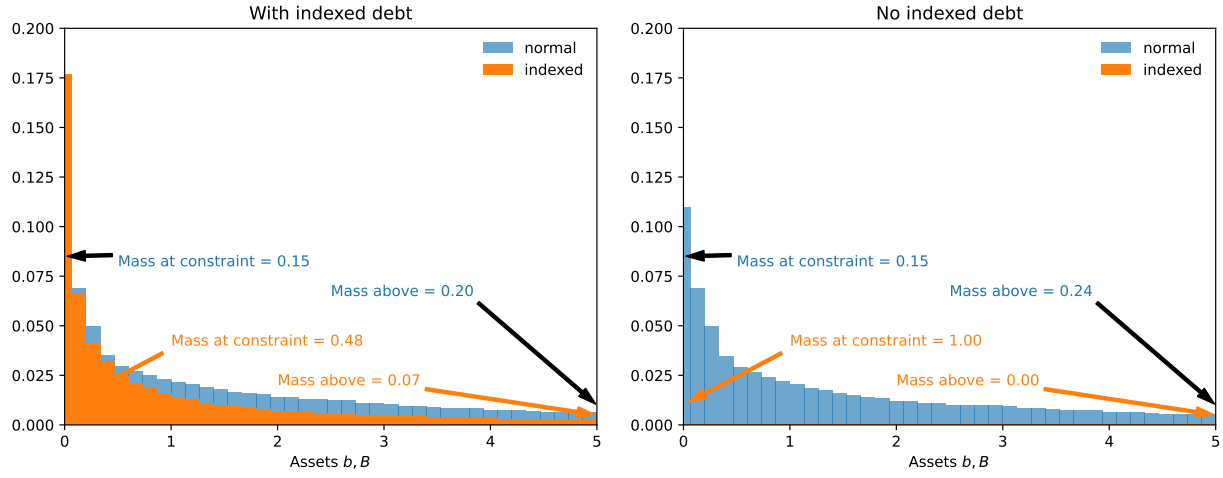
At $\theta^{const.} = 1$, the corresponding active-market condition is

$$P_t = \frac{q_{ss} b_t}{\mathcal{A}^I}.$$

Equation (B.13) is consistent with both expressions after the inactive supply and its equilibrium demand are set to zero. The proposition does not, however, establish existence or uniqueness of the corresponding boundary real allocation.

C Further simulation results

Even though government debt aggregates are exogenously supplied in steady-state for all calibrations, the distribution of debt across households cannot be deduced immediately from the calibration itself, as it is generally dependent on the properties of the idiosyncratic process to income. Figure C.1 plots the distribution of debt holdings across households in two cases - once for the standard calibration to the U.K., and once for the counterfactual calibration where steady-state issuance of indexed government debt is set to 0. For the calibration without indexed debt supply, plotted in panel (b), no households hold indexed debt, while indexed bonds are held by 52% of all households in the main calibration.



(a) Both types of debt (U.K. calibration)

(b) Non-indexed government debt only

Figure C.1: Steady-state distributions of asset holdings across households in the calibrated steady-state

Policy mix	U.K. portfolio	U.S. portfolio	No indexed debt	U.K. − no-index
Cumulative first-year price change (*percent*)				
Fiscally-led	0.434	0.283	0.204	+0.230
Monetary-led	0.073	0.077	0.077	−0.004
Annualized quarterly inflation on impact (*percentage points*)				
Fiscally-led	0.441	0.288	0.208	+0.234
Monetary-led	0.073	0.077	0.077	−0.003

Table C.1: Price and inflation responses to an initially unfunded increase in government purchases of 1% of quarterly steady-state GDP on impact, $\rho_G = 0.50$. The first-year price change sums the first four quarterly inflation responses. The last column reports differences in percentage points, calculated before rounding.

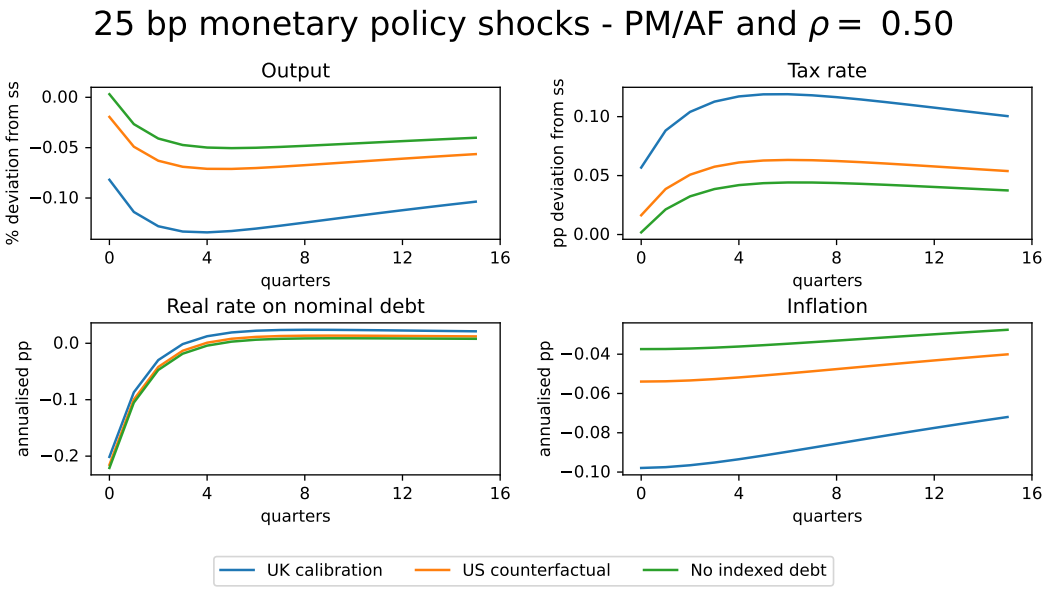


Figure C.2: IRFs to a 25bps expansionary monetary shock - under a fiscally-led policy mix and $\rho = 0.5$.

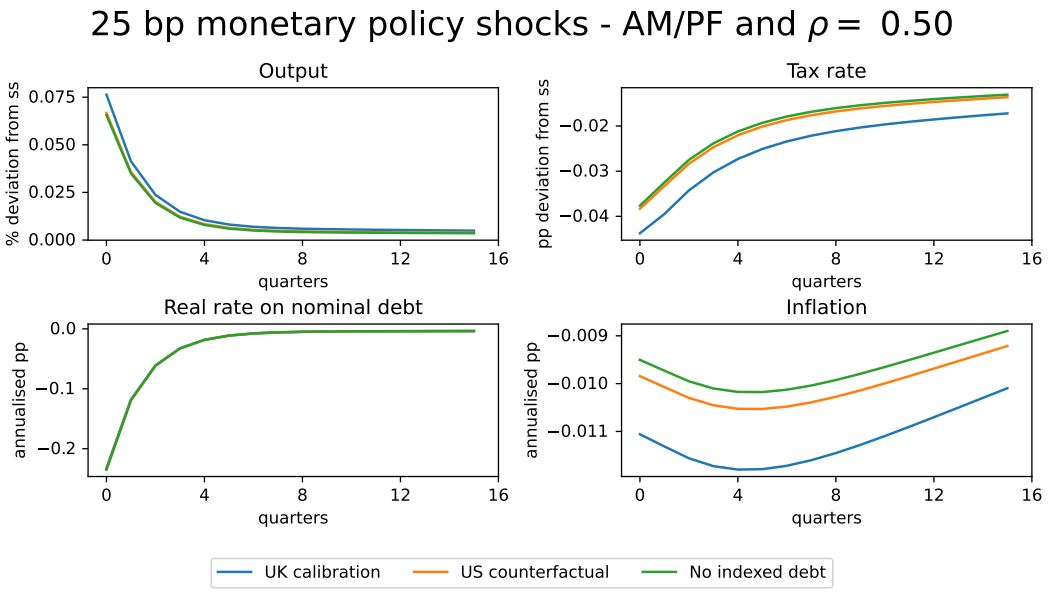


Figure C.3: IRFs to a 25bps expansionary monetary shock - under a monetary-led policy mix and $\rho = 0.5$.

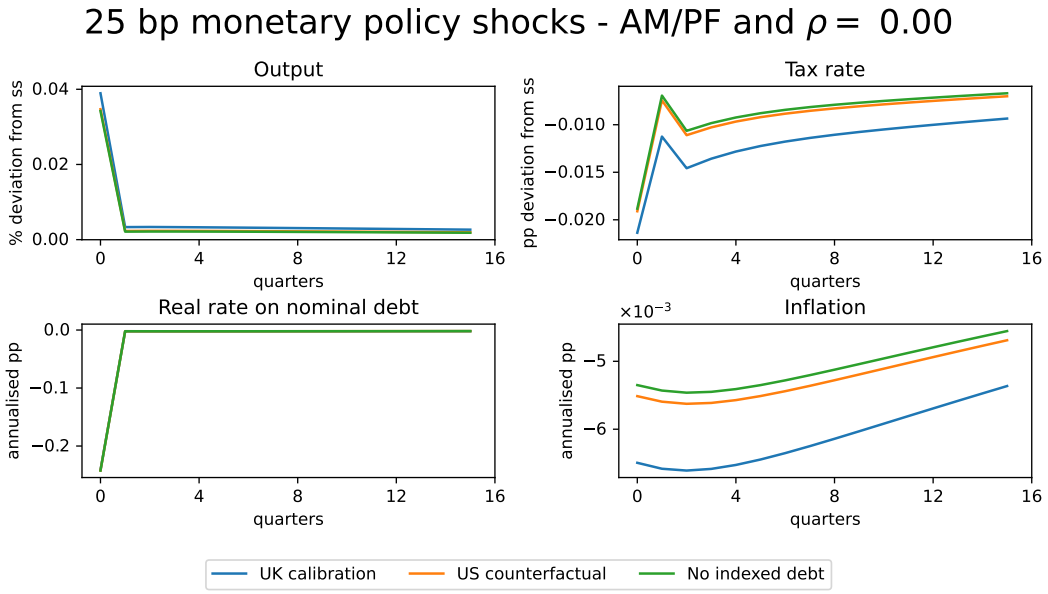


Figure C.4: IRFs to a 25bps expansionary monetary shock - under a monetary-led policy mix and $\rho = 0$.

For the main policy scenario (the 'fiscally-led policy mix'), I furthermore provide additional evidence on changes of quantities directly informing the intertemporal government budget constraint (13).

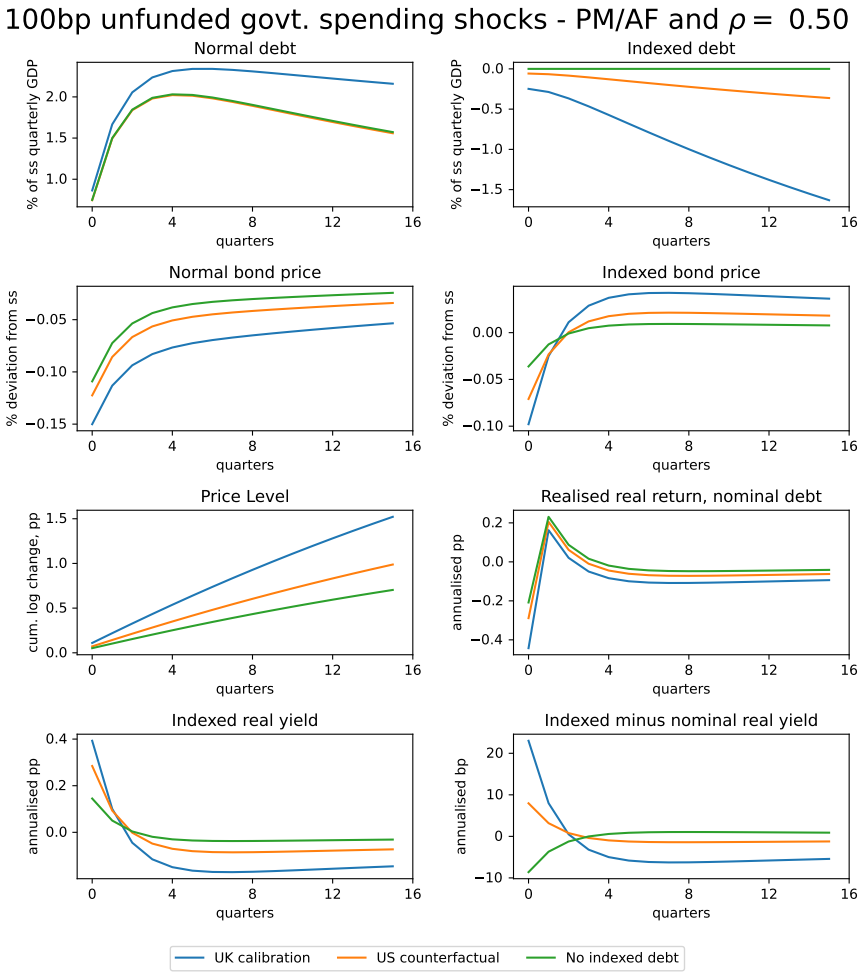


Figure C.5: Further IRFs to a 100bps expansionary fiscal spending shock - under a fiscally-led policy mix and $\rho = 0.5$.

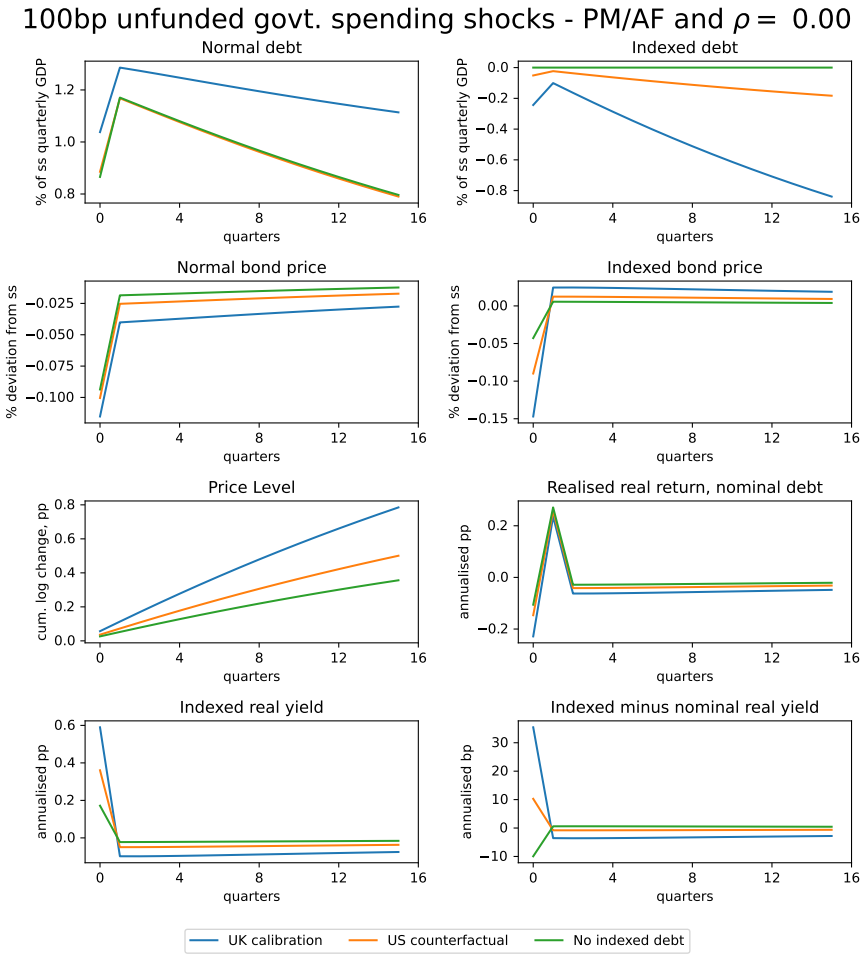


Figure C.6: Further IRFs to a 100bps expansionary fiscal spending shock - under a fiscally-led policy mix and $\rho = 0$.

D Distributional effects in the HANK model

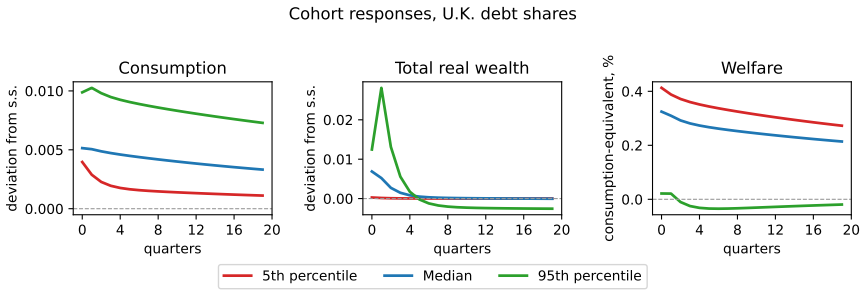


Figure D.1: Impulse-responses following an unfunded government spending shock of 1% of quarterly GDP for wealth percentiles in the model calibrated to U.K. debt shares. Consumption and wealth are in model units, with quarterly steady-state output normalized to one. Welfare is in consumption-equivalent percent.

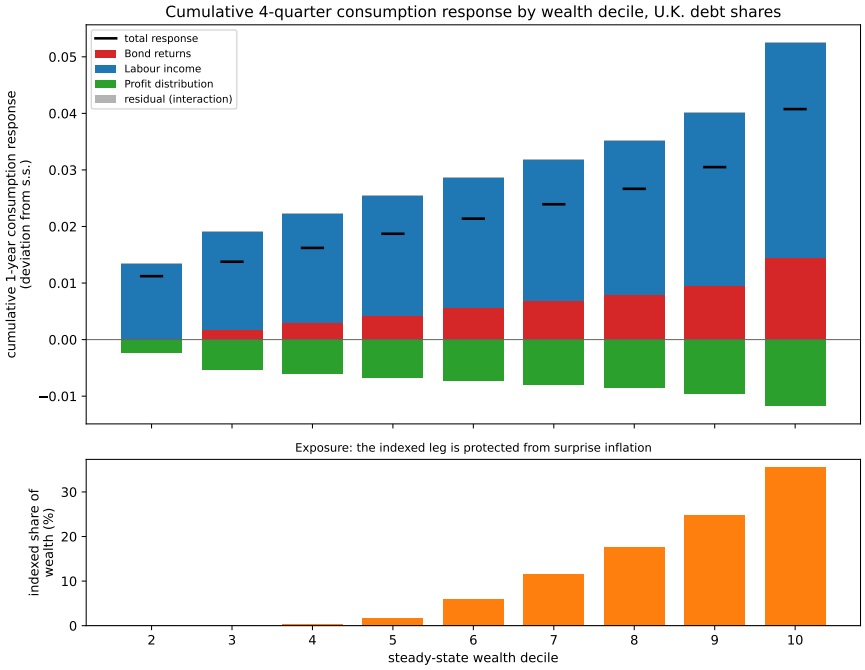


Figure D.2: Sum of the first four quarterly consumption responses to an unfunded government spending shock of 1% of quarterly GDP, for wealth deciles 2–10 in the model calibrated to the U.K. debt shares. Responses are in model units, with quarterly steady-state output normalized to one.

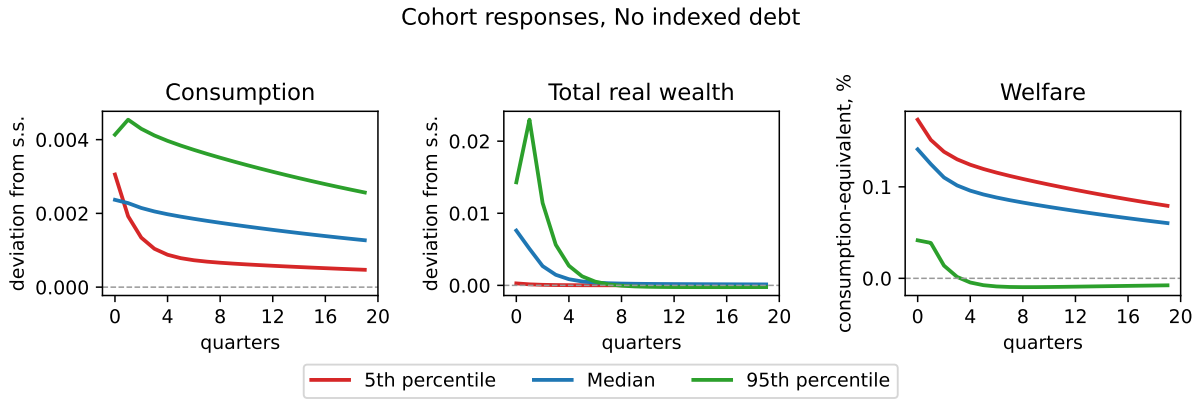


Figure D.3: Impulse-responses following an unfunded government spending shock of 1% of quarterly GDP for wealth percentiles in the model calibrated to the 'no indexed debt' scenario. Consumption and wealth are in model units, with quarterly steady-state output normalized to one. Welfare is in consumption-equivalent percent.

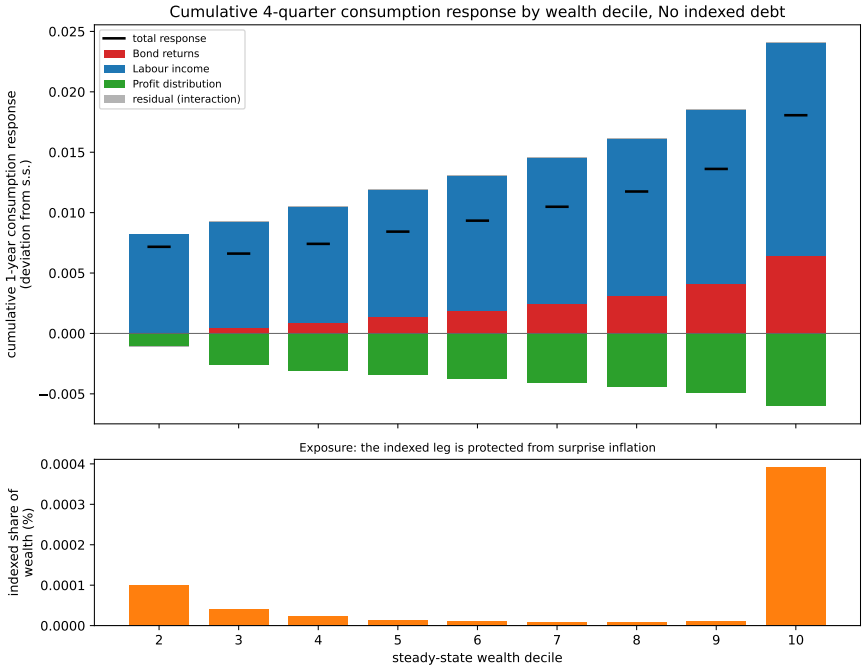


Figure D.4: Sum of the first four quarterly consumption responses to an unfunded government spending shock of 1% of quarterly GDP, for wealth deciles 2–10 in the model calibrated to the 'no indexed debt' scenario. Responses are in model units, with quarterly steady-state output normalized to one.

E A tractable exposition of the effects of inflation-indexed debt

This appendix derives the equilibrium system underlying section 7 and proves proposition 2. Uppercase variables denote steady-state levels and lowercase variables denote deviations from the deterministic steady-state, while debt, taxes and household wealth are measured in absolute deviations from steady-state and are scaled by Y^{SS}. Throughout I suppress the economy index j wherever an expression applies to both economies, and I write ω for the survival probability with the understanding that $\omega = 1$ delivers the representative-agent economy.

E.1 Household block and aggregate demand

I retain the aggregate representation of the economy of Angeletos et al. (2024) with finite household horizons. Government debt is the only aggregate financial asset held by households, who survive from one period to the next with probability ω and who receive their labor income through unions, such that individual labor supply and taxation do not differ across households. Conditional on total real financial wealth a_t and on the proportional deviation of the expected gross real return on the government debt portfolio, which is given by equation (27), aggregate consumption satisfies

$$\begin{aligned} c_t = (1-\beta\omega)\left(a_t + \mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k (y_{t+k} - t_{t+k})\right) \\ -\beta\left[\sigma\omega - \frac{1-\beta\omega}{\beta}\frac{D^{SS}}{Y^{SS}}\right](\phi+\theta\psi)\,\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\omega)^k y_{t+k}. \end{aligned} \tag{E.1}$$

Market clearing requires $c_t = y_t$ and $a_t = d_t$. Taking then the difference of equation (E.1) with its expected value one period later (after pre-multiplying the latter by $\beta\omega$), and using market clearing gives

$$y_t - \beta\omega\mathbb{E}_t y_{t+1} = (1-\beta\omega)\left(d_t - \beta\omega\mathbb{E}_t d_{t+1} + y_t - t_t\right) - \beta\left[\sigma\omega - \frac{1-\beta\omega}{\beta}\frac{D^{SS}}{Y^{SS}}\right](\phi+\theta\psi)\, y_t, \quad \text{(E.2)}$$

which can be rearranged into the forward-looking relation

$$\mathbb{E}_t y_{t+1} = y_t + \frac{\sigma\omega - \frac{1-\beta\omega}{\beta}\frac{D^{SS}}{Y^{SS}}}{\omega}(\phi+\theta\psi)\, y_t - \frac{1-\beta\omega}{\beta\omega}(d_t - t_t) + (1-\beta\omega)\,\mathbb{E}_t d_{t+1}. \quad \text{(E.3)}$$

Equation (E.3) is the aggregate demand block of the model. Mortality risk enters through the discounting of future household income and taxes and through the coefficient on the portfolio return, and it does not by itself alter the contractual indexation of government debt.

E.2 Long-term inflation-indexed debt

Let J_t denote the units of indexed debt outstanding at the beginning of period t. A share of indexed debt pays one unit of the consumption good in period t and leaves δ units outstanding after the payment, so that the real market value of the indexed stock before paying out the coupon is $J_t(1+\delta Q_t)$ and the gross real holding return between t and $t+1$ is

$$\frac{1+\delta Q_{t+1}}{Q_t}. \quad \text{(E.4)}$$

All payments in equation (E.4) are expressed in consumption units, so that no price level term enters the indexed payoff and the claim cannot be devalued by an inflation surprise. Requiring the expected holding return to equal the required gross real return $\beta^{-1}(1+r_t+\psi y_t)$ delivers equation (24), whose deterministic steady-state is $Q^{SS} = \beta\left(1+\delta Q^{SS}\right)$ and hence $Q^{SS} = \beta/(1-\beta\delta)$. Log-linearizing around that point and using $\frac{\delta Q^{SS}}{1+\delta Q^{SS}} = \beta\delta$ gives equation (25). Imposing then that $\lim_{T\to\infty}(\beta\delta)^T\,\mathbb{E}_t q_{t+T} = 0$ and solving equation (25) forward, while substituting the monetary rule (26), yields

$$q_t = -(\phi+\psi)\,\mathbb{E}_t\sum_{k=0}^{\infty}(\beta\delta)^k\, y_{t+k}. \quad \text{(E.5)}$$

The market price of the outstanding indexed debt stock is a geometrically discounted sum of the future output path, discounted at the rate $\beta\delta$, whereas inflation is a geometrically discounted sum of the same path at the rate β.

E.3 The government budget constraint

Denote by d_t the real market value of the consolidated government debt portfolio. The ex-post real return on the one-period nominal instrument equals β^{-1} times one plus its proportional expected-return deviation less the inflation surprise. By equations (E.4) and $\frac{\delta Q^{SS}}{1+\delta Q^{SS}} = \beta\delta$, the ex-post real return on the indexed instrument equals β^{-1} times one plus its proportional expected-return deviation and the proportional revaluation term $\beta\delta\left(q_{t+1} - \mathbb{E}_t q_{t+1}\right)$. Consequently, the expected-return and nominal inflation-surprise terms enter the debt law with coefficient β^{-1}, while the indexed price surprise enters it with coefficient δ. Linearizing the government flow budget constraint accordingly delivers equation (29).

Taking conditional expectations of equation (29) and substituting the fiscal rule (28) then gives a recursive equation pinning down the expected value of government debt:

$$\mathbb{E}_t d_{t+1} = \frac{1}{\beta}\left[\left(1-\tau_d\right)\left(d_t+\varepsilon_t\right) - \left(\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)\right) y_t\right]. \tag{E.6}$$

E.4 The recursive system and its roots

Substituting the fiscal rule (28), the identity $d_t = \left(d_t+\varepsilon_t\right) - \varepsilon_t$, and the expected debt recursion (E.6) into the aggregate demand block (E.3) delivers, after collecting terms, the system

$$\begin{aligned}
\mathbb{E}_t d_{t+1} &= \frac{1-\tau_d}{\beta}\left(d_t+\varepsilon_t\right) - \frac{1}{\beta}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)\right) y_t, \\
\mathbb{E}_t y_{t+1} &= -\frac{\left(1-\beta\omega\right)\left(1-\omega\right)}{\beta\omega}\left(1-\tau_d\right)\left(d_t+\varepsilon_t\right) \\
&\quad + \left[1+\sigma\left(\phi+\theta\psi\right) + \frac{\left(1-\beta\omega\right)\left(1-\omega\right)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)\right)\right] y_t.
\end{aligned} \tag{E.7}$$

The determinant of the coefficient matrix of the system (E.7) simplifies considerably, since the two products involving $\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)$ cancel, leaving $\left(1-\tau_d\right)\left[1+\sigma\left(\phi+\theta\psi\right)\right]/\beta$. Its characteristic polynomial is therefore equation (31), and its roots are

$$\begin{aligned}
\rho^{\pm} = \frac{1}{2}\Bigg\{ & \frac{1-\tau_d}{\beta} + 1 + \sigma\left(\phi+\theta\psi\right) + \frac{\left(1-\beta\omega\right)\left(1-\omega\right)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)\right) \\
& \pm \Bigg[\left(\frac{1-\tau_d}{\beta} + \frac{\left(1-\beta\omega\right)\left(1-\omega\right)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)\right) - 1 - \sigma\left(\phi+\theta\psi\right)\right)^2 \\
& \quad + \frac{4\left(1-\beta\omega\right)\left(1-\omega\right)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}\left(\phi+\theta\psi\right)\right)\left(1+\sigma\left(\phi+\theta\psi\right)\right)\Bigg]^{\frac{1}{2}}\Bigg\},
\end{aligned} \tag{E.8}$$

and I write $\rho_A(\tau_d) \equiv \rho^-$ for the smaller root when $\omega < 1$ and $\bar{\rho}_A \equiv \lim_{\tau_d \to 0^+} \rho_A(\tau_d) = \rho_A(0)$ for its limiting value. The equality follows from the continuity of the smaller root in τ_d under (32).

Lemma 5 *Under the restrictions in (32) the following hold:*

1. $\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi) \geq \tau_y > 0$.
2. *For $\omega < 1$, the system (E.7) has exactly one root inside and one root outside the unit circle, so that the equilibrium is unique and bounded and $\rho_A(\tau_d) \in (0,1)$, for every*

$$\tau_d < 1 - \beta + \frac{(1-\beta\omega)(1-\omega)}{\sigma\omega|\phi + \theta\psi|}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi)\right) \tag{E.9}$$

   when $\phi + \theta\psi < 0$, and for every $\tau_d \in [0,1)$ when $\phi + \theta\psi = 0$. The threshold in (E.9) strictly exceeds $1-\beta$, so that the restriction $\tau_d \in [0, 1-\beta)$ maintained in (32) is sufficient but not necessary for economy A. For $\omega = 1$ and $\tau_d > 1-\beta$, by contrast, both roots lie inside the unit circle when $\phi + \theta\psi < 0$. When $\phi + \theta\psi = 0$, the roots are $(1-\tau_d)/\beta \in (0,1)$ and one. Economy B is indeterminate in both cases, although at the real-rate peg this conclusion follows from the unit root rather than from the presence of two strictly stable roots.
3. *For $\omega = 1$, the characteristic polynomial factorizes into $\left(\rho - \frac{1-\tau_d}{\beta}\right)(\rho - 1 - \sigma(\phi + \theta\psi))$. At $\tau_d = 0$, the intertemporal government budget condition selects the unique fiscally-led equilibrium with $\rho_B = 1 + \sigma(\phi + \theta\psi) \in (0,1]$.*
4. *$\rho_A(\tau_d) < \rho_B$ strictly for every $\omega \in (0,1)$.*

Proof. The first claim follows from $\tau_y > 0$ and $\phi + \theta\psi \leq 0$. For the second claim, denote by $p(\rho)$ the characteristic polynomial (31). Evaluating it at one and collecting terms gives

$$p(1) = \frac{\sigma(\phi + \theta\psi)(1 - \beta - \tau_d)}{\beta} - \frac{(1-\beta\omega)(1-\omega)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi)\right). \tag{E.10}$$

Under (32) the first term is weakly negative, because $\sigma(\phi + \theta\psi) \leq 0$ and $1 - \beta - \tau_d > 0$, and the second term is strictly negative for $\omega < 1$ by the first claim, so that $p(1) < 0$. More generally, when $\phi + \theta\psi < 0$, the inequality $p(1) < 0$ is equivalent to (E.9), while when $\phi + \theta\psi = 0$, the first term vanishes and $p(1) = -\frac{(1-\beta\omega)(1-\omega)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi)\right) < 0$ for every τ_d. Since $p(0) = (1-\tau_d)[1 + \sigma(\phi + \theta\psi)]/\beta > 0$ and the sum of the roots is positive, both roots are positive and exactly one of them lies in $(0,1)$. With one predetermined variable in the system (E.7), this delivers a unique bounded equilibrium. For $\omega = 1$, the terms involving $1 - \omega$ vanish. When $\phi + \theta\psi < 0$ and $\tau_d > 1 - \beta$, both roots therefore lie strictly inside the unit circle. If instead $\phi + \theta\psi = 0$, the factorization in the third claim gives the roots $(1-\tau_d)/\beta \in (0,1)$ and one. The unit root leaves the initial value of the jump variable unrestricted among a continuum of bounded paths, so that economy B is indeterminate at this boundary as well.

For the third claim, note that under $\omega = 1$ the terms involving $(1-\omega)$ vanish, and $1 + \sigma(\phi + \theta\psi) \in (0,1]$ follows from (32). For the fourth claim, evaluate p at $1 + \sigma(\phi + \theta\psi)$, which gives

$$p(1 + \sigma(\phi + \theta\psi)) = -\frac{(1-\beta\omega)(1-\omega)}{\beta\omega}\left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi)\right)[1 + \sigma(\phi + \theta\psi)] < 0 \tag{E.11}$$

for every $\omega \in (0,1)$, so that $1+\sigma(\phi+\theta\psi)$ lies strictly between the two roots of p and therefore strictly above $\rho_A(\tau_d)$. ■

Lemma 6 *Under the restrictions in (32), for a one-off unanticipated deficit shock, the bounded equilibrium of economy A for $\tau_d \in [0, 1-\beta)$ and the fiscally-led equilibrium of economy B at $\tau_d = 0$ can be expressed as*

$$y_t = \chi(d_t + \varepsilon_t), \qquad \mathbb{E}_t d_{t+1} = \rho(d_t + \varepsilon_t), \qquad \chi \equiv \frac{1-\tau_d-\beta\rho}{\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)}, \tag{E.12}$$

where $\rho = \rho_A(\tau_d)$ for $\omega < 1$ and $\rho = \rho_B = 1+\sigma(\phi+\theta\psi)$ for $\omega = 1$ and $\tau_d = 0$.

Proof. For $\omega < 1$, the expression follows as the standard eigenvector representation of the stable subspace of the system (E.7), with the third expression in (E.12) obtained from the first row of that system, as in Rachel and Ravn (2025). For $\omega = 1$ and $\tau_d = 0$, the second row of the system (E.7) is decoupled and yields $\mathbb{E}_t y_{t+1} = [1+\sigma(\phi+\theta\psi)] y_t$, so that $y_{t+k} = \rho_B^k y_t$. Iterating the first row of (E.7) forward along the no-Ponzi condition gives $d_t + \varepsilon_t = \left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)\right) \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k y_{t+k} = \left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)\right) y_t / (1-\beta\rho_B)$, which is the third expression in (E.12), and $\mathbb{E}_t d_{t+1} = \rho_B(d_t+\varepsilon_t)$ then follows. ■

In the expressions that follow, I set $\rho = \bar{\rho}_A$ for economy A, obtained by taking $\tau_d \to 0^+$, and $\rho = \rho_B$ for economy B, evaluated at $\tau_d = 0$. The third expression of (E.12) then yields

$$1-\beta\rho = \left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)\right)\chi, \tag{E.13}$$

and, for a one-off date zero deficit shock with no subsequent disturbances, the forward-iterated Phillips curve together with (E.12) and (E.13) gives

$$\pi_0 = \frac{\kappa\chi}{1-\beta\rho}(d_0+\varepsilon_0) = \frac{\kappa}{\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)}(d_0+\varepsilon_0), \tag{E.14}$$

while equation (E.5) together with (E.12) gives

$$q_0 = -\frac{(\phi+\psi)\chi}{1-\beta\delta\rho}(d_0+\varepsilon_0). \tag{E.15}$$

E.5 Proof of proposition 2

We can now establish proposition 2 with the help of the two previous lemmas, which establish the respective equilibrium selections and the ordering of the roots. Still left open are the derivations of equation (33), the sign of the two denominators, and the equations (34) and (35).

Substituting equations (E.14) and (E.15) into (30) gives

$$(d_0+\varepsilon_0)\left[1 + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\frac{\kappa}{\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi+\theta\psi)} + \theta\delta\frac{D^{SS}}{Y^{SS}}\frac{(\phi+\psi)\chi}{1-\beta\delta\rho}\right] = \varepsilon_0. \tag{E.16}$$

Multiplying through by $\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi)$, using the identity (E.13) to replace $\left(\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi)\right)\chi$ by $1 - \beta\rho$, and substituting the result back into equation (E.14) then gives

$$\pi_0 = \frac{\kappa\,\varepsilon_0}{\tau_y - \frac{D^{SS}}{Y^{SS}}(\phi + \theta\psi) + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa + \theta\delta\frac{D^{SS}}{Y^{SS}}(\phi + \psi)\frac{1-\beta\rho}{1-\beta\delta\rho}}, \tag{E.17}$$

which is equation (33) once ρ is set to $\bar{\rho}_A$ and to ρ_B respectively, using lemma 6.

To sign the denominator of equation (E.17), write it as $\tau_y + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa + \frac{D^{SS}}{Y^{SS}}\Lambda$ with

$$\Lambda \equiv -(\phi + \theta\psi) + \theta\delta(\phi + \psi)\frac{1-\beta\rho}{1-\beta\delta\rho}. \tag{E.18}$$

Since $\rho \in (0,1]$ and $\delta \in [0,1)$, the ratio $(1-\beta\rho)\,/\,(1-\beta\delta\rho)$ lies in $(0,1]$, and $\theta\delta < 1$. If $\phi + \psi > 0$, then $\Lambda \geq 0$ because both terms are weakly positive. Because $\psi \geq 0$ implies $\phi + \theta\psi \leq \phi + \psi \leq 0$ and hence $|\phi + \psi| \leq |\phi + \theta\psi|$, if $\phi + \psi \leq 0$, the second term of Λ is bounded in absolute value by $\theta\delta|\phi + \theta\psi| \leq |\phi + \theta\psi|$, so that $\Lambda \geq 0$ again. Both denominators therefore weakly exceed $\tau_y + \frac{1}{\beta}(1-\theta)\frac{D^{SS}}{Y^{SS}}\kappa$ and are strictly positive, which also establishes the claim that the peg weakly maximizes impact inflation.

Finally, the two denominators differ only through the last term, so that

$$\Delta = \theta\delta\frac{D^{SS}}{Y^{SS}}(\phi + \psi)\left[\frac{1-\beta\bar{\rho}_A}{1-\beta\delta\bar{\rho}_A} - \frac{1-\beta\rho_B}{1-\beta\delta\rho_B}\right]. \tag{E.19}$$

For any x and y in $(0,1]$,

$$\frac{1-\beta x}{1-\beta\delta x} - \frac{1-\beta y}{1-\beta\delta y} = \frac{\beta(1-\delta)(y-x)}{(1-\beta\delta x)(1-\beta\delta y)}, \tag{E.20}$$

so that, setting $x = \bar{\rho}_A$ and $y = \rho_B = 1 + \sigma(\phi + \theta\psi)$,

$$\Delta = \frac{\theta\beta\delta(1-\delta)\frac{D^{SS}}{Y^{SS}}(\phi + \psi)\left[1 + \sigma(\phi + \theta\psi) - \bar{\rho}_A\right]}{(1-\beta\delta\bar{\rho}_A)\left(1-\beta\delta\left[1 + \sigma(\phi + \theta\psi)\right]\right)}, \tag{E.21}$$

which is equation (34). The denominator of Δ is strictly positive because $\beta\delta\rho < 1$ for both roots. By lemma 5, the bracket $1 + \sigma(\phi + \theta\psi) - \bar{\rho}_A$ is strictly positive for every $\omega \in (0,1)$ and equals zero at $\omega = 1$. The remaining factors are θ, which is zero if and only if the government issues no indexed debt, the net maturity exposure $\delta(1-\delta)$, which is zero at zero maturity and converges to zero at the infinite geometric-duration boundary, and $\phi + \psi$, which is zero if and only if the required real return on the indexed instrument does not respond to the deficit-induced boom. Since both denominators of equation (33) are strictly positive, $\pi^A_{\varepsilon,0} > \pi^B_{\varepsilon,0}$ if and only if $\Delta < 0$, and fixing $\omega \in (0,1)$, $\delta \in (0,1)$ and $\phi + \psi \neq 0$ yields equation (35). Setting $\phi = \psi = 0$ in equation (E.17) proves corollary 1. ■

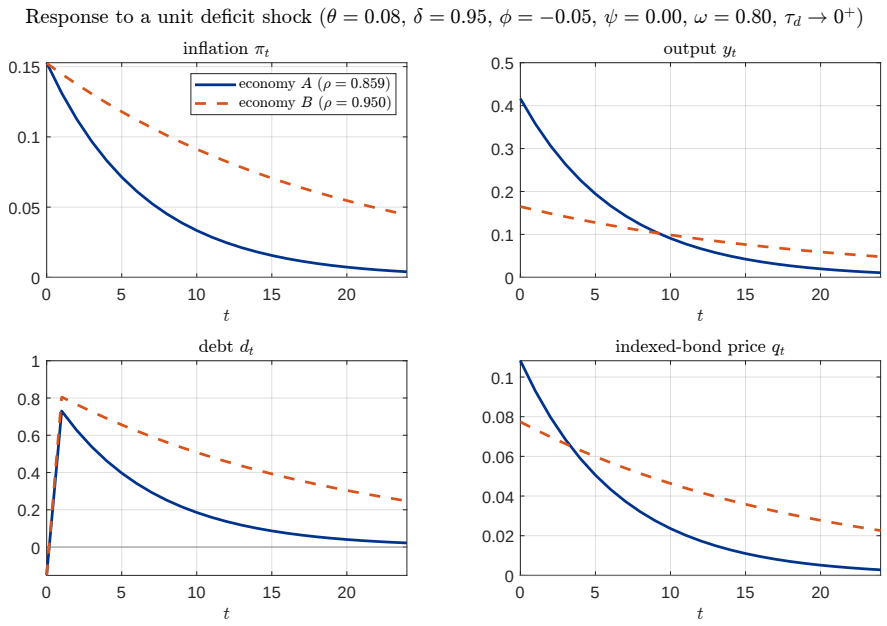


Figure E.1: Comparison of the responses of some variables in economies A and B to a 1% impulse in deficits relative to GDP.

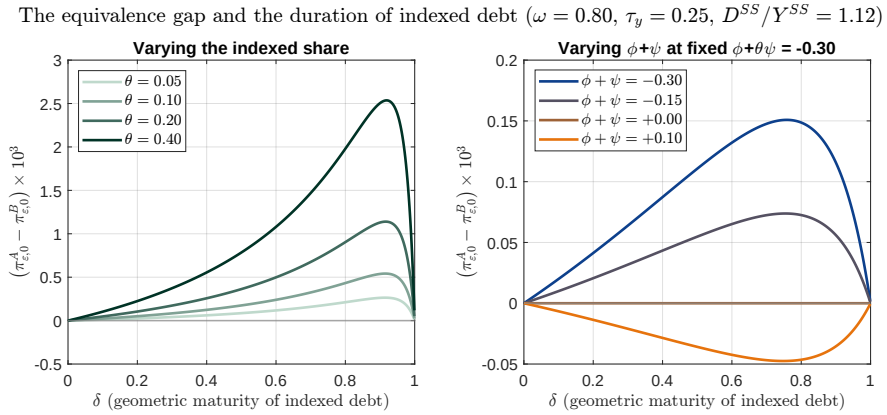


Figure E.2: Determinants of variation in the impact inflation gap between economies A and B. In the right panel, let $h_1, \ldots, h_n$ denote the reported values of $h = \phi + \psi$ and define $\bar{g} \equiv \min\{0, h_1, \ldots, h_n\}$. The expected portfolio-return response is held fixed at the admissible value $\phi + \theta\psi = \bar{g} \leq 0$, while $\phi + \psi$ is varied across the curves. For each curve, $\psi = [h - \bar{g}] / (1 - \theta)$ and $\phi = [\bar{g} - \theta h] / (1 - \theta)$, which also ensures $\psi \geq 0$.

F On how to define inflation-indexed government debt

In this part of the appendix, I establish formally the inflation-indexation convention used in the paper. In particular, I show that if b_t is the base principal of an indexed bond measured in currency units at the date of issuance, then its payment at $t + 1$ is exactly $(P_{t+1}/P_t)b_t$. I also prove that this formulation is economically equivalent to a real-debt contract after the change of variables $d_t = b_t/P_t$. From this, I conclude that the specification of the indexed payment in the paper does not define a different security from the indexed debt considered in Cochrane (2023). Instead, the indexed-debt formulation records the same security in base-principal units, which is closer to the manner in which the principal of an actual indexed bond is quoted and subsequently adjusted.

The indexation contract and its units

Let the monetary unit be called one unit of currency and let $P_t > 0$ denote the number of currency units required to purchase one unit of the composite consumption good. Hence, P_t has units of currency per good. Let $I_t > 0$ be the reference price index written in any fixed base-year normalization. In the ideal case in which the reference basket coincides with the model's composite good, there is a constant $\zeta > 0$ such that $I_t = \zeta P_t$, and therefore

$$\frac{I_{t+1}}{I_t} = \frac{P_{t+1}}{P_t} = \Pi_{t+1}. \tag{F.1}$$

Consider an indexed bond issued at time t with base principal b_t currency units. Indexation means that its time-$(t+1)$ currency payment is its base principal multiplied by the change in the reference index. Its payment X^b_{t+1} is therefore

$$X^b_{t+1} = b_t \frac{I_{t+1}}{I_t} = b_t \frac{P_{t+1}}{P_t} = \Pi_{t+1} b_t. \tag{F.2}$$

The purchasing power of this payment at time $t+1$ is

$$\frac{X^b_{t+1}}{P_{t+1}} = \frac{b_t}{P_t}. \tag{F.3}$$

Thus, the bond repays at $t+1$ the number of goods that its base principal could purchase at the time of issuance. In particular, a bond issued at $t-1$ pays $\Pi_t b_{t-1}$ at t. This timing also implies that the payment of a bond issued at t is $\Pi_{t+1} b_t$, not $\Pi_t b_t$. Equation (F.2) is the idealized one-period counterpart of the adjustment of coupon and principal payments on inflation-indexed government bonds, abstracting from indexation lags, deflation floors, tax treatment, and differences between the model price index and the contractual reference index.

The level assigned to I_t is immaterial because only the ratio I_{t+1}/I_t enters the contract. This observation separates the arbitrary normalization of an index number from the economically meaningful promise to preserve purchasing power.

Money, the unit of account, and the cashless limit

Likewise, it is possible to clarify the indexation convention by considering a monetary economy before taking its cashless limit. Let a representative household have preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[u(c_t) + \chi \log\left(\frac{M_t}{P_t}\right)\right], \qquad \chi > 0, \tag{F.4}$$

and the nominal budget constraint

$$P_t c_t + M_t + Q_t B_t + q_t b_t \leq P_t y_t - P_t \tau_t + M_{t-1} + B_{t-1} + \Pi_t b_{t-1}. \tag{F.5}$$

Here, M_t denotes non-interest-bearing money carried from t to $t+1$, B_t denotes a one-period nominal bond paying one unit of currency at $t+1$, and b_t denotes the base principal of the indexed bond in (F.2). Consolidation of the treasury and the issuer of money gives rise to the government flow budget constraint

$$M_{t-1} + B_{t-1} + \Pi_t b_{t-1} = P_t s_t + M_t + Q_t B_t + q_t b_t, \tag{F.6}$$

where s_t is the primary real surplus excluding seigniorage. Every term has units of currency.

Define the real stochastic discount factor by

$$\mathcal{M}_{t,t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}. \tag{F.7}$$

The first-order conditions for nominal and indexed debt are

$$Q_t = \mathbb{E}_t \left[\frac{\mathcal{M}_{t,t+1}}{\Pi_{t+1}} \right], \qquad q_t = \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \right]. \tag{F.8}$$

The cancellation of Π_{t+1} from the indexed-bond pricing equation follows from the nominal payment $\Pi_{t+1} b_t$ and the conversion of nominal marginal utility at $t+1$ into real marginal utility. The first-order condition for money is

$$\frac{M_t}{P_t} = \frac{\chi}{u'(c_t)(1 - Q_t)}. \tag{F.9}$$

Suppose that $1 - Q_t \geq \underline{\iota} > 0$ and that $u'(c_t)$ is bounded away from zero. Equation (F.9) then implies $M_t / P_t \to 0$ as $\chi \downarrow 0$. Assume also that initial real balances belong to the same limiting family. Then $M_{t-1}/P_t = (M_{t-1}/P_{t-1})/\Pi_t \to 0$ whenever gross inflation remains finite and strictly positive. Divide (F.6) by P_t, use $d_t \equiv b_t / P_t$, and take this limit to obtain

$$\frac{B_{t-1}}{P_t} + d_{t-1} = s_t + Q_t \frac{B_t}{P_t} + q_t d_t. \tag{F.10}$$

Multiplying (F.10) by P_t and substituting $d_{t-1} = b_{t-1}/P_{t-1}$ and $d_t = b_t / P_t$ yields

$$B_{t-1} + \frac{P_t}{P_{t-1}} b_{t-1} = P_t s_t + Q_t B_t + q_t b_t. \tag{F.11}$$

Thus, the flow constraint used in the paper is the cashless limit of a monetary economy with a

suitably defined indexation contract. The limit removes equilibrium real money balances. It does not remove the monetary unit in which prices and nominal contracts are stated. Here, the currency unit is the unit of account, while the nominal bond is a claim promising one unit of that currency. One may use the nominal bond as a financial numéraire, but that is a separate choice from the definition of the monetary unit.

That being said, it is possible to link my indexation choice to the usual formulation of indexed or real government debt. Define $d_t \equiv \frac{b_t}{P_t}$. Then, $\Pi_{t+1} b_t = P_{t+1} d_t$, $q_t b_t = P_t q_t d_t$. The first equality says that a base principal of b_t currency units indexed from t to $t+1$ is exactly the same promise as d_t units of real debt, each of which pays P_{t+1} currency units at $t+1$, which is related to the convention used in chapter 8 of Cochrane (2023).

Initial conditions and the Niepelt critique

The preceding distinction also clarifies what can be established concerning the initial-value objection of Niepelt (2004). Setting $P_{-1} = 1$ is a permissible normalization of the monetary unit. It is not, by itself, an equilibrium condition that explains the absolute value of the initial unit of account. The economically meaningful initial objects are real quantities such as B_{-1}/P_0 and $d_{-1} = b_{-1}/P_{-1}$, together with relative price levels and inflation rates.

To examine consistency with debt issuance, suppose a state z is realized at time 0 and was assigned strictly positive conditional probability at time -1. Let $\mathcal{S}_0(z)$ denote the state-contingent present value of primary surpluses, calculated using the equilibrium stochastic discount factor from time 0 onward. The time-0 government debt valuation equation is

$$\frac{B_{-1}}{P_0(z)} + d_{-1} = \mathcal{S}_0(z). \tag{F.12}$$

If the inherited bonds were issued or traded at time -1 under the same stochastic policy regime, their pricing equations are

$$Q_{-1} \frac{B_{-1}}{P_{-1}} = \mathbb{E}_{-1}\left[M_{-1,0} \frac{B_{-1}}{P_0(z)} \right], \qquad q_{-1} d_{-1} = \mathbb{E}_{-1}\left[M_{-1,0} d_{-1} \right]. \tag{F.13}$$

Adding these equations and using (F.12) gives

$$Q_{-1} \frac{B_{-1}}{P_{-1}} + q_{-1} d_{-1} = \mathbb{E}_{-1}\left[M_{-1,0} \mathcal{S}_0(z) \right]. \tag{F.14}$$

Equation (F.14) is sn ex-ante consistency condition that is absent when an inherited nominal liability is simply inserted after its issuance date. It shows that state-contingent revaluations of nominal debt need not violate household optimality. The realized return may differ across states, but the security is priced using the same probability distribution and policy rule that generate those returns. Indexed debt does not receive a surprise revaluation in this calculation because its real payoff is

the predetermined amount d_{-1}.

If time -1 is itself the first issuance date, there are no older liabilities, and money creation is absent, the time-(-1) flow constraint also requires the issuance proceeds to finance that period's primary deficit. Equations (F.14) and the time-(-1) flow constraint then jointly imply

$$-s_{-1} = Q_{-1}\frac{B_{-1}}{P_{-1}} + q_{-1}d_{-1} = \mathbb{E}_{-1}\left[\mathcal{M}_{-1,0}\mathcal{S}_0(z)\right]. \tag{F.15}$$

This is the ex-ante intertemporal restriction emphasized by Niepelt (2004). The construction avoids a surprise-return inconsistency by satisfying that restriction. It does neither prove that an arbitrary non-Ricardian policy can be appended after the issuance date, nor does it settle the terminological dispute over whether a policy satisfying (F.15) should be called Ricardian ex ante.

The relevant consistency result therefore holds only under explicit conditions. The initial debt must have been issued or willingly held under the same regime, the time-0 fiscal innovation must belong to the support of the shock distribution perceived at time -1, and the asset-pricing equations, government flow constraint, and transversality condition must all hold under that regime. A deterministic experiment in which a fiscal change assigned zero probability at the issuance date is imposed unexpectedly does not meet these conditions. Such an experiment can be used as a local comparative exercise, but it cannot on its own be described as a rational-expectations equilibrium extending through the issuance date.

Introducing money makes the monetary denomination and the initial nominal liability a bit clearer, but money alone does not mechanically overturn the critique. Daniel (2007) establishes the important result that a zero level of initial government debt does not render fiscal price-level restrictions ineffective when the government's asset menu is restricted to standard nominal debt and money and policy is stochastic. That result does not depend on treating indexed debt as different from real debt, and it does not imply that an arbitrary surprise after debt issuance is innocuous. In the cashless limit considered here, the safest conclusion is therefore that the model does not exhibit the surprise-issuance inconsistency if $P_{-1} = 1$ is identified as a normalization, the fiscal shock process is specified ex ante, inherited debt satisfies (F.13), and any first issuance satisfies (F.15). If these conditions are not imposed, the indexation formula $\Pi_t b_{t-1}$ cannot by itself resolve the initial-value objection.

In sum, therefore, the payment $\Pi_t b_{t-1}$ is the correct nominal payment on indexed base principal and is invariant to the normalization of the price index and the monetary unit.

G Long-term debt and debt indexation

In this part of the appendix, I briefly derive the debt valuation equation under complete markets with long-term debt.

Under a deterministic real discount factor and following the exposition in the main body of the

text, bond pricing kernels for long-term assets maturing at time $(t+j)$ evaluated at time t are given by:

$$Q_t^{(t+j)} = \mathbb{E}_t\left(\beta^j \frac{P_t}{P_{t+j}}\right) \qquad q_t^{(t+j)} = \beta^j,$$

reflecting that the price of inflation-indexed debt depends on its remaining maturity alone and not on the state, as its face value accounts for changes to the price level between issuance and redemption.

Throughout, $b_t^{(t+j)}$ denotes face value in date t currency units, so that a unit of indexed debt issued at t pays $b_t^{(t+j)} P_{t+j}/P_t$ at maturity. In this context, the government flow budget condition is given by:

$$B_{t-1}^{(t)} + \Pi_t b_{t-1}^{(t)} = P_t s_t + \sum_{j=1}^{\infty} Q_t^{(t+j)} \left(B_t^{(t+j)} - B_{t-1}^{(t+j)}\right) + \sum_{j=1}^{\infty} q_t^{(t+j)} \left(b_t^{(t+j)} - \Pi_t b_{t-1}^{(t+j)}\right).$$

This condition states that in each period t, the payout of maturing debt (left-hand side) must be equal to the nominal surpluses raised *plus* the possible income from issuing additional debt maturing as a later point in the future (relative to what had already been issued before). Governments can also redeem more bonds than they issue, in which case either of the sums on the right-hand side can also be negative.

That flow condition keeps track of mounting payments on inflation-indexed debt by adjusting the prospective cost of serving indexed debt in each period by the accumulated face value payments, given by $\left(b_t^{(t+j)} - \Pi_t b_{t-1}^{(t+j)}\right)$. In sum, the net proceeds from indexed issuance on the right-hand side of the previous equation get diminished when current inflation Π_t has been high, as that inflation is reflected in the obligations that the government will have as that long-term inflation-indexed debt matures.

Grouping terms in the previous equation yields:

$$\sum_{j=0}^{\infty} Q_t^{(t+j)} B_{t-1}^{(t+j)} + \Pi_t \sum_{j=0}^{\infty} q_t^{(t+j)} b_{t-1}^{(t+j)} = P_t s_t + \sum_{j=1}^{\infty} Q_t^{(t+j)} B_t^{(t+j)} + \sum_{j=1}^{\infty} q_t^{(t+j)} b_t^{(t+j)}, \tag{G.1}$$

where I have used that $Q_t^{(t)} = q_t^{(t)} = 1$. Let the real value of debt now be defined as:

$$V_t^d := \sum_{j=0}^{\infty} Q_t^{(t+j)} \frac{B_{t-1}^{(t+j)}}{P_t} + \sum_{j=0}^{\infty} q_t^{(t+j)} \frac{b_{t-1}^{(t+j)}}{P_{t-1}}.$$

Focus on the right-hand side of equation (G.1). I now rewrite the two summative terms again to obtain $\mathbb{E}_t V_{t+1}^d$. Dividing those terms by P_t:

$$\sum_{j=1}^{\infty} Q_t^{(t+j)} \frac{B_t^{(t+j)}}{P_t} + \sum_{j=1}^{\infty} q_t^{(t+j)} \frac{b_t^{(t+j)}}{P_t}.$$

Shifting the index from $j = 1$ to $j = 0$ gives:

$$\sum_{j=0}^{\infty} Q_t^{(t+j+1)} \frac{B_t^{(t+j+1)}}{P_t} + \sum_{j=0}^{\infty} q_t^{(t+j+1)} \frac{b_t^{(t+j+1)}}{P_t}.$$

$q_t^{(t+j+1)} = \beta q_{t+1}^{(t+j+1)}$ by the bond pricing kernels defined previously, and $Q_t^{(t+j+1)}/P_t = \beta\, \mathbb{E}_t \left(Q_{t+1}^{(t+j+1)}/P_{t+1} \right)$ by the law of iterated expectations. Thus, the previous expression becomes

$$\beta\, \mathbb{E}_t \left[\sum_{j=0}^{\infty} Q_{t+1}^{(t+j+1)} \frac{B_t^{(t+j+1)}}{P_{t+1}} + \sum_{j=0}^{\infty} q_{t+1}^{(t+j+1)} \frac{b_t^{(t+j+1)}}{P_t} \right] = \beta\, \mathbb{E}_t V_{t+1}^d$$

by the definition of V_t^d. Dividing equation (G.1) through by P_t therefore delivers the recursion $V_t^d = s_t + \beta\, \mathbb{E}_t V_{t+1}^d$. Now, iterating that recursion forward and applying a transversality condition of the form

$$\lim_{T \to \infty} \beta^T\, \mathbb{E}_t \left[\sum_{j=0}^{\infty} Q_{t+T}^{(t+j+T)} \frac{B_{t+T}^{(t+j+T)}}{P_{t+T}} + \sum_{j=0}^{\infty} q_{t+T}^{(t+j+T)} \frac{b_{t+T}^{(t+j+T)}}{P_{t+T}} \right] = 0,$$

yields the debt valuation equation with inflation-indexed debt:

$$\sum_{j=0}^{\infty} Q_t^{(t+j)} \frac{B_{t-1}^{(t+j)}}{P_t} + \sum_{j=0}^{\infty} q_t^{(t+j)} \frac{b_{t-1}^{(t+j)}}{P_{t-1}} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j}, \tag{G.2}$$

which is a straightforward generalization of the government debt valuation equation exposed, for instance, in Cochrane (2001).